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Design and Implementation of a Bidirectional Onboard EV Charger for V2X Systems

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يَرْفَعِ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ وَاللَّهُ بِمَا تَعْمَلُونَ خَبِيرٌ

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Abstract

As global efforts to reduce CO_2 emissions and improve energy efficiency intensify, Electric Vehicles (EVs) are evolving from conventional transportation systems into active components of the modern energy grid. This project, titled" Design and Implementation of a Bidirectional Onboard EV Charger for V2X Systems", addresses the limitations of traditional unidirectional charging by enabling the Vehicle-to-Everything (V2X) paradigm. Through bidirectional power flow, EVs can operate as distributed energy storage units, supporting Vehicle-to-Grid (V2G), Vehicle-to-Home (V2H), and Vehicle-to-Load (V2L) applications.

The proposed system adopts a two-stage bidirectional power conversion architecture. The front-end AC/DC stage operates as a Totem-Pole Power Factor Correction (PFC) converter during battery charging and as a grid-synchronized single-phase inverter during energy discharge. The second stage is an isolated bidirectional DC/DC converter based on the Dual Active Bridge (DAB) topology, operating in buck mode for battery charging and boost mode during discharging, ensuring efficient and flexible energy transfer.

Model-Based Development (MBD) is employed for implementing the control algorithms, enabling a seamless transition from simulation to real-time hardware execution. Furthermore, IoT-based communication is integrated to allow real-time monitoring of the State of Charge (SOC) and remote system supervision, enhancing compatibility with smart grid infrastructure.

The project demonstrates a complete workflow from system modeling and control design to practical hardware implementation, highlighting the role of bidirectional onboard EV chargers as key enablers for transforming EVs into intelligent energy assets that contribute to a sustainable and resilient power system.

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Chapter 1
Introduction

1.1 Background and Motivation

1.1.1 Overview of Global EV Transition

1.1.1.1 The Global Energy Landscape and Environmental Drivers

Transportation is undergoing a fundamental transformation as the world shifts from fossil fuels to cleaner alternatives. For decades, gasoline and diesel vehicles have dominated but contributed significantly to air pollution and greenhouse gas emissions. With growing climate concerns, electric vehicles (EVs) and plug-in hybrids (PHEVs) have emerged as sustainable solutions, representing not just new vehicles but a new approach to mobility and energy.

Governments and industry leaders worldwide are investing heavily in EV technology. Since 2009, collaborative efforts between Europe, the United States, and Japan have established international charging standards, promoting global compatibility and reliability. This standardization is crucial for widespread adoption and user convenience. Technologically, EVs offer superior efficiency—electric motors achieve 80–95% efficiency compared to less than 20% for combustion engines. This translates to significant energy savings and lower operating costs. Projections indicate explosive growth: by 2030, the global EV fleet is expected to reach approximately 245 million vehicles, increasing electricity demand for charging from 20 TWh in 2020 to about 280 TWh. This growth necessitates smarter grids, expanded renewable energy integration, and enhanced charging infrastructure.

Beyond environmental benefits, EV adoption drives innovation in battery technology, creates clean energy jobs, and encourages cities to develop cleaner, quieter urban environments. For many nations, EVs represent a critical pathway toward carbon neutrality and reduced oil dependency.

1.1.1.2 Paradigm Shift: From Passive Consumption to V2X Participation

Traditionally, EVs functioned solely as energy consumers, drawing power from the grid without contributing back. The Vehicle-to-Everything (V2X) concept transforms this relationship, enabling EVs to serve as distributed energy resources through bidirectional power flow.

Key V2X applications include:

- **Vehicle-to-Load (V2L)**: Powering external devices and equipment directly from the EV battery.
- **Vehicle-to-Home (V2H)**: Supplying household electricity during outages or peak pricing periods.
- **Vehicle-to-Grid (V2G)**: Injecting power back into the grid to support stability and manage demand.

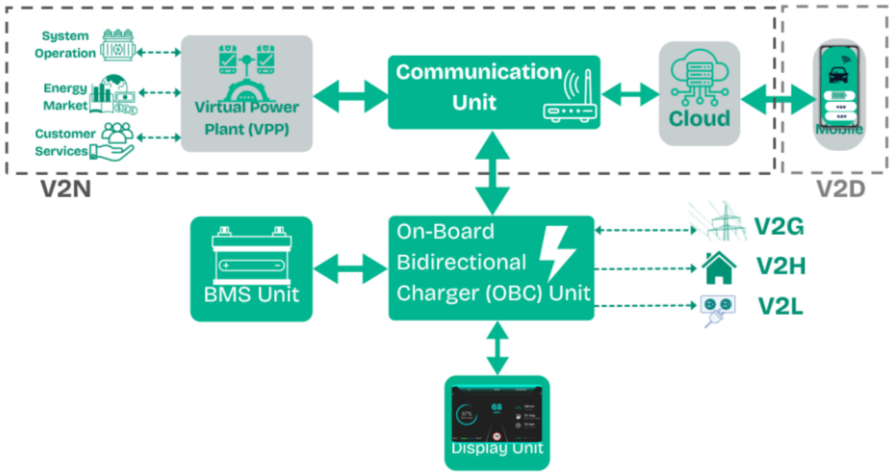


Figure 1.1: Integrated V2X ecosystem showing energy flows (V2G, V2H, V2L, V2D, V2N)

This evolution aligns with the broader CASE Mobility framework (Connected, Autonomous, Shared, and Electric), where vehicles become integrated components of smart energy ecosystems. The potential impact is substantial: millions of EV batteries could collectively stabilize grids, provide emergency backup, and optimize energy costs for owners.

1.1.1.3 Bidirectional Onboard EV Chargers

The onboard charger (OBC) is central to enabling V2X functionality, managing power conversion between external supplies and the vehicle battery. While traditional OBCs are unidirectional, V2X applications require bidirectional designs that support energy exchange in both directions.

Bidirectional onboard chargers integrate advanced power electronics, control algorithms, and protection systems to facilitate safe, efficient operation across multiple modes. Compared to external bidirectional chargers, onboard solutions offer greater vehicle integration, accessibility, and reduced infrastructure dependency. They enable closer coordination with vehicle systems like the Battery Management System (BMS), enhancing overall performance and safety.

As the interface between vehicles, grids, homes, and loads, bidirectional OBCs must balance efficiency, reliability, compactness, and regulatory compliance to realize the full potential of V2X energy interactions.

1.2 Problem Statement

Most existing on-board electric vehicle (EV) charging systems are designed to operate with unidirectional power flow, where electrical energy is transferred only from the grid to the vehicle battery. In this conventional approach, electric vehicles function solely as energy consumers, without the ability to interact dynamically with the electrical grid or local energy systems. As the number of EVs continues to increase worldwide, this limitation becomes more significant, as a large amount of available battery storage capacity remains unused for most of the vehicle's lifetime.

The lack of bidirectional capability restricts the utilization of EV batteries as distributed energy storage units. Consequently, electric vehicles are unable to support grid stability, assist in peak load

management, or provide backup power during grid disturbances and outages. This limitation reduces the potential benefits of EV integration within smart grids and weakens their role in modern energy management strategies.

In addition to functional limitations, many traditional on-board charger designs rely on power factor correction (PFC) and DC/DC converter topologies that are not optimized for bidirectional operation. When these conventional topologies are extended to support reverse power flow, they often suffer from increased switching losses, reduced overall efficiency, and higher thermal stress on power electronic components. Such drawbacks negatively impact system reliability, efficiency, and suitability for compact on-board applications.

Furthermore, conventional charger architectures frequently lack accurate sensing and monitoring mechanisms required for safe and controlled bidirectional operation. Insufficient voltage and current measurement can compromise system protection, control accuracy, and compliance with grid power quality requirements. These challenges highlight the need for advanced on-board EV charger architectures that combine high efficiency, bidirectional power flow, galvanic isolation, accurate sensing, and compact design, while remaining practical for real-world implementation in V2X-enabled energy systems.

1.3 Motivation

The rapid growth in electric vehicle (EV) adoption is reshaping modern transportation and energy systems. While this growth supports environmental sustainability and emission reduction goals, it also introduces several challenges to existing electrical grids. Large-scale EV charging can lead to increased peak demand, power quality degradation, and inefficient utilization of available energy resources, especially when charging is concentrated during specific hours of the day.

Conventional unidirectional on-board chargers restrict electric vehicles to passive energy consumption from the grid. As a result, the substantial energy storage capacity available in EV batteries remains largely underutilized. This limitation prevents EVs from actively contributing to grid support functions such as peak load reduction, voltage regulation, and emergency power supply, as well as participating in local energy management at the residential or community level.

Vehicle-to-Everything (V2X)-enabled charging systems provide a promising solution to these challenges by allowing electric vehicles to operate as distributed energy storage units. Through bidirectional power flow, EVs can not only draw energy from the grid but also return stored energy when needed. However, achieving reliable and efficient bidirectional operation places stringent requirements on the charger design, including high power factor, low harmonic distortion, galvanic isolation, and precise control.

These requirements motivate the use of advanced power electronic converter topologies that outperform traditional solutions. Topologies such as the Totem-Pole Power Factor Correction (PFC) converter and the Dual Active Bridge (DAB) DC-DC converter offer improved efficiency, enhanced power quality, and inherent bidirectional capability. The motivation of this project lies in translating these advanced power electronic concepts into a practical, circuit-level implementation suitable for on-board EV charging systems, with a focus on enabling V2X functionality in real-world applications.

1.4 Aims and Research Objectives

1.4.1 Aims

The primary aim of this project is to design and implement a high-efficiency bidirectional on-board electric vehicle (EV) charger capable of supporting both grid-to-vehicle (G2V) and vehicle-to-grid/vehicle-to-home (V2G/V2H) operation. The proposed system enables controlled bidirectional power flow across all power conversion stages, allowing energy exchange between the utility grid, the DC link, and the EV battery.

The system architecture consists of a bidirectional AC/DC stage and a bidirectional isolated DC/DC stage.

The AC/DC stage operates as a bridgeless Totem-Pole Power Factor Correction (PFC) converter during charging mode to ensure near-unity power factor and low total harmonic distortion, and as a single-phase grid-connected inverter during discharging mode to inject synchronized AC power into the grid or local loads.

The isolated DC/DC stage is implemented using a Dual Active Bridge (DAB) topology, which enables bidirectional power transfer between the DC link and the EV battery. During charging operation, the DAB operates in buck mode to regulate battery charging, while during discharging operation, it operates in boost mode to raise the battery voltage to the DC-link level required for inverter operation. This configuration ensures flexible energy management and galvanic isolation under both operating modes.

In addition to bidirectional power conversion, the project aims to bridge the gap between theoretical power electronic design and practical implementation. The design emphasizes high efficiency, power density, thermal reliability, and accurate sensing to ensure stable and safe operation under varying grid and load conditions.

To support intelligent energy management and smart grid integration, the system incorporates a model-based digital control framework and an IoT-based communication layer, enabling real-time monitoring, diagnostics, and remote control of the charging and discharging processes.

1.4.2 Research Objectives

To achieve the stated aims, the following objectives are defined:

1. Design and implement a bidirectional AC/DC power stage capable of operating as a bridgeless Totem-Pole PFC during G2V mode and as a single-phase grid-connected inverter during V2G/V2H mode, ensuring compliance with grid standards related to power factor, harmonic distortion, and synchronization.

2. Design and analyze a bidirectional Dual Active Bridge (DAB) DC/DC converter to provide galvanically isolated power transfer between the DC link and the EV battery, operating in buck mode during charging and boost mode during discharging.

3. Develop custom PCB layouts for all power and control stages, with emphasis on thermal management, optimized component placement, and reinforced isolation.

4. Develop precision voltage and current sensing circuits using isolated amplifiers and Hall-effect sensors to enable accurate closed-loop control, protection, and fault detection.

5. Implement digital control algorithms using a Model-Based Development (MBD) approach, allowing systematic simulation, automatic code generation, and real-time execution on the TI C2000 F28379D microcontroller.

6. Incorporate an IoT-based communication and monitoring system to enable real-time data acquisition, state-of-charge estimation, remote diagnostics, and supervisory control of bidirectional charging operations.

7. Validate the proposed system through detailed simulation and experimental testing, evaluating efficiency, stability, power quality, and bidirectional power flow performance across both G2V and V2G operating modes.

Chapter 2
Literature Review

2.1 Introduction to Literature Review

This chapter presents a comprehensive review of existing research related to bidirectional on-board electric vehicle (EV) chargers and Vehicle-to-Everything (V2X) systems. The literature is analyzed across four interrelated domains: (1) AC/DC converter topologies for grid interfacing, (2) bidirectional DC/DC converter architectures for battery integration, (3) digital control and sensing methodologies, and (4) communication frameworks enabling V2X functionality. By critically examining prior studies, highlighting their advantages and limitations, and comparing alternative approaches, this chapter establishes the technical background of the project and identifies the key research gaps that motivate the proposed work.

2.2 AC/DC Converter Topologies

The AC/DC conversion stage plays a critical role in achieving high efficiency, power quality, and bidirectional power flow in EV charging systems. Early studies primarily focused on diode-bridge rectifiers combined with boost Power Factor Correction (PFC) converters due to their simplicity and robustness. However, several comparative analyses reported that this approach suffers from increased conduction losses caused by the diode bridge, leading to a reduction in overall system efficiency by approximately 1–2% when compared to more advanced topologies.

To overcome these limitations, the literature has increasingly shifted toward bridgeless PFC configurations. Among them, the Totem-Pole PFC converter has been widely identified as a promising solution for bidirectional on-board chargers. Multiple studies report that eliminating the input diode bridge significantly reduces conduction losses and component count, thereby improving efficiency and power density. In grid-to-vehicle (G2V) operation, the Totem-Pole PFC is commonly controlled to achieve near-unity power factor and low input current harmonic distortion.

Furthermore, prior research demonstrates that the same power stage can be reconfigured to operate in vehicle-to-grid (V2G), vehicle-to-home (V2H), or vehicle-to-load (V2L) modes, effectively functioning as a grid-synchronized inverter. In this operating mode, a major design challenge highlighted in the literature is the implementation of an appropriate output filter. Several studies emphasize the use of LCL filters to attenuate high-frequency switching harmonics and ensure compliance with grid standards such as IEEE 1547 and IEC 61000-3-2. Experimental results reported in the literature indicate that a properly designed LCL filter can reduce current Total Harmonic Distortion (THD) from typical values of 5–8% to below 3%, which is essential for safe and compliant grid interaction.

2.3 DC/DC Converter Topologies

The isolated DC/DC conversion stage is responsible for providing galvanic isolation, voltage matching between the DC link and the battery pack, and efficient bidirectional power transfer. Among the various topologies discussed in the literature, the Dual Active Bridge (DAB) converter has received significant attention for bidirectional EV charging applications. Researchers consistently highlight its symmetrical structure, capability for bidirectional operation, and suitability for high-frequency switching, which contributes to reduced transformer size and increased power density.

Comparative studies between the DAB converter and conventional phase-shifted fullbridge converters indicate a clear efficiency advantage for the DAB topology. Reported peak efficiencies typically range between 96% and 97% in both charging and discharging modes, exceeding those of traditional isolated converters by approximately 2–3% under comparable operating conditions. The fundamental operating principle of the DAB relies on phase-shift modulation between the primary and secondary bridges, enabling controllable power flow.

2.4 Control Methods and Sensing

The adoption of digital control techniques has fundamentally transformed the design and implementation of modern power electronic systems. The literature shows a clear trend toward Model-Based Development (MBD), enabled by powerful digital signal processors (DSPs) and microcontrollers. Numerous studies report that MBD significantly streamlines the development process by allowing control algorithms to be designed, simulated, verified, and automatically deployed to hardware. Quantitative analyses indicate that this approach can reduce development time by approximately 30–40% compared to conventional manual coding methodologies.

Accurate and reliable sensing is identified in the literature as a prerequisite for effective control and system protection. For current measurement, many implementations favor Hall-effect sensors due to their inherent galvanic isolation and non-intrusive nature. Prior studies report bandwidths exceeding 200 kHz and typical measurement accuracy within ±1–2%, making these sensors suitable for high-frequency switching applications. For high-voltage measurements, isolated amplifiers with reinforced insulation are commonly employed to ensure user safety and signal integrity. Reported devices provide isolation levels of several kilovolts with gain errors below 0.5%, enabling precise DC-link voltage monitoring in compliance with safety standards.

2.5 IoT and V2X Communication

Recent literature increasingly frames bidirectional EV chargers as integral components of the smart grid through the integration of Internet of Things (IoT) communication. One of the most extensively studied functions in this context is State of Charge (SOC) estimation. Research indicates that hybrid approaches combining coulomb counting with voltage-based correction can achieve SOC estimation accuracy in the range of 3–5% under practical operating conditions.

From a grid interaction perspective, effective V2G operation requires reliable and lowlatency communication between the vehicle, the charger, and grid operators. Studies

evaluating wireless communication technologies such as Wi-Fi and cellular networks (4G/5G) report achievable control-command latencies below 100 ms, which are sufficient for many grid-support services, including peak shaving and frequency regulation. Moreover, simulation and field studies demonstrate that coordinated V2G participation, supported by load forecasting algorithms, has the potential to reduce residential peak demand by approximately 10–15%, highlighting the value of bidirectional chargers for grid stabilization. To ensure interoperability and cybersecurity, the literature emphasizes adherence to standardized communication protocols. Standards such as ISO 15118 and IEEE 2030.5 are frequently cited as key enablers for secure data exchange, authentication, and energy management in V2X systems.

2.6 Summary of Research Gaps

Despite the significant progress reported in the literature, several gaps remain and justify the practical orientation of this project:

1. **Limited Practical Implementation:** A large portion of published work focuses on simulation or theoretical analysis, with limited documentation of complete hardware realization, including PCB design, thermal considerations, and experimental validation.

2. **Hardware–Software Co-Design:** Few studies provide a detailed end-to-end workflow demonstrating the transition from model-based control design to deployed firmware on real-time hardware with full peripheral integration.

3. **Incomplete V2X Ecosystem Demonstration:** Existing research often addresses individual subsystems in isolation, rather than presenting a fully integrated charger supporting multiple V2X modes (V2G, V2H, and V2L) combined with IoT-based communication.

2.7 Conclusion of Literature Review

This chapter has systematically reviewed the key technological pillars underlying bidirectional EV chargers. The literature identifies the Totem-Pole PFC as a high-efficiency AC/DC interface and the Dual Active Bridge converter as a suitable isolated DC/DC topology for bidirectional power transfer. Model-Based Development and digital control platforms are established as dominant trends in modern power electronics design, while IoT-enabled communication is recognized as a critical enabler for V2X functionality. The primary research gap lies in the lack of holistic, experimentally validated implementations. Accordingly, the present project is positioned to address this gap by detailing the complete engineering process from simulation to a fully integrated, IoT-enabled hardware prototype.

Chapter 3
System Overview

3.1 Introduction

This chapter presents a high-level overview of the proposed bidirectional on-board electric vehicle (EV) charger integrated with Vehicle-to-Everything (V2X) capabilities. The objective is to describe the overall system architecture, identify the functional role of each major subsystem, and explain the interaction between these subsystems before introducing the detailed design and implementation aspects in subsequent chapters. This top-down approach establishes a clear conceptual framework that illustrates how advanced power conversion stages, digital control strategies, and communication interfaces are integrated to form a cohesive and functional V2X energy system.

3.2 System Architecture Overview

The proposed system is designed as a multi-layered architecture that combines power electronics, intelligent control, and networked communication. The architecture can be divided into three main layers:

1. **Power Conversion Layer:** This layer includes the bidirectional AC/DC and DC/DC power converters responsible for efficient and reliable energy transfer between the grid and the EV battery.

2. **Control and Sensing Layer:** This layer contains the digital controller, sensing elements, and gate driver circuits required to regulate power flow, ensure system protection, and manage operational modes.

3. **Communication and Integration Layer:** This layer enables V2X functionality through IoT-based communication, allowing interaction with external entities such as the electrical grid, residential systems, and user interfaces.

The layered architecture enhances modularity and scalability, allowing improvements or modifications in one layer—such as adopting a new converter topology or communication protocol—without significantly affecting the remaining layers.

3.3 Functional Description of Core Components

The overall system functionality is achieved through the coordinated operation of several key components, each fulfilling a distinct role within the architecture described in Section 3.2.

3.3.1 Power Conversion Unit

The power conversion unit provides the physical pathway for all energy transfer within the system.

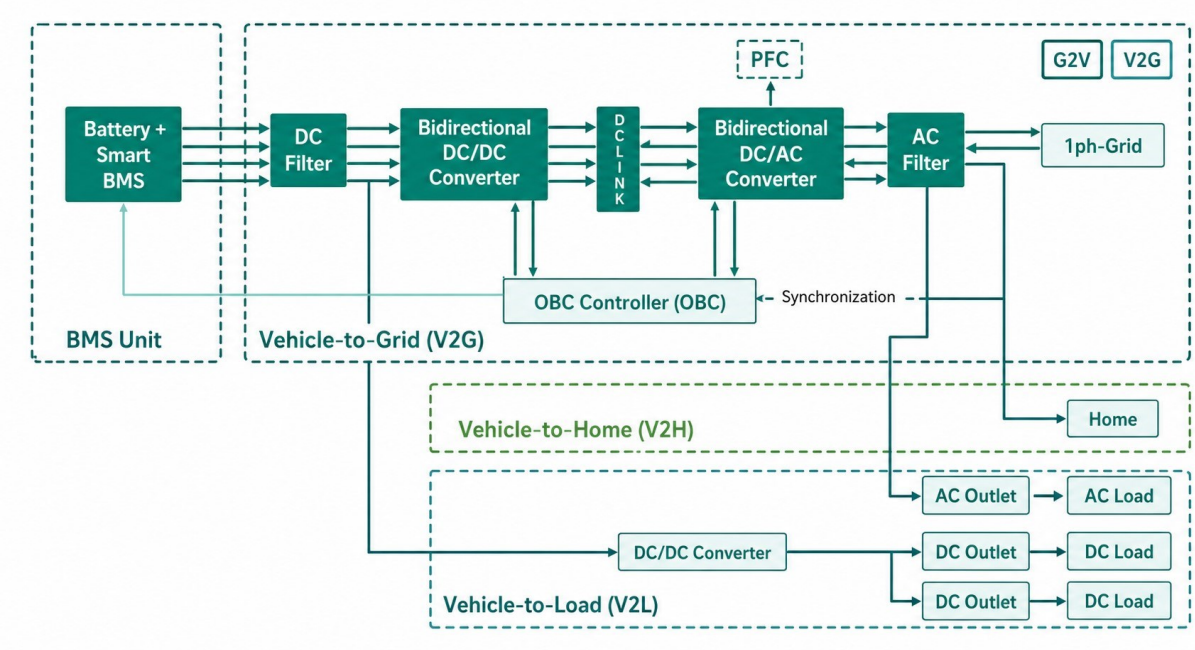


Figure3.1: System architecture of the bidirectional on-board EV charger.

Bidirectional AC/DC Converter: This converter serves as the primary interface with the single-phase AC grid (230 V, 50 Hz). During Grid-to-Vehicle (G2V) operation, it functions as a Totem-Pole Power Factor Correction (PFC) rectifier, drawing sinusoidal current with a near-unity power factor, high efficiency (above 95%), and low total harmonic distortion (THD below 5%). In Vehicle-to-Grid (V2G) or Vehicle-to-Home (V2H) modes, the same stage operates as a grid-tied single-phase inverter, injecting synchronized sinusoidal current back to the grid or supplying local loads through an LCL output filter.

Bidirectional DC/DC Converter (Dual Active Bridge – DAB): The isolated DC/DC stage provides galvanic isolation and voltage adaptation between the high-voltage DC link (approximately 400 V) and the EV battery pack (typically 48 V). During battery charging, the DAB operates in buck mode, while during discharging it operates in boost mode, enabling controlled bidirectional power transfer with peak efficiencies exceeding 96%.

3.3.2 Control, Sensing, and Actuation Unit

This unit represents the system's central intelligence and is responsible for closed-loop control, protection, and mode management.

Digital Controller: The system is controlled using a Texas Instruments TMS320F28379D microcontroller implemented on a C2000 LaunchPad platform. The controller executes all real-time control algorithms for the AC/DC and DC/DC converters, manages system protection, and coordinates transitions between operating modes. The control firmware is developed using a Model-Based Design (MBD) approach, enabling seamless migration from control algorithm design and simulation in MATLAB/Simulink to automatic code generation and real-time execution on the target hardware.

Sensing Network: The sensing subsystem consists of isolated voltage and current sensors that provide accurate feedback signals for control and monitoring purposes. Voltage sensing is implemented using isolated amplifiers, while Hall-effect or fluxgate-based current sensors are used to measure converter currents, ensuring safe operation and reliable feedback under high-voltage conditions.

Gate Driver Circuits: Gate driver circuits interface the low-voltage PWM signals generated by the controller with the high-power semiconductor switches of the converters. These drivers ensure fast and reliable switching while providing the necessary isolation and protection features.

3.3.3 Communication and V2X Integration Unit

This unit enables the charger to operate as a smart, network-connected V2X node.

IoT Communication Module: The communication module enables real-time data exchange between the charger and a cloud-based platform or mobile application. Parameters such as State of Charge (SOC), operating mode, power flow direction, and fault status can be monitored remotely, while control commands can be issued by the user or an energy management system.

V2X Service Interface: This logical interface supports various V2X services, including Vehicle-to-Grid (V2G), Vehicle-to-Home (V2H), and Vehicle-to-Load (V2L). It abstracts the underlying power electronics, allowing higher-level energy management systems to utilize the EV as a distributed energy resource.

3.4 Operational Modes and Power Flow

The charger operates in multiple predefined modes depending on grid conditions, user commands, and battery status. Table 3.1 summarizes the main operational modes, corresponding power flow directions, and active converter configurations.

Table 3.1: Operational modes of the bidirectional EV charger system

Mode	Symbol	Power Flow	Converter Configuration	Key Application
Grid-to-Vehicle	G2V	Grid→ Battery	AC/DC: Totem-Pole PFC (Rectifier) DC/DC: DAB in Buck Mode	Standard and smart charging
Vehicle-to-Grid	V2G	Battery→ Grid	AC/DC: Grid-tied inverter with LCL filter DC/DC: DAB in Boost Mode	Peak shaving, grid support
Vehicle-to-Home	V2H	Battery→ Home	AC/DC: Standalone inverter (off-grid) DC/DC: DAB in Boost Mode	Home backup during outages
Vehicle-to-Load	V2L	Battery→ Load	DC/DC: DAB in Boost Mode	Supplying external DC loads

Transitions between modes are supervised by the digital controller, which reconfigures control algorithms, enables or disables specific converter stages, and ensures safe operation. For grid-interactive modes, synchronization with grid voltage and frequency is performed prior to power injection.

3.5 System Integration and Data Flow

System integration is achieved through well-defined electrical and communication interfaces between the three layers. The power conversion layer supplies conditioned power to the control and sensing circuits, while the controller generates PWM signals for the gate drivers based on real-time sensor feedback, thereby closing the voltage and current control loops.

The IoT communication module exchanges data with the controller through a serial interface, relaying operational data to the cloud and forwarding external commands to the control system. This bidirectional data flow—from sensors to controller to cloud and user interface, and back to the power stage—enables the charger to function as an intelligent and responsive V2X energy asset capable of participating in advanced energy management schemes such as virtual power plants.

3.6 Summary

This chapter has presented a comprehensive system-level overview of the proposed bidirectional on-board EV charger with V2X capabilities. The architecture integrates a highefficiency two-stage power conversion structure based on a Totem-Pole PFC AC/DC converter and an isolated DAB DC/DC converter, coordinated by a digital control system with precise sensing and enhanced through IoT-based communication. By supporting multiple operating modes—G2V, V2G, V2H, and V2L—the system transforms the EV into an active participant in the energy ecosystem. The following chapters provide detailed design, implementation, and validation of the individual subsystems introduced in this overview.

Chapter 4
Power Electronics Components and Hardware Interfaces

4.0 Introduction

The performance of a bidirectional on-board EV charger is strongly influenced by the selection of its power electronic components and hardware interfaces. In high-frequency, high-efficiency converter applications such as the proposed Totem-Pole PFC stage and inverter-based bidirectional structures, semiconductor device choice plays a critical role in determining switching losses, thermal behavior, and overall system reliability.

In this project, MOSFET devices are employed as the main switching elements in both the Totem-Pole PFC stage and the bidirectional power conversion circuits. Silicon (Si) MOSFETs offer cost-effectiveness and proven reliability, while Silicon Carbide (SiC) MOSFETs provide superior switching performance, lower conduction losses, and higher efficiency at elevated switching frequencies. The use of these devices supports compact design and improved energy conversion efficiency, which are essential for on-board charger applications.

To ensure proper and reliable operation of the MOSFETs, dedicated gate driver circuits are integrated to provide fast switching, adequate isolation, and protection against overcurrent and voltage transients. The gate driver stage is crucial for maintaining switching stability and minimizing losses, especially in high-frequency bidirectional operation.

Additionally, voltage and current sensing circuits are incorporated at key points within the system to enable accurate feedback, control, and protection. These sensors provide real-time monitoring of system parameters, ensuring safe operation during both charging and discharging modes.

Together, the selected power devices, gate drivers, and sensing interfaces form the hardware foundation of the proposed bidirectional OBC system, directly impacting its efficiency, controllability, and overall performance.

4.1 Power Semiconductor Devices

4.1.1 Silicon (Si) MOSFETs – Low Voltage Side

Si MOSFETs are employed in the low-voltage (LV) sections of the converter. These devices are particularly suitable for circuits operating at moderate voltage levels where extreme high-voltage blocking capability is not required.

The main reasons for selecting Si MOSFETs for the LV side include:

- **Cost-effectiveness:** Si MOSFETs are significantly more affordable compared to wide bandgap devices, making them practical for low-voltage stages.

- **Wide availability:** They are widely available in the market with mature manufacturing technology and proven reliability.
- **Suitability for low-to-medium power applications:** Their electrical characteristics are well-matched to LV converter sections where switching stress and voltage levels are relatively lower.
- **Ease of implementation:** Gate driving requirements are less demanding compared to SiC devices, simplifying hardware design.

For these reasons, Si MOSFETs provide a reliable and economical solution for the LV parts of the bidirectional OBC without compromising system stability.

4.1.2 Silicon Carbide (SiC) MOSFETs – High Voltage Side

Silicon Carbide (SiC) MOSFETs are selected for the high-voltage (HV) sections of the system. High-voltage operation requires devices capable of handling elevated blocking voltages while maintaining high efficiency and thermal stability.

The key advantages of using SiC MOSFETs in the HV side include:

- **Higher switching speed:** SiC devices support faster switching transitions, which reduces switching losses and enables higher operating frequencies.
- **Lower conduction and switching losses:** This improves overall system efficiency, especially in high-power and high-frequency applications.
- **Superior thermal performance:** SiC MOSFETs can operate at higher temperatures with improved thermal conductivity, reducing cooling requirements.

By employing Si MOSFETs in the LV stage and SiC MOSFETs in the HV stage, the proposed bidirectional OBC achieves an optimized balance between cost, efficiency, and performance, ensuring reliable operation in both G2V and V2G modes.

Table 4.1: Comparison Between Si and SiC MOSFETs for the Proposed OBC

Parameters	Si MOSFETS	SiC MOSFETS
Semiconductor Material	Silicon	Silicon Carbide
Voltage Rating (VDS)	Up to 600 – 900V	650 V – 1700 V and higher
Best Application Range	Suitable for low-voltage stages	Ideal for high-voltage stages
Switching Frequency Capability	Moderate switching frequency	Higher switching frequency
Switching Losses	Higher at high frequency	Significantly lower
Conduction Losses	Acceptable at low voltage	Lower at high voltage ratings
Thermal Performance	Moderate tolerance	Higher capability
Efficiency Contribution	Moderate	High under high stress
Gate-Driving Requirements	Simpler drive requirements	Requires careful gate control and isolation
Cost	Lower cost	Higher cost

4.2 Isolated Four-Channel Gate Driver

The gate driver stage provides the interface between the F28379D-LaunchPad controller and the four power MOSFETs used in power stages. Its main function is to convert low-voltage control signals into high-current gate drive pulses capable of switching the MOSFETs reliably at high frequency (up to 500 kHz).

The implemented architecture is based on a four-channel isolated gate driver configuration, where each switching device is driven independently. The system includes both High-Side (HS) and Low-Side (LS) drivers, ensuring proper synchronization between the two legs of the Totem-Pole structure and preventing shoot-through conditions.

4.2.1 Power Supply Arrangement

The driver board operates using two external supply rails:

- **3.3V** for logic-level interface with the LaunchPad
- **15V** for gate driving power

This separation ensures clean logic control while providing sufficient gate overdrive for both Si and SiC MOSFETs.

4.2.2 Control and Interface Signals

The driver board interfaces directly with the LaunchPad through the following control pins:

• H-PWM / L-PWM
• PWM1, PWM2, PWM3, PWM4
• EN (Enable)
• RST (Reset)
• FLT (Fault)

These signals allow independent control of each switching channel. The EN and RST pins provide safe startup and reset control, while the FLT signal enables real-time fault feedback to the controller to ensure safe shutdown under abnormal conditions.

4.2.3 Gate Driver IC Implementation

The implemented architecture integrates high-speed buffering, isolation, and gate driving stages:

- **PO74G126A**: is used as a high-speed buffer between the MCU and the isolation stage to ensure clean PWM signal transitions and reduce noise coupling.
- **UCC27211**: is utilized as a high-speed half-bridge driver for fast switching and tight propagation delay matching.
- **ISO5852S-Q1**: reinforced isolated gate driver is employed for high-voltage switching devices, providing strong isolation, high CMTI performance, and integrated protection features such as desaturation detection and UVLO.
- **UCC53x0**: single-channel isolated drivers are used where independent isolated channels are required.

For high-speed signal transmission across isolation boundaries, pulse transformers are incorporated in the design. Transformer-based coupling provides strong immunity against high dv/dt stress and enhances robustness during high-frequency switching, particularly when driving SiC MOSFETs.

4.2.4 Output Stage Configuration

Each gate driver channel is connected to a MOSFET through four essential pins:

• Gate (G)
• Source (S)
• DSAT (Desaturation detection)
• Kelvin Source (S reference)

The use of desaturation detection enhances protection against overcurrent conditions, particularly in high-voltage switching devices.

The outputs are divided into:

• High-Side (HS) channels

• Low-Side (LS) channels

This configuration ensures independent and synchronized control of the four switching devices forming the Totem-Pole PFC.

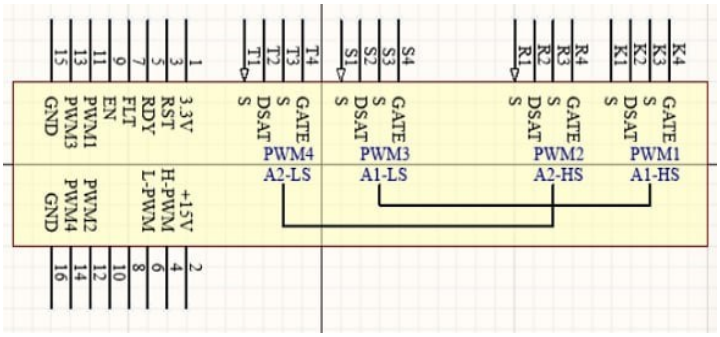


Figure 4.1: Schematic of the Isolated Four-Channel Gate

4.2.5 Isolation and Protection

Galvanic isolation is implemented between the low-voltage control domain and the high voltage power stage to protect the MCU from voltage transients and high common-mode dv/dt stress. The combination of reinforced isolated gate driver ICs and transformer-based signal coupling ensures reliable operation at switching frequencies up to 500 kHz.

Protection mechanisms integrated within the driver stage include:

• Undervoltage Lockout (UVLO)

• Desaturation Detection

• Fault Reporting (FLT signal)

These features guarantee safe and stable operation during both charging (G2V) and discharging (V2G) modes.

The implemented isolated four-channel gate driver board ensures precise high-frequency switching, proper isolation, and reliable protection, forming a robust hardware interface for the bidirectional Totem-Pole PFC stage.

4.2.6 Isolated 4-Channel Gate Driver Strategy

The physical implementation of this system is consolidated into a 4-channel gate driver architecture. This integrated strategy is vital because the gate driver serves as more than just a switch; it is a power amplifier that translates low-power 3.3V logic signals into the high-energy

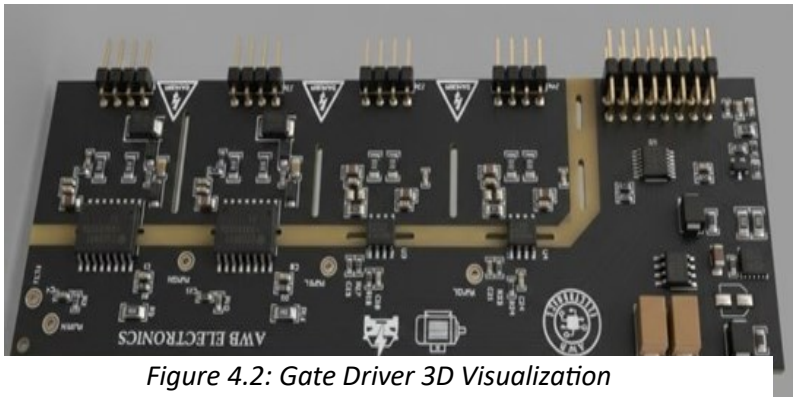


Figure 4.2: Gate Driver 3D Visualization

pulses required to fully saturate MOSFET gates. The 4-channel board also features a comprehensive interface for Reset (**RST**), Ready (**RDY**), and Fault (**FLT**) signals, allowing the MCU to monitor the health of all power stages in real-time, ensuring reliability and performance.

4.3 Microcontrollers as Interface Hardware

In the proposed bidirectional V2X charger system, two dedicated microcontrollers are employed to manage different functional layers of the system. The control architecture separates real-time power conversion control from communication and IoT services to enhance modularity, reliability, and deterministic performance.

- The bidirectional charger consists of:
- An AC–DC stage (implemented using a Totem-Pole PFC topology)
- An isolated DC–DC stage
- Protection and monitoring circuits
- Communication and IoT interface

To efficiently manage these subsystems, the following controllers are utilized:

- F28379D LaunchPad
- ESP32 Module

Each controller is selected based on its technical suitability for the assigned tasks.

4.3.1 F28379D LaunchPad – Power Stage Interface Controller

The F28379D LaunchPad functions as the primary real-time digital controller of the bidirectional charger. It manages the switching operations and feedback control of the complete power conversion system, including both the AC–DC and DC–DC stages.

The controller directly interfaces with the isolated multi-channel gate driver board and acquires feedback signals from voltage and current sensors distributed across the system.

4.3.1.1 Functional Role

The LaunchPad performs the following critical tasks:

- Generates synchronized high-frequency PWM signals (up to 500 kHz) for:
    - The AC–DC Totem-Pole PFC stage
    - The isolated bidirectional DC–DC converter
- Controls complementary switching signals with accurate dead-time insertion.
- Interfaces with gate driver control pins (H-PWM, L-PWM, PWM1–PWM4, EN, RST, FLT).
- Acquires real-time analog measurements from:
    - Grid-side voltage and current sensors
    - DC-link voltage sensor

    - Battery-side voltage and current sensors
- Executes digital control algorithms for power flow regulation.
- Implements protection mechanisms such as overcurrent, overvoltage, and fault detection.
- Ensures deterministic timing behavior to prevent shoot-through and ensure stable operation.

All analog feedback signals are directly connected to the high-resolution ADC modules of the LaunchPad, enabling accurate and high-speed data acquisition required for closed-loop control.

4.3.1.2 Justification for Selecting F28379D over General-Purpose Controllers

The F28379D was selected instead of general-purpose microcontrollers (e.g., Arduino platforms) due to the demanding requirements of high-frequency bidirectional power conversion systems.

While Arduino boards are suitable for low-speed embedded applications and educational purposes, they are not optimized for real-time digital control of high-frequency switching converters.

The key technical advantages of the F28379D include:

- Dedicated enhanced PWM (ePWM) modules designed for power electronics applications, supporting high-frequency, synchronized, and complementary PWM generation.
- Precise dead-time control required for safe operation of the Totem-Pole topology.
- High-resolution, high-speed ADC modules suitable for current and voltage feedback loops.
- High processing capability optimized for real-time control.
- Advanced peripheral integration tailored for motor control and power conversion systems.
- Deterministic timing performance critical for safe bidirectional operation.

In contrast, general-purpose platforms:

- Operate at lower clock frequencies.
- Lack hardware-synchronized PWM modules.
- Provide limited ADC sampling performance.
- Are not intended for high-frequency switching control applications.

Therefore, the F28379D provides a robust and reliable platform for managing the real-time control requirements of a bidirectional V2X charger system.

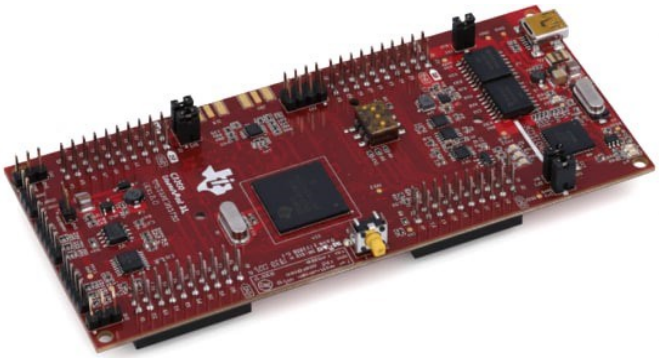


Figure 4.3: F28379D – LaunchPad kit

4.3.2 ESP32 – IoT and V2X Communication Controller

The ESP32 module is dedicated to the communication and IoT layer of the system. It does not directly control the power electronics stage but operates as a wireless data acquisition and communication interface.

4.3.2.1 Functional Role

The ESP32 performs the following functions:

- Receives system parameters from the main controller (voltage, current, DC-link level, power flow direction, operating mode).
- Transmits operational data to cloud platforms.
- Enables remote monitoring and supervision for:
  - Vehicle-to-Grid (V2G) mode
  - Vehicle-to-Home (V2H) mode
  - Vehicle-to-Load (V2L) mode
- Supports IoT-based energy management integration.
- Provides a scalable interface for future smart grid expansion.

4.3.2.2 Justification for Selection

The ESP32 was selected due to:

- Integrated Wi-Fi and Bluetooth connectivity.
- Adequate processing capability for communication and data handling.
- Low cost and compact physical size.
- Strong ecosystem support for IoT development and cloud integration.

By delegating communication tasks to the ESP32, the real-time performance of the main control loop



Figure 4.4: ESP32 Module

4.4 Sensing and Monitoring System

4.4.1 Introduction

The Sensing and Monitoring System acts as a critical feedback loop for the Bidirectional Onboard Charger. It enables the precise control of power delivery by continuously measuring the necessary electrical parameters, ensuring they meet the specific battery charging needs. Through the integration of sensors and DC voltage modules with an F28379D Launchpad, the system manages the State of Charge (SOC) and provides data essential for real-time monitoring and remote diagnostics via IoT.
For optimal safety and high-precision accuracy, the system utilizes the TMCS1133 current sensor and AMC1311 isolated amplifier, both of which offer reinforced galvanic isolation to protect against high-voltage transients. Designed with high immunity to external magnetic fields, these components ensure reliable operation across all operational modes, helping the Battery Management System (BMS) prevent battery damage

4.4.2 Current Sensing Implementation

This section details the current monitoring subsystem, designed to provide high-speed, accurate feedback for the F28379D controller. The implementation focuses on maintaining stable power flow and system safety through precision sensing components.

4.4.2.1 Isolated Hall-Effect Sensing (TMCS1133)

4.4.2.1.1 Overview and Functional Description

The bidirectional charger utilizes the TMCS1133, a galvanically isolated Hall-effect sensor, to measure AC and DC currents. It produces an analog output voltage linearly proportional to the sensed current, featuring a coreless construction that eliminates magnetic concentrators and reduces system bulk.

The sensing mechanism relies on an internal lead frame with a low resistance of 0.67 mΩ, minimizing power loss and thermal stress on the PCB. By employing differential Hall sensors, the system inherently rejects stray external magnetic fields—a critical feature for maintaining signal integrity in compact power electronic layouts.

For precise control feedback, the device incorporates on-chip drift compensation. This ensures a maximum total error of <2.5% over the operating temperature and lifetime. With a one-time calibration, this error can be further reduced to <1.5%.

From a safety standpoint, the TMCS1133 provides reinforced isolation up to 5000 VRMS and a working voltage of 1100 VDC. The 8.1 mm creepage and clearance distances comply with high-voltage EV safety standards. Additionally, the integrated shielding provides high common-mode transient immunity (CMTI), ensuring reliable performance during high-frequency switching.

4.4.2.1.2 Pin Configuration and Hardware Interface

The TMCS1133 is integrated into the design using a 10-pin SOIC (DVG) package. This package is specifically selected for its ability to handle high surface-mount currents while maintaining the 8.1 mm clearance required for reinforced isolation between the high-power path (Pins 1-2) and the low-voltage control side (Pins 3-10).

The hardware interface is categorized into three functional groups:

- **Power Path (IN+, IN-):** Robust pins designed to interface with heavy copper pours for continuous current flow.
- **Analog Interface (VOUT, VOC, VS):** The sensor is powered by a 5V supply, with V_{OUT} providing the feedback signal to the F28379D ADC. V_{oc} is used to set the hardware-level overcurrent threshold.
- **Diagnostic Outputs (ALERT, OC):** These active-low, open-drain pins are utilized for real-time fault detection, providing an immediate hardware interrupt to the controller during thermal or overcurrent events.

4.4.2.1.3 High-Power Performance and Thermal Management

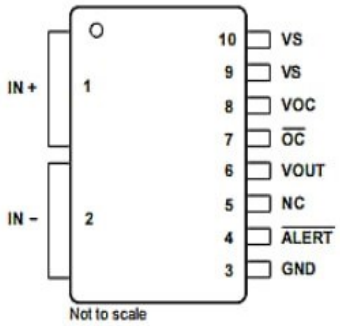


Figure 4.5: TMCS1133 - Top View

Thermal stability is critical when handling the system's continuous current requirement of $75A_{RMS}$.To mitigate self-heating, the TMCS1133 utilizes a low-ohmic lead frame 0.67 mOhm, which minimizes resistive power loss $P = I^2 R$ at the sensing node.
In the PCB design, the Safe Operating Area (SOA) shown in the following figure was a primary constraint. Since current-carrying capacity derates as ambient temperature increases, heavy copper pours and strategic via placement were implemented to ensure the sensor operates within its thermal limits during peak G2V and V2G transitions. This prevents thermal runaway and maintains measurement accuracy under high-load conditions.

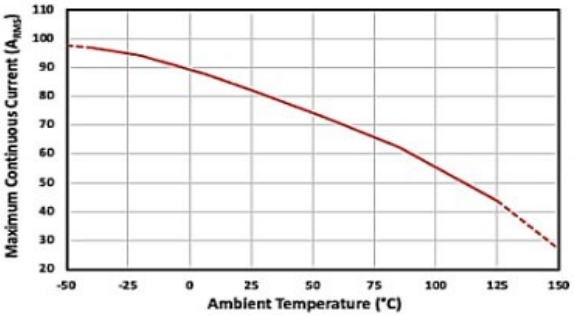


Figure 4.6: Maximum Continuous RMS Current vs. Ambient Temperature (SOA Graph))

4.4.2.1.4 Reinforced Isolation and Dynamic Response

The TMCS1133 ensures system safety through reinforced galvanic isolation, rated for 1100 VDC working voltage and $5000V_{RMS}$ transients. This isolation effectively shields the F28379D controller from high-voltage spikes.

Optimized for high-speed converters like the Totem Pole PFC, the sensor features a 1MH bandwidth and a low 80 ns propagation delay. These characteristics, combined with a High Common-Mode Transient Immunity (CMTI of 75-150 kV/µs), prevent signal distortion during the rapid dv/dt transitions typical of SiC-based power stages, enabling precise real-time PWM control.

4.4.2.1.5 Ambient Field Rejection (AFR)

In compact PCB designs, magnetic interference from adjacent inductors or high-current traces can compromise measurement accuracy. The TMCS1133 mitigates this using Ambient Field Rejection (AFR) via internal differential Hall sensors. By measuring the magnetic field at two distinct points and subtracting the common-mode component, the device ignores external stray fields. This ensures that the sensor output only reflects the current flowing through its internal lead frame, maintaining high fidelity in the dense environment of the bidirectional charger.

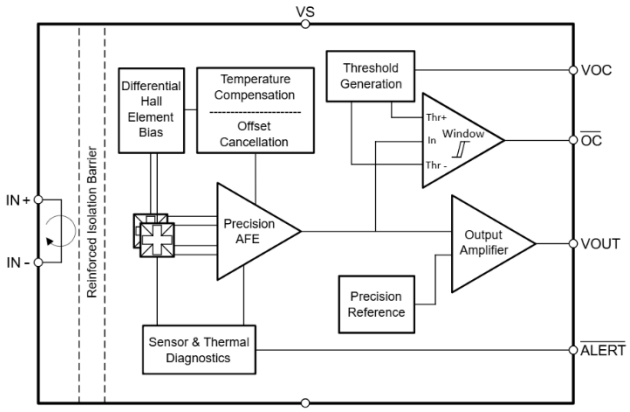


Figure 4.7: Functional Block Diagram of TMCS1133

4.4.2.1.6 Integrated Overcurrent Detection (OCD)

The TMCS1133 features an integrated Overcurrent Detection (OCD) circuit that provides a rapid digital trigger via the OC pin in only 500 ns. This hardware-level protection is independent of the microcontroller's ADC processing speed, allowing for near-instantaneous shutdown of the converter to protect the MOSFETs from short-circuit damage.

The trip threshold I_{OC} is user-programmable via the voltage applied to the VOC pin, governed by the following relationship:

$$V_{OC} = S \times I_{OC}$$

Where:

- S: Device sensitivity (mV/A).
- I_{OC}: Target overcurrent threshold (A).
- V_{OC}: Applied threshold-setting voltage (V).

4.4.2.1.7 Diagnostic Monitoring

To ensure system reliability, the sensor features an ALERT pin that provides real-time status updates through a PWM signal. This pin warns the controller of two critical conditions:

1. **Thermal Alert**: Triggered if the junction temperature approaches its 165°C limit.
2. **Sensor Alert**: Triggered if internal sensitivity or offset drift is detected.

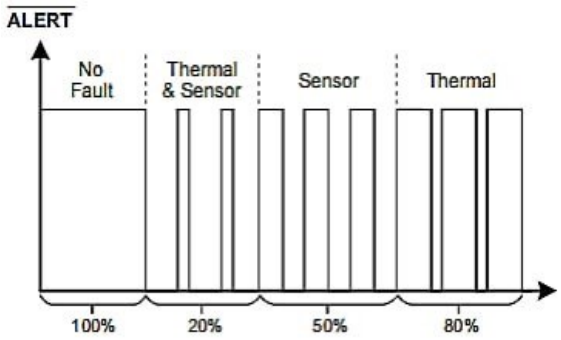


Figure 4.8: Sensor Diagnostics Waveform

4.4.2.1.8 Variant Selection and Dynamic Range Optimization

The selection of the TMCS1133 variant is critical to ensure the sensor captures the peak currents of G2V and V2G modes without reaching saturation (clipping). The measurable range is defined by the supply voltage V_S, sensitivity S, and the zero-current reference V_{REF}.

1. Quiescent Point Selection:

To optimize the dynamic range for our bidirectional application, we utilized Variant A (V_{REF} = 2.5V) with a 5V supply. This configuration provides symmetrical headroom for both charging and discharging currents, centered at the mid-point of the ADC's input range.

2. Sensitivity and Resolution Trade-off:

There is an inverse relationship between sensitivity and measurable range. In our design, we selected the highest sensitivity variant that accommodates the full-scale current of the PFC and DAB stages. This ensures:

Maximum ADC Resolution: Leveraging the full dynamic range of the F28379D ADC.

Safety Margin: Avoiding $V_{CLIPout}$ during transient current spikes.

The maximum measurable currents are calculated using:

$$I_{IN,max+} = \frac{V_{OUT,max} - V_{REF}}{S}$$

$$I_{IN,max-} = \frac{V_{REF} - V_{OUT,min}}{S}$$

For example, using the TMCS1133A1 (25 mV/A) allows a range of approx. ±96 A, which provides a sufficient safety buffer for our system's 75 A_{RMS} requirement.

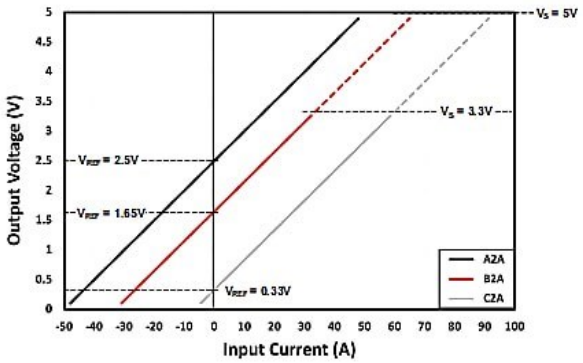


Figure 4.9: Output Voltage Relationship to Input Current

4.4.2.2 Circuit Design Description

4.4.2.2.1 Power Integrity and Output Filtering

To maintain sensor accuracy, a dual-capacitor decoupling strategy was implemented. A 0.1 µF and 10 µF bypass capacitors (C1, C3) are placed in parallel as close as possible to the supply pins to suppress high-frequency switching noise. Additionally, an RC low-pass filter (R2, C2) footprint is integrated at the output to attenuate transients originating from the high-frequency switching of the Totem-Pole PFC stage.

4.4.2.2.2 Overcurrent Set Point Configuration

To ensure robust hardware protection, a flexible threshold configuration was implemented using a 1x3 selection header (J1). This allows the system to define the overcurrent trip point (V_{OC}). through three distinct modes:

1. **Manual Tuning:** Utilizing a trimmer potentiometer (R_5) in series with a 5.6 kΩ fixed resistor (R4). The inclusion of R_4 is a critical design choice to ensure V_{OC} remains above a 0.3 V minimum floor, preventing nuisance or accidental triggers at low reference voltages.

$$V_{OC} = V_S \times \frac{R_4}{R_{POT} + R_4}$$

2. **Fixed Threshold:** A precision resistor divider (R_6 , R_7) can be used for a stable, predefined trip point.

$$V_{OC} = V_S \times \frac{R_6}{R_6 + R_7}$$

3. **External Reference:** The header allows V_{OC} to be driven by an external source for dynamic threshold adjustment.

Upon exceeding the set limit, the active-low pin initiates a digital system shutdown in 500 ns, providing a near-instantaneous safeguard for the power MOSFETs.

4.4.2.2.3 High-Power Interface and Load Connectors

The power path utilizes high-current lug connectors (IN+, IN-) to interface with the grid and battery. These connectors are mechanically fastened to the PCB to minimize contact resistance and ensure thermal stability at currents up to 90 A.
The design ensures that the high-current flow through the internal lead-frame remains within the sensor's Safe Operating Area (SOA). This mechanical and electrical integration allows for precise magnetic field sensing while maintaining the structural integrity of the PCB under continuous 75 A_{RMS} load conditions.

4.4.2.3 Hardware Design and Circuit Implementation

The physical implementation of the current sensing module was developed in **KiCad**, focusing on three engineering pillars: power stability, thermal management, and precision protection logic. The design strictly adheres to high-power routing guidelines to ensure both signal integrity and operator safety.

4.4.2.3.1 PCB Layout and Thermal Management

Handling a continuous current of 75 A_{RMS} requires specialized PCB geometry. The high-current path (IN+/IN-) utilizes heavy copper pours (4-oz) to minimize resistive heating. To further enhance heat dissipation, thermal via farms were placed strategically around the isolated current inputs. These vias act as thermal conduits, transferring heat from the top layer to the bottom copper planes, effectively maintaining the sensor within its Safe Operating Area (SOA) during peak operation.

4.4.2.3.2 Isolation and Diagnostic Routing

To comply with reinforced isolation standards, an 8.1 mm to 8.5 mm exclusion zone was strictly maintained beneath the sensor package. This ensures that no conductive material or ground plane crosses the isolation barrier, preventing high-voltage arcing. Additionally, the ALERT (Pin 4) trace was routed to provide the F28379D with real-time PWM status updates on thermal limits and sensor health, completing the hardware-level diagnostic loop.

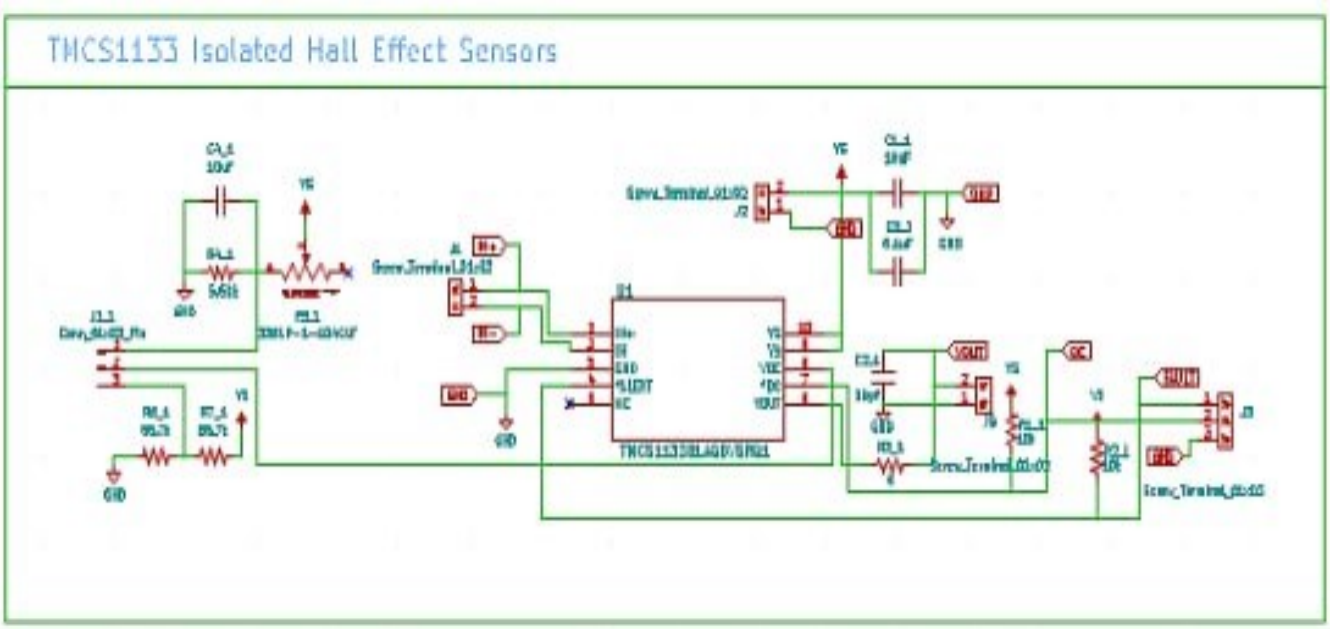


Figure 4.10: KiCad Schematic of TMCS1133 Sensing Circuit

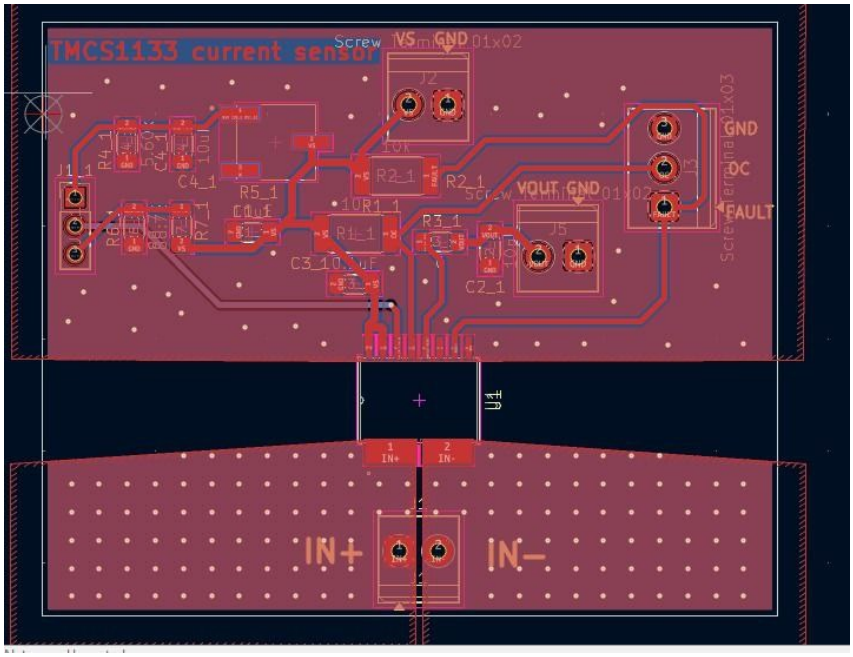


Figure 4.11: PCB Layout of Current Sensing Circuit

4.4.2.4 Final Hardware Integration

The 3D model illustrates the final physical arrangement of the sensing module, showcasing the reinforced isolation barrier and the lug connectors designed for high-power grid-to-vehicle (G2V) and vehicle-to-grid (V2G) current paths.

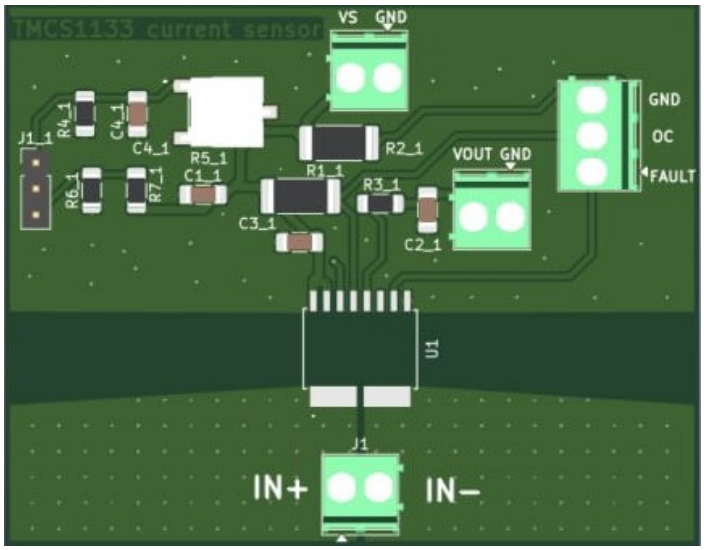


Figure 4.12: 3D Visualization of Current Sensing Module

4.4.2.4.1 Physical Hardware Implementation

The final fabricated PCB for the current sensing stage is shown in figure. The physical module demonstrates the successful implementation of the heavy copper pours and the high-current lug connectors. During assembly, special care was taken to ensure the TMCS1133 sensor was perfectly centered to maintain magnetic field symmetry, while the low-ESR decoupling capacitors were soldered in close proximity to the power pins to maximize noise suppression.

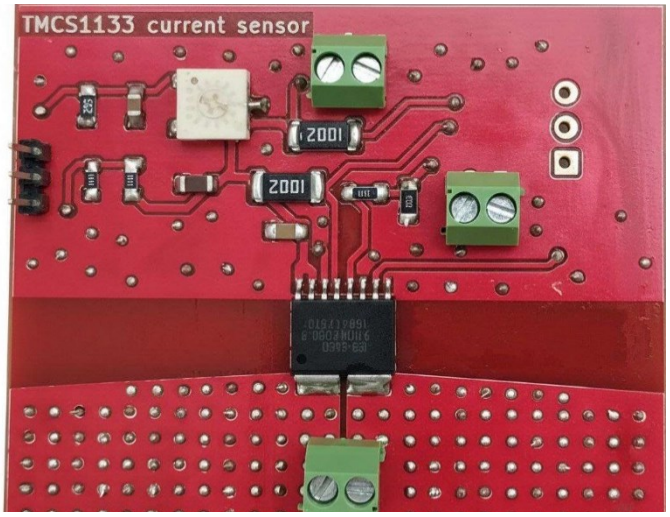


Figure 4.13: Fabricated and Assembled TMCS1133 Current Sensing Module

4.4.2.5 Laboratory Implementation and Testing

To verify the performance of the sensing stage, the module was tested under controlled conditions. The sensor was powered by a regulated **5 VDC** supply, with the high-current load path galvanically isolated from the measurement circuitry.

1. Zero-Current Calibration:

Prior to loading, the output voltage V_{OUT} was measured at **2.5V**. This aligns with the nominal V_{REF} (half of V_S), confirming stable biasing and a precise baseline for bidirectional sensing.

2. Linearity and Sensitivity Test:

A controllable DC source injected current in discrete steps. The measured response demonstrated excellent linearity across the test range.

$$V_{OUT} = V_{REF} + (S \times I_{IN})$$

- At **2A,** the output shifted by **50 mV** (25 mV/A sensitivity).
- At **10A**, the output reached **2.75 V** (250mV shift).

3. Bidirectional Verification:

Reversing the load current resulted in a symmetrical voltage drop below the 2.50 V reference. This confirmed the system's ability to distinguish between G2V (Charging) and V2G (Discharging) modes without signal clipping or offset drift.

4.4.2.6 Bill of Materials - Current Sensing

Table 4.2: Bill of Materials - TMCS1133 Current Sensing Circuit

Ref. Designator	Component Type	Description
C1, C4	Ceramic Capacitor	10 µF, 10 V, ±10%, X7R
C2	Ceramic Capacitor	10 pF, 10 V, ±10%, X7R
C3	Ceramic Capacitor	0.1 µF, 50 V, ±10%, X7R
R1, R2, R7	Resistor	10 kΩ, 1%, 0.125 W, AEC-Q200 Grade 0
R3	Resistor	0 Ω, 1%, 0.1 W, AEC-Q200 Grade 0
R4	Resistor	5.6 kΩ, 1%, 0.1 W, AEC-Q200 Grade 0
R5	Trimmer Potentiometer	100 kΩ, 0.5 W, Cermet, 1-Turn, Top Adjustment
R6	Resistor	88.7 kΩ, 1%, 0.1 W, AEC-Q200 Grade 0
U1	Current Sensor IC	TMCS1133A1QDVGR
J1	Pin Header Connector	3-Position, 2.54 mm Pitch, Through-Hole

4.4.3 Voltage Sensing Implementation

The 400V DC-link and battery terminals require high-precision monitoring to ensure control stability. For this, we selected the AMC1311, a reinforced isolated amplifier optimized for high-impedance voltage sensing. This device acts as the critical safety barrier between the high-voltage power stage and the low-voltage F28379D Launchpad.

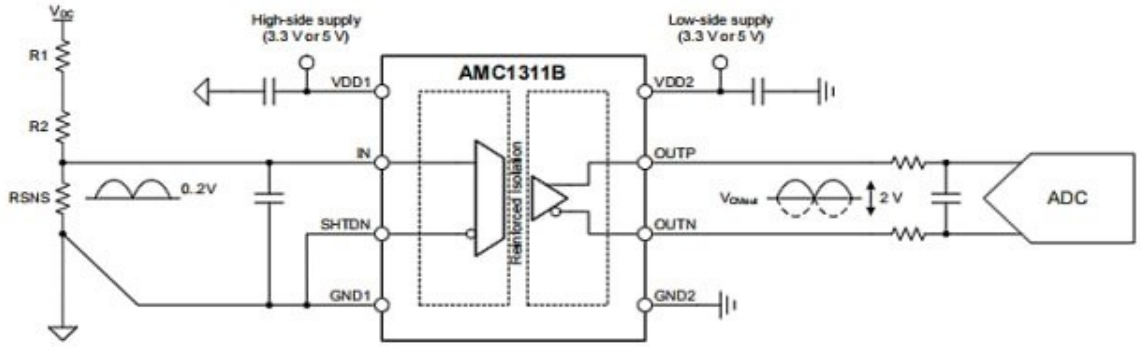


Figure 4.14: Simplified Schematic of AMC1311 Voltage Sensing

4.4.3.1 Overview and Functional Description

The AMC1311 is utilized for precise monitoring of high-voltage nodes, specifically the 400V DC-link and battery terminals. This high-precision isolation amplifier provides a safe interface between the power conversion stages and the F28379D controller by maintaining a robust galvanic barrier.

4.4.3.1.1 Isolation and Signal Transmission

The device employs a capacitive silicon-dioxide (SiO_2) isolation barrier. The analog input is digitized and transmitted across this barrier using an On-Off Keying (OOK) modulation scheme. This method is highly resistant to magnetic interference, which is critical in the high-speed switching environment of the Totem-Pole PFC.

For this design, the AMC1311B variant was selected due to its superior DC accuracy, including lower offset error and drift, ensuring stable voltage feedback across the extended industrial temperature range (–55°C to +125°C).

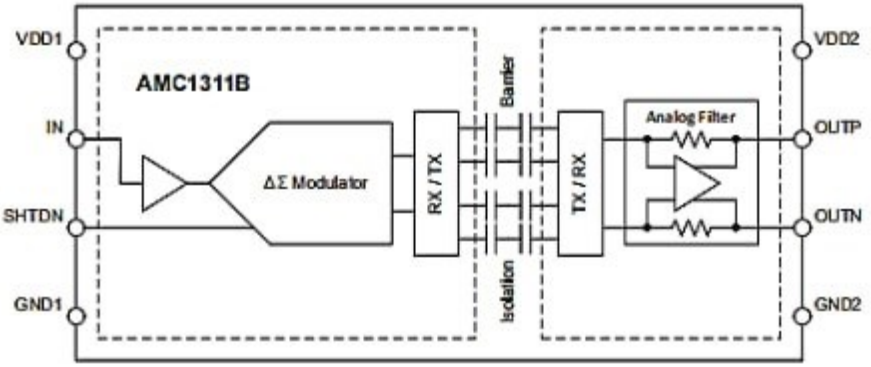


Figure 4.15: Functional Block Diagram of AMC1311

4.4.3.2 Pin Configuration and Hardware Interface

The AMC1311 is housed in an 8-pin SOIC (DWV) package, specifically designed to maintain clear galvanic separation between high-voltage and low-voltage domains. The hardware interface is split into two distinct sides:

- **The High-Side (Input):** This section is powered through **V_DD1** (Pin 1) and referenced to the isolated ground **GND_1** (Pin 4). The high-voltage signal is fed into the high-impedance analog input **IN** (Pin 2). Additionally, the **SHTDN** (Pin 3) provides a digital shutdown feature; its active-high logic, supported by an internal 100 kΩ pull-up resistor, allows the system to disable the amplifier during fault conditions to conserve power or protect the signal chain.
- **The Low-Side (Output):** Operating on the controller's voltage domain, this side is powered by **V_DD2** (Pin 8) and shared ground **GND_2** (Pin 5). The processed signal is delivered as a differential output through **OUTP** and **OUTN** (Pins 7 and 6). This differential signaling is crucial in our design as it significantly improves noise immunity and ensures that the F28379D Launchpad receives a clean, proportional representation of the high-voltage nodes.

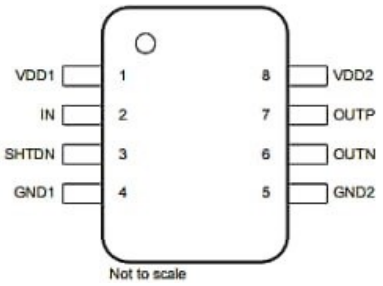


Figure 4.16: AMC1311 (Top View)

4.4.3.3 Technical Specifications and Isolation Safety

In EV power electronics, the choice of the AMC1311 is driven by the need for high noise immunity and operator safety. The following parameters were critical for our hardware integration:

- **High Input Impedance**: It features a typical input resistance of 1 GΩ, which significantly minimizes loading errors when connected to high-voltage resistive dividers.
- **Robust Reinforced Isolation**: The device is certified for reinforced galvanic isolation up to 7000 VPK (per DIN EN IEC 60747-17) and 5000 VRMS (per UL1577). This ensures that the Launchpad remains electrically shielded from voltages up to 1500 VRMS during continuous operation.
- **High CMTI**: With a minimum Common-Mode Transient Immunity (CMTI) of 100 kV/µs, the amplifier maintains signal integrity and prevents output disturbances during the high-speed switching transitions characteristic of bidirectional power stages.

• **Linear Input Range**: It is optimized for a 2-V linear full-scale input, allowing for high-resolution measurement of scaled DC-link voltages.

4.4.3.4 Precision Analog Output and Diagnostic Logic

The AMC1311 provides a highly predictable signal path for the F28379D Launchpad, categorized into three operational regions:

• **Linear Range**: The device maintains a nominal gain of 1 for input voltages (V_{IN}) between –0.1 V and +2 V. This 1:1 ratio ensures that a 2-V input directly results in a 2-V differential output, simplifying firmware scaling.

• **Saturation and Clipping**: For inputs between 2 V and 2.516 V, the device continues to operate with reduced linearity. If the input exceeds the $V_{Clipping}$ threshold (2.516V), the output saturates at a fixed level ($V_{CLIPout}$) to prevent unpredictable signal spikes.

• **Fail-Safe Mechanism**: To enhance system diagnostics, the amplifier features an intelligent fail-safe mode that generates a negative differential output voltage (–2.5V). This distinct signal allows the controller to instantly distinguish a hardware fault from a legitimate zero-volt measurement.

The fail-safe diagnostic is triggered by three specific hardware events:

1. **Missing supply**: If the high-side power (VDD1) is completely disconnected.
2. **Undervoltage**: If falls below the critical threshold (VDD1UV).
3. **Active Shutdown**: If the SHTDN pin is pulled high to intentionally disable the

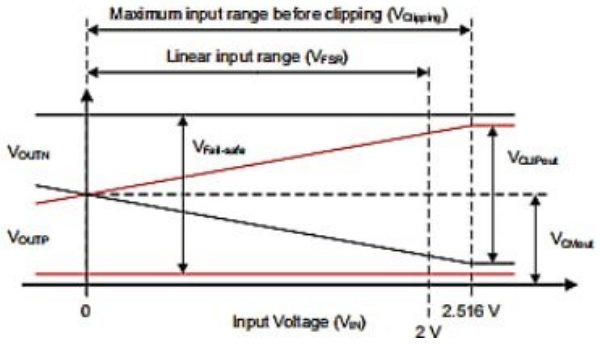


Figure 4.17: Output Behavior of the AMC1311

amplifier.

4.4.3.5 Circuit Design Description

The voltage sensing circuit is designed to scale the high-voltage DC-link (0–400V) to a low-voltage differential signal suitable for the ADC.

4.4.3.5.1 Analog Input and Voltage Divider Calculation

The high-voltage input is interfaced via a screw terminal J1.1. To scale the voltage, a resistive divider network R1, R2 and R3 is used in series with the input resistor R_4. This network reduces the measured node voltage to a maximum of **2V** at the IN pin, as per the AMC1311 linear range.

The resistors are chosen according to the following voltage divider equation:

$$R_{total} = R_1 + R_2 + R_3 = \left(\frac{V_{peak} - V_{INP}}{V_{INP}}\right) \times R_4$$

Where:

- **V_{peak}**: Maximum DC-link voltage to be measured (400V).
- **V_{INP}**: Maximum input voltage to the AMC1311 (2V).
- **R_4**: Fixed input resistance (10 kΩ).

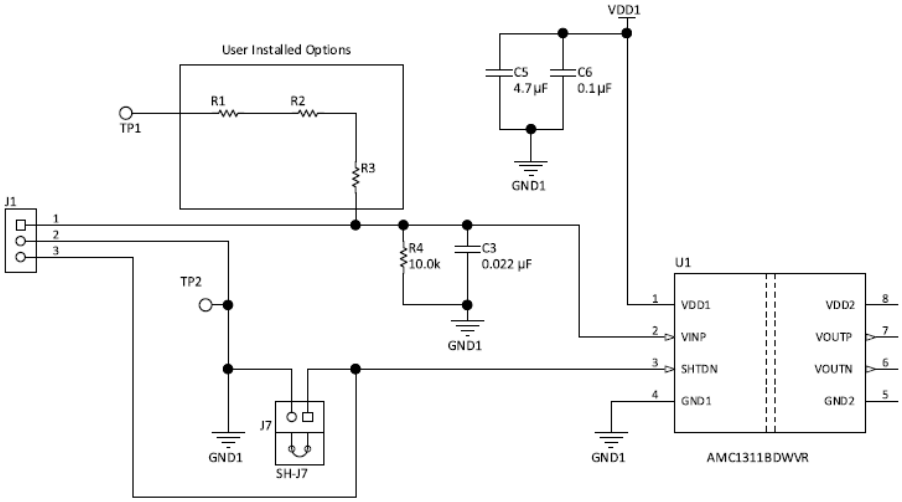


Figure 4.18: AMC1311EVM Schematic – Analog Input Section

4.4.3.5.2 Signal Conditioning and Filtering

An anti-aliasing filter formed by R_4 and C_3 is implemented at the input stage. With the default values, it provides a corner frequency of 723 Hz, effectively filtering out high-frequency switching noise from the power stages before it reaches the ΔΣ modulator.

4.4.3.5.3 Isolated Power Supply Strategy

The AMC1311 requires two independent, galvanically isolated power rails to maintain the integrity of the isolation barrier:

1. **High-Side Supply V_{DD1}:** In this design, V_{DD1} is generated using an SN6501 push-pull driver and a transformer T_1. This creates an isolated 5V rail from the low-side supply,

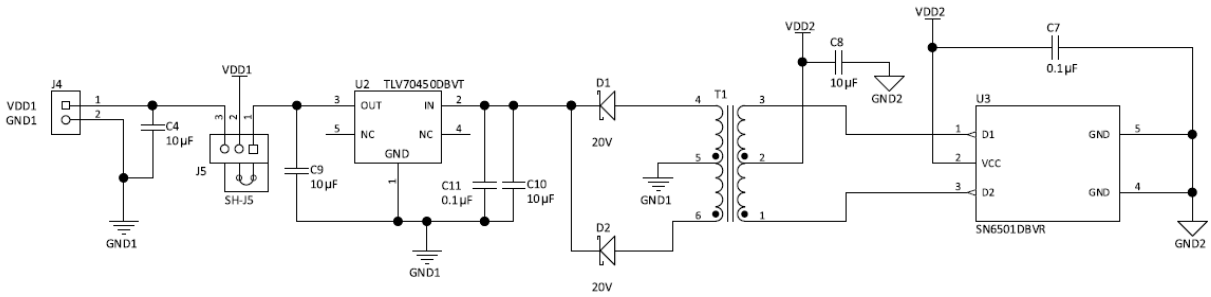


Figure 4.19: VDD1 Input

ensuring no electrical path exists between the 400V side and the controller.

2. **Low-Side Supply V_{DD2}:** Powered directly from the controller's 3.3V or 5V rail, supporting the output differential stage.

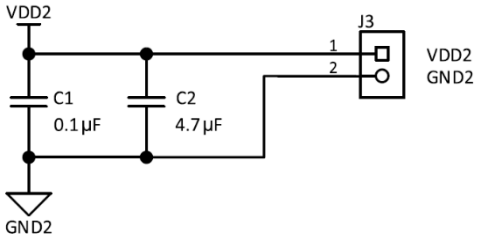


Figure 4.20: VDD2 Input Connector

4.4.3.5.4 Differential to Single-Ended Conversion

The AMC1311 provides a fully differential output centered at 1.44V. To interface with the F28379D ADC, an operational amplifier stage U_4, U_5 is used to convert this differential signal into a single-ended voltage. This stage is configured for unity gain, providing an output swing from $0.5V to 2.5V$, which perfectly matches the unipolar input range of the Launchpad's ADC.

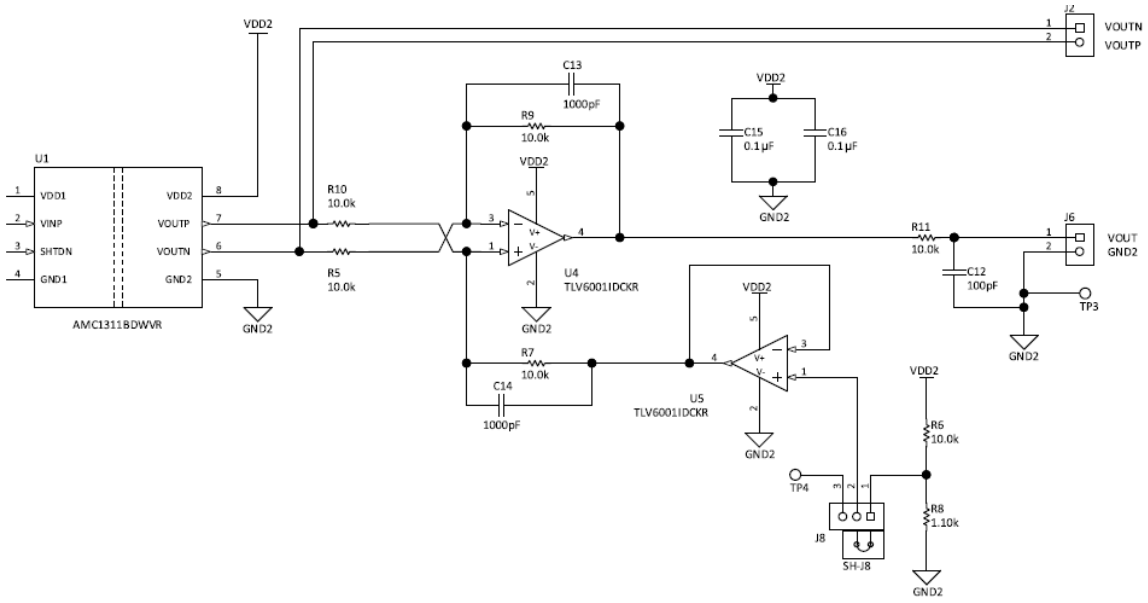


Figure 4.21: AMC1311EVM Schematic – Analog Output Section

4.4.3.6 Hardware Design and Circuit Implementation

The transition from theoretical scaling to physical hardware was executed in **KiCad**, with a design priority on maintaining reinforced galvanic isolation and ensuring high-speed signal integrity.

4.4.3.6.1 Schematic Design and Power Integrity

The schematic integrates the high-impedance resistive divider string R_1 and R_3 with the AMC1311 isolated amplifier. To safeguard against power-rail transients and high-frequency switching noise, a dual-capacitor decoupling strategy was employed. A 100nF ceramic capacitor is placed in parallel with a 1 µF capacitor as close as possible to both the V_{DD1} and supply pins. This configuration provides a low-impedance path for noise across a wide frequency spectrum, ensuring

stable operation of the **ΔΣ** modulator.

4.4.3.6.2 High-Voltage Layout Considerations

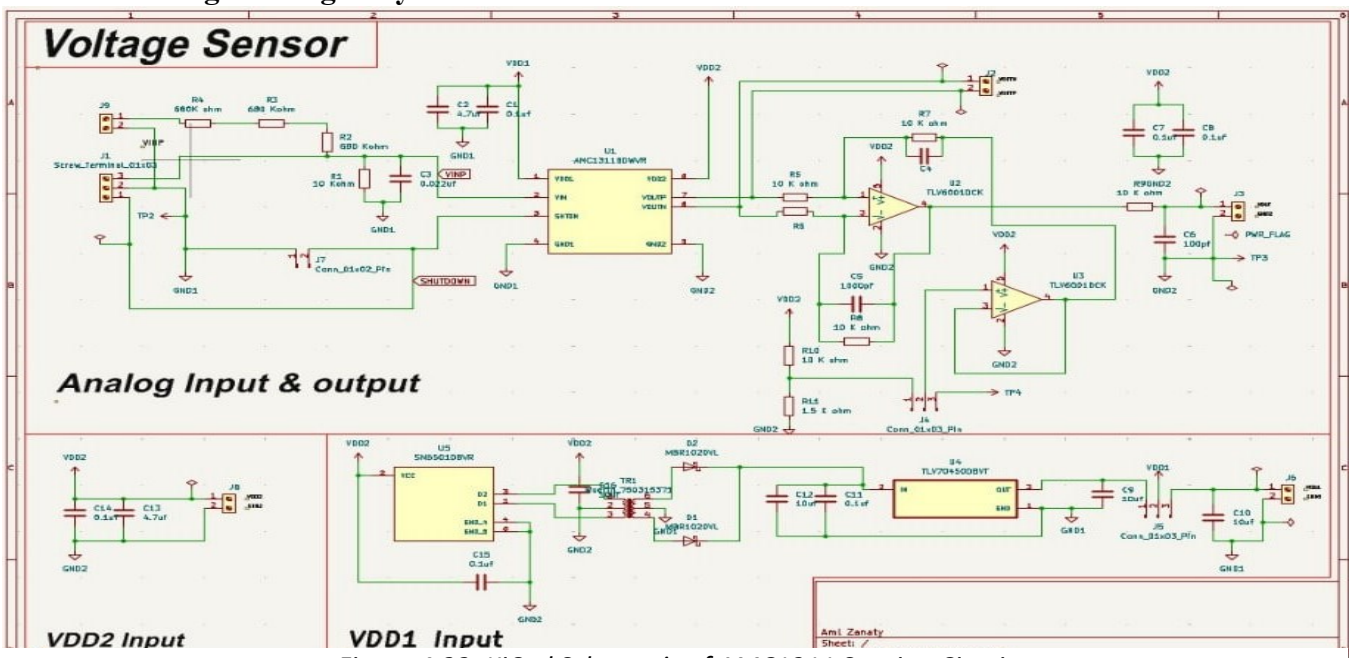


Figure 4.22: KiCad Schematic of AMC1311 Sensing Circuit

In the PCB layout, specific attention was paid to creepage and clearance distances to handle the 400V DC-link safely. The high-voltage input traces are physically separated from the low-voltage output signals by an 8.5 mm isolation gap beneath the AMC1311 package. Additionally, the resistive divider string was laid out in a linear fashion to distribute the voltage stress across multiple SMD packages (1206 size), preventing arcing and ensuring long-term reliability under continuous high-voltage operation.

4.4.3.6.3 KiCad Layout and 3D Visualization

The PCB layout implements a strict isolation moat to maintain an 8.5 mm clearance under the AMC1311, ensuring safety at 400V. As seen in the 3D model, the resistive divider is arranged linearly to prevent arcing by distributing voltage stress across multiple components.

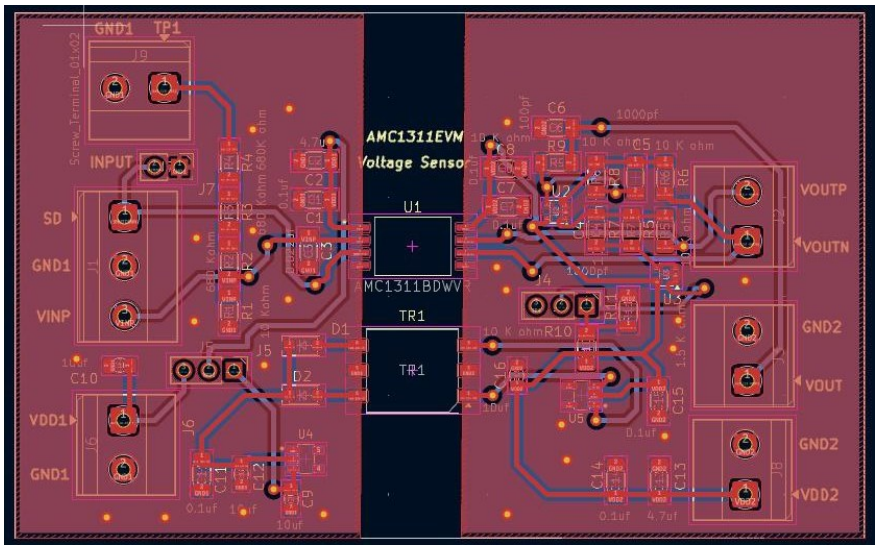


Figure 4.23: PCB Layout of AMC1311 Voltage Sensing Circuit

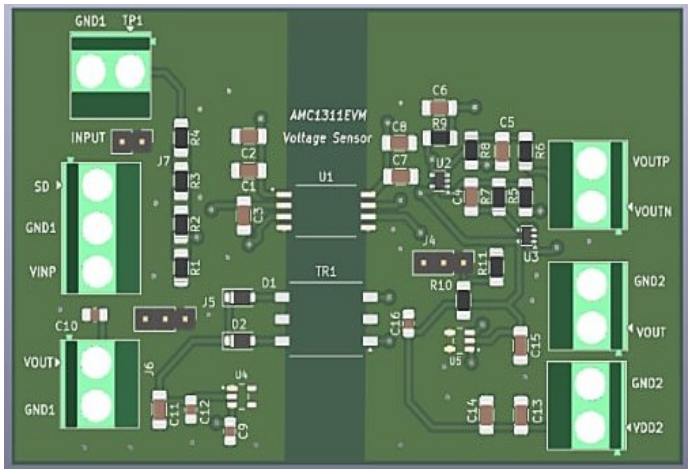


Figure 4.24: 3D Visualization of AMC1311 Voltage Sensing Module

4.4.3.7 Final Hardware Fabrication

Following the design and routing phase, the PCB was fabricated and assembled. Figure 4.19 shows the actual physical implementation of the voltage sensing module. The final board confirms the successful integration of the high-voltage isolation gap and the precise placement of the SMD components, matching the initial design specifications.

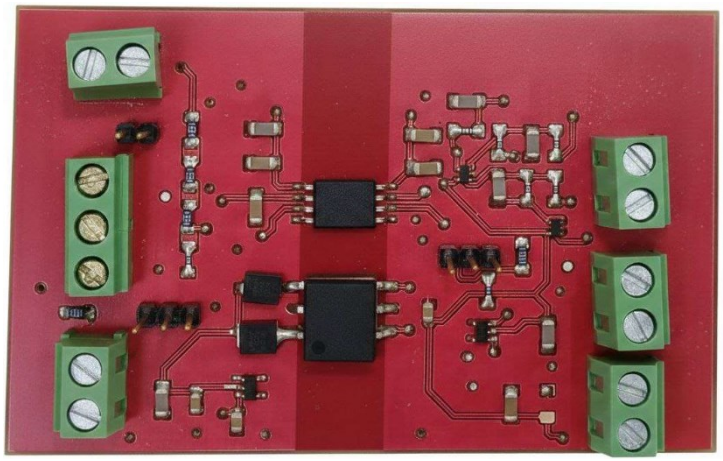


Figure 4.25: Fabricated and Assembled AMC1311 Voltage Sensing PCB

4.4.3.8 Laboratory Implementation and Testing

The voltage sensing stage was validated using the assembled AMC1311 module. The primary objective was to verify the accuracy of the resistive divider and the integrity of the galvanic isolation.

- **Setup:** The high-side $\boldsymbol{V_{DD1}}$ was powered by an isolated **5V** supply, while the low-side $\boldsymbol{V_{DD2}}$ was integrated with the **3.3V** rail of the F28379D Launchpad.
- **Measurement:** A variable DC source was applied to the high-voltage input. The differential output was monitored using a digital multimeter and an oscilloscope to check for noise and common-mode stability.
- **Results:** The output showed a linear differential swing centered around **1.44V**, as expected. The isolation barrier successfully protected the controller side from the high-voltage input, with no observed saturation or signal clipping within the 0–400V equivalent range.

4.4.4.9 Bill of Materials - Voltage Sensing

Table 4.3: Bill of Materials - AMC1311 Voltage Sensing Circuit

Ref. Designator	Component Type	Description
C1, C6	Ceramic Capacitor	0.1 µF, 50 V, ±10%, X7R
C2, C5	Ceramic Capacitor	4.7 µF, 25 V, ±20%, X5R
C3	Ceramic Capacitor	0.022 µF, 50 V, ±1%, C0G/NP0
C4, C8, C9, C10	Ceramic Capacitor	10 µF, 16 V, ±10%, X5R
C7, C15, C16	Ceramic Capacitor	0.1 µF, 50 V, ±10%, X7R
C11	Ceramic Capacitor	0.1 µF, 25 V, ±10%, X7R

C12	Ceramic Capacitor	100 pF, 10 V, ±10%, X7R
C13, C14	Ceramic Capacitor	1000 pF (1 nF), 10 V, ±10%, X7R
D1, D2	Schottky Diode	20 V, 0.5 A
R1, R2, R3	Chip Resistor	680 kΩ, ±5%, 250 mW
R4	Resistor	10 kΩ, 1%, 0.25 W
R5, R6, R7, R9, R10, R11	Resistor	10 kΩ, 1%, 0.1 W
R8	Resistor	1.1 kΩ, 1%, 0.1 W
T1	Power Transformer	340 µH, SMT
TP1, TP2, TP3, TP4	Turret Terminal	Double, Through-Hole
J1	Terminal Block	3-Position, 3.5 mm Pitch
J2, J3, J4, J6	Terminal Block	2-Position, 3.5 mm Pitch
J5, J8	Pin Header	3-Position, 2 mm Pitch, Tin
J7	Pin Header	2-Position, 50 mil Pitch, Gold
U1	Isolated Amplifier	Precision 2-V Input Reinforced Isolated Amplifier
U2	LDO Regulator	5 V, 150 mA, Ultra-Low IQ
U3	Transformer Driver	Isolated Power Supply Driver
U4, U5	Operational Amplifier	1 MHz, Rail-to-Rail I/O

4.5 Signal Conditioning

4.5.1 Scaling for Microcontroller ADC

To ensure accurate interpretation of measured parameters by the TMS320F28379D LaunchPad, the raw analog outputs from the sensing circuits must be conditioned and scaled to match the 0–3.3 V input range of the microcontroller's ADC.

This process includes:

- Filtering high-frequency switching noise generated by the power stage.
- Converting differential signals into single-ended signals compatible with the ADC hardware.
- Adjusting signal amplitude and offset to maximize ADC resolution.

4.5.2 Current Signal Conditioning (TMCS1133)

The TMCS1133 Hall-effect sensor provides a fixed-sensitivity analog output proportional to the input current (e.g., 25 mV/A or 100 mV/A depending on the variant).

Dynamic Range Alignment

The output voltage swing is maintained between approximately 10 mV above ground and 100 mV below the supply voltage (VS), ensuring full utilization of the linear operating region.

Zero-Current Offset

For bidirectional operation in V2X mode, the sensor integrates an internal reference voltage (typically 2.5 V or 1.65 V) representing the zero-current condition. This allows the ADC to distinguish between:

- Charging mode (G2V)
- Discharging mode (V2G)

Noise Suppression

To ensure signal integrity:

- Low-ESR bypass capacitors (0.1 µF and 10 µF) are placed close to the supply pins.
- An optional RC output filter (e.g., 10 pF with a configurable series resistor footprint) may be added to attenuate high-frequency transients generated by the Totem-Pole PFC switching stage.

Hardware Overcurrent Protection

The TMCS1133 includes a programmable VOC pin that defines a digital overcurrent threshold. This feature enables a fast hardware response (~500 ns), independent of ADC sampling or firmware execution speed.

4.5.3 Voltage Signal Conditioning (AMC1311)

The AMC1311 isolated amplifier is used to measure the high-voltage DC-link through a resistive divider scaled to its 2-V differential input range.

Differential-to-Single-Ended Conversion

The AMC1311 provides differential outputs (OUTP and OUTN) centered around a 1.44 V common-mode voltage. Since the microcontroller ADC is single-ended, an external operational amplifier such as the TLV6001 is configured to convert the differential signal into a single-ended output.

Scaling and Level Shifting

The conditioning circuit:

- Converts the differential signal to single-ended form.
- Offsets the signal around a reference voltage (VREF).
- Produces a typical output swing from 0.5 V to 2.5 V.

This range is ideal for 3.3 V ADC systems and provides high measurement resolution for monitoring the 400 V DC-link.

Anti-Aliasing Filter

A simple RC low-pass filter (e.g., 10 Ω and 0.022 μF) is implemented at the amplifier input to:

- Reduce high-frequency noise
- Prevent aliasing before ADC sampling

Fail-Safe Feature

The AMC1311 integrates a fail-safe mechanism: If high-side power is lost, the output transitions to approximately −2.5 V, allowing immediate firmware detection of isolation or power faults.

4.6 Isolated Gate Driver Modules for DAB Converter

These modules act as the vital bridge between the low-voltage control signals and the high-frequency power stage. To meet the specific requirements of the power switches, two distinct types of gate drivers are utilized, each tailored for either Si or SiC MOSFETs.

These drivers provide the precise timing, robust fault protection, and high-speed signal conditioning essential for efficient DAB operation. By ensuring reliable switching and complete electrical isolation, they are fundamental to maintaining system stability and achieving maximum power conversion efficiency.

4.6.1 Gate Driver for Si MOSFETs

Power Supply Arrangement

The module operates using a single supply rail:

- **5V DC**: For input supply (VCC1), with current consumption ranging from 50 mA to 90 mA.

Control and Interface Signals

The module interfaces with the controller using the following pin:

- **PWM:** TTL/CMOS compatible signal input for switching control.

Input High-level Signal Range: 2 V to 5 V.

This module offers a reliable and straightforward control solution for standard switching tasks.

Output Stage Configuration

Each gate driver channel connects directly to the MOSFET through two essential pins:

- **Gate (G):** Provides the switching voltage to control the MOSFET state.
- **Emitter/Source (S):** Serves as the reference point for the gate drive signal.

The output stage features built-in gate resistance, supporting switching frequencies up to 50 kHz. The voltage levels for controlling the MOSFET switching states are defined as follows:

- **Turn-on voltage:** 12 V (typical), ranging from 11.5 V to 12.5 V.
- **Turn-off voltage:** 0 V.

This configuration eliminates the need for external passive components, ensuring a compact and efficient design for the power stage.

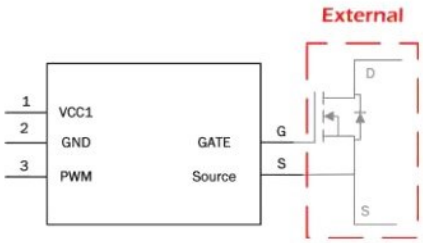


Figure 4.26: Symbol model of the Si MOSFET gate driver module.

Isolation and Protection

The gate driver modules protect the MCU from voltage spikes and noise by isolating the control domain from the high-power HV and LV stages. Key features include:

- **Galvanic Isolation:** Provides 3 kV isolation to ensure signal integrity and circuit safety.
- **Active Miller Clamping:** Prevents parasitic turn-on of the MOSFET during high-speed switching transitions.
- **Input ESD Protection:** Ensures robustness against electrostatic discharge at the control terminals.

1-Channel Isolated Si Gate Driver Strategy

The operational strategy for this driver is to function as a high-performance power amplifier, bridging the gap between the low-power logic signals of the microcontroller and the high-energy requirements of the power switches. By consolidating switching, isolation, and protection into a single, compact architecture, the system achieves precise MOSFET saturation and ensures consistent timing across all operational cycles.

By offloading signal conditioning and protection tasks to the gate drivers, this design simplifies the control architecture, ensures high noise immunity, and maintains stable switching performance even under demanding operating conditions.

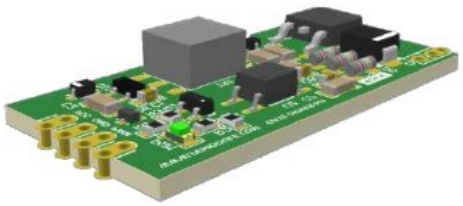


Figure 4.27: 3D visualization of the Si gate driver module.

4.6.2 Gate Driver for SiC MOSFETs

Power Supply Arrangement

The module operates using a single supply rail:

- **5V DC:** For input supply (VCC1), with current consumption ranging from **150 mA** to **300 mA**.

Control and Interface Signals

The module interfaces with the controller using the following pins:

- **PWM:** TTL/CMOS compatible signal input for switching control.
- **FLT:** Fault control output signal for real-time monitoring.

Input High-level Signal Range: 2 V to 5 V.

Output Stage Configuration

Each gate driver channel connects to the SiC MOSFET through:

- **Collector (C):** Monitors the collector-emitter voltage, essential for the integrated desaturation (DESAT) detection functionality.
- **Gate (G):** Provides the switching voltage to control the device state.
- **Emitter (E):** Serves as the reference point for the gate drive signal.

The module features integrated gate resistance and supports high-speed switching up to 50 kHz, eliminating the need for external components. It provides a precise bipolar drive (+15V/-5V), ensuring optimal switching performance for the SiC power stage.

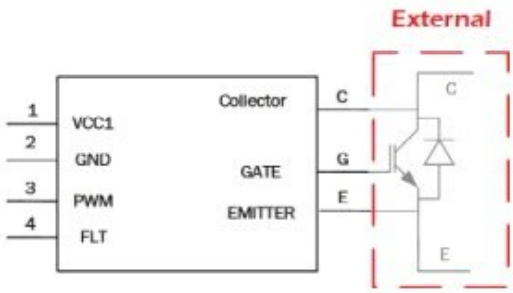


Figure 4.28: Symbol model of the SiC MOSFET gate driver module.

Isolation and Protection

The module is designed for high-voltage, fast-switching applications, providing advanced protection:

- **Galvanic Isolation:** Provides up to 3 kV isolation to ensure circuit safety.
- **Short-Circuit Protection:** Features integrated desaturation (DESAT) detection with immediate fault reporting via the FLT pin.
- **Active Miller Clamping:** Prevents parasitic turn-on during high-speed transitions.
- **Robustness:** High common-mode transient immunity (50 kV/µs) and ESD input protection.

1-Channel Isolated SiC Gate Driver Strategy

The control strategy employs an intelligent gate driver architecture that shifts the system beyond simple signal amplification. By integrating real-time fault diagnostics—including desaturation monitoring, short-circuit detection, and under-voltage protection—the driver acts as a proactive, closed-loop safety layer for the SiC power stage.

This architecture ensures high-speed, reliable switching while enabling the MCU to actively monitor system health, ultimately maximizing both the performance and operational lifespan of the power electronics.



Figure 4.29: 3D visualization of the SiC gate driver module.

4.7 Isolated Voltage Sensing System for DAB Converter

4.7.1 Sensor Board Overview

The system utilizes the Plugy Discovery sensor board, a high-performance, compact platform designed to host up to two independent transducer modules. This board facilitates precise voltage measurement for DC waveforms, ensuring high accuracy through its flexible, hot-swappable configuration.

4.7.2 Transducer Module Specifications

The system employs two fully isolated differential transducer modules to precisely monitor different voltage ranges: model **V.050** for **+/- 50 V** and model **V.400** for **+/- 400 V**.
Key characteristics include:

- **Accuracy:** Delivers precise and reliable measurements that accurately reflect the power circuit voltage levels.
- **Dynamic Response:** A 200 kHz bandwidth ensures fast and accurate tracking of DC waveforms with minimal distortion.
- **Isolation and Immunity:** Features 3 kV galvanic isolation and 15 kV/µs transient immunity to effectively block electrical noise.
- **Signal Integrity:** Differential output signals maintain stable, noise-free transmission to the controller, even in high-frequency environments.
- **Output Characteristics:** The modules are calibrated to provide specific analog feedback signals (VOA and VOB) that are linearly proportional to the sensed voltage (Vsens):

Model V.050 (+/- 50 V):

$$V_{OA} = 0.016 \times V_{sens} + 1.28 \tag{4.6}$$

$$V_{OB} = -0.016 \times V_{sens} + 1.28 \tag{4.7}$$

Model V.400 (+/- 400 V):

$$V_{OA} = 0.002 \times V_{sens} + 1.28 \tag{4.8}$$

$$V_{OB} = -0.002 \times V_{sens} + 1.28 \tag{4.9}$$



Figure 4.30: Differential transducer module for isolated voltage sensing.

4.7.3 Interface and Connectivity

The sensor board provides a straightforward integration with the control system through standardized terminals:

- **Power Supply:** Operates on a +5V DC control power supply, with a common ground terminal for system stability.
- **Signal Output:** Features dedicated analog outputs (V1A/V1B and V2A/V2B) for each transducer module, allowing the MCU to monitor two independent DC voltage signals simultaneously.
- **Input Terminals:** Utilizes spring terminal connectors (S1+/S1- and S2+/S2-) for secure and easy wiring of the voltage inputs.

4.7.4 System Implementation and Design

The implementation strategy prioritizes a compact, isolated design that prevents ground loops and ensures MCU protection. By integrating both transducers and their signal conditioning circuits onto a single board, the design eliminates the need for external components, resulting in a streamlined, robust, and highly reliable measurement path.



Figure 4.31: 3D visualization of the Plugy Discovery sensor board with transducer modules.

4.8 Integrated Overcurrent Detection (TMCS1133)

The TMCS1133 features an integrated Overcurrent Detection (OCD) circuit that triggers an alert via the OC pin in just 500 ns. This hardware-level protection enables nearinstantaneous system shutdown to safeguard MOSFETs from short-circuit damage.

The trip threshold (I_{OC}) is user-programmable by adjusting the voltage at the VOC pin, governed by the following relationship:

$$V_{OC} = S \times I_{OC} \tag{4.10}$$

Where:

- S: Device sensitivity (mV/A).
- I_{OC}: Target overcurrent threshold (A).
- V_{OC}: Applied threshold-setting voltage (V).

4.8.1 Variant Selection and Dynamic Range Optimization

To optimize the TMCS1133 for bidirectional (G2V/V2G) applications:

1. **Quiescent Point Selection:** We selected Variant A ($V_{REF} = 2.5V$) to provide symmetrical headroom for charging and discharging currents within the 5V supply range.

2. **Sensitivity vs. Resolution:** We chose the highest sensitivity variant that accommodates the full-scale current of the PFC and DAB stages. This ensures maximum ADC resolution while maintaining a safety margin to prevent signal clipping during current transients.

The maximum measurable currents are calculated using:

$$I_{I_{IN,MAX+}} = \frac{V_{OUT,MAX} - V_{REF}}{S}$$

$$I_{I_{IN,MAX-}} = \frac{V_{REF} - V_{OUT,MIN}}{S}$$

For example, using the TMCS1133A1 (25 mV/A) allows a range of approx. ±96 A, which provides a sufficient safety buffer for our system's 75 ARMS requirement.

4.8.2 Circuit Design Description

Power Integrity and Output Filtering

To ensure sensor accuracy, we implemented a dual-capacitor decoupling strategy (0.1 μF and 10 μF) near the supply pins to suppress high-frequency noise. Additionally, an integrated RC low-pass filter at the output attenuates voltage transients caused by the Totem-Pole PFC stage's high-frequency switching.

To ensure robust protection, a 1x3 header (J1) allows for flexible overcurrent threshold (VOC) configuration. For manual tuning, a trimmer potentiometer (R5) is used in series with a 5.6 kΩ resistor (R4); this fixed resistor sets a 0.3 V minimum floor to prevent nuisance triggers.

4.9 Summary

This chapter has established the comprehensive hardware infrastructure of the bidirectional on-board EV charger. By strategically selecting 650V-rated Si MOSFETs for low-frequency applications and SiC MOSFETs for high-frequency switching, the system achieves an optimized balance between cost, efficiency, and performance. The implementation of reinforced isolated gate drivers with critical protection features—including desaturation detection, active Miller clamping, and undervoltage lockout—ensures reliable switching at frequencies up to 500 kHz.

The dual-controller architecture further enhances system robustness by separating demanding real-time control tasks on the F28379D LaunchPad from IoT communication

functions on the ESP32 module. This separation guarantees deterministic power stage control while maintaining flexible connectivity for V2X applications.

The precision sensing and monitoring system, built around the TMCS1133 Hall-effect current sensor and AMC1311 isolated amplifier, provides accurate feedback with reinforced galvanic isolation up to 5000 VRMS. Comprehensive laboratory testing validated the linear response of both sensors, confirming their suitability for bidirectional current measurement and high-voltage DC-link monitoring.

The signal conditioning circuits effectively interface these sensors with the microcontroller's ADC, ensuring reliable data acquisition across all operating modes. Hardwarelevel protection mechanisms—including programmable overcurrent detection and fail-safe

diagnostic outputs—provide immediate response to fault conditions independent of software execution.

Together, these integrated hardware components form a robust and modular foundation for the bidirectional AC/DC and DC/DC conversion stages discussed in the subsequent chapters, directly enabling the high efficiency, precise control, and reliable V2X functionality required for modern electric vehicle charging systems.

Chapter 5
Bidirectional AC/DC Converter Design and Implementation

5.1 Introduction

This chapter presents the comprehensive design, modeling, and implementation of the bidirectional AC/DC converter, which serves as the primary grid interface in the proposed EV charging system. The converter is engineered to operate efficiently in both power flow directions: as a high-power-factor rectifier during Grid-to-Vehicle (G2V) charging and as a grid-synchronized inverter during Vehicle-to-Grid (V2G) discharging. The focus is on translating the theoretical advantages of the Totem-Pole topology into a practical, hardware-ready design through systematic component selection, control strategy development, simulation validation, and PCB implementation. The design targets a 1 kW power rating with efficiency exceeding 98% and input current Total Harmonic Distortion (THD) below 5%, ensuring compliance with international standards such as IEC 61000-3-2 and IEEE 1547.

5.2 Fundamentals of Power Factor Correction (PFC)

Power electronics converters connected to the utility grid must comply with power quality standards to ensure efficient energy transfer and minimal harmonic distortion. One of the most important power quality indicators is the Power Factor (PF). Modern electronic loads such as rectifiers, switch-mode power supplies, and battery chargers inherently introduce non-sinusoidal currents, which degrade the power factor and increase harmonic distortion. Power Factor Correction (PFC) techniques are therefore implemented to improve the relationship between input voltage and current, reduce harmonics, and enhance overall system efficiency.

5.2.1 Introduction to Power Factor Correction (PFC)

Power Factor (PF) is defined as the ratio between the real (active) power and the apparent power delivered to a load:

$$PF = \frac{P}{S}$$

In an ideal system with a purely resistive load and perfectly sinusoidal voltage and current waveforms, the power factor equals unity. However, in practical power electronic systems—particularly those employing diode bridge rectifiers followed by bulk capacitors—the input current becomes highly distorted and non-sinusoidal.

A low power factor leads to several undesirable effects:

• Increased RMS current in the supply lines

• Higher conduction losses in cables and components

• Additional voltage drops in distribution systems

• Poor utilization of electrical infrastructure

To mitigate these issues, Power Factor Correction (PFC) circuits are used to shape the input current so that it follows the input voltage waveform, ensuring sinusoidal current and minimal phase displacement.

5.2.2 Limitations of the Conventional Bridge Rectifier

A conventional diode bridge rectifier followed by a bulk capacitor is widely used for AC-to-DC conversion due to its simplicity and low cost. However, this configuration suffers from poor power factor and high harmonic distortion.

Figure 1 illustrates the relationship between the input voltage and input current in such a system. While the input voltage (red) is sinusoidal, the input current (blue) is highly distorted and flows only in narrow pulses near the voltage peaks. This occurs because the capacitor charges only when the instantaneous input voltage exceeds the capacitor voltage, resulting in short bursts of high current.

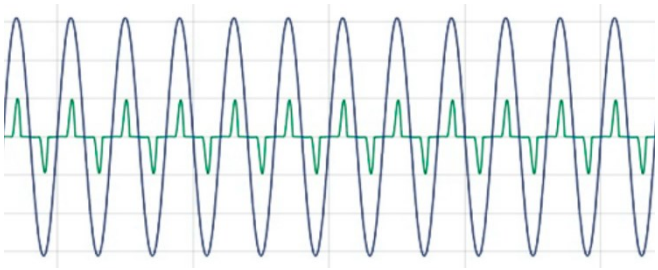


Figure 5.1: Input voltage and input current of a bridge rectifier with capacitor filter

This pulsating current causes several significant issues:

• **Low Power Factor:** The non-sinusoidal current and phase displacement reduce the overall power factor, typically to values between 0.6 and 0.7.

• **High Total Harmonic Distortion (THD):** Sharp current pulses inject significant harmonic components into the grid, degrading power quality and potentially violating regulatory standards.

• **Reduced Efficiency and Reliability:** High peak currents increase thermal and electrical stress on components such as diodes and capacitors, reducing their operational lifetime.

To overcome these drawbacks, active Power Factor Correction (PFC) circuits are employed to shape the input current to follow the input voltage waveform. This improves efficiency, reduces harmonic distortion, and ensures compliance with power quality standards.

In this work, a Totem-Pole PFC topology is adopted due to its high efficiency and reduced conduction losses compared to conventional PFC implementations.

5.3 Topology Selection and Justification

The selection of the power converter topology is a critical design decision that directly influences system efficiency, bidirectional power capability, control complexity, and overall power density of the on-board EV charger. Since the proposed charger is designed to support both Grid-to-Vehicle (G2V) and Vehicle-to-Grid (V2G) operation, the converter must function efficiently in both rectification and inversion modes.

After evaluating several candidate architectures, the Totem-Pole Bridgeless PFC topology was selected as the main AC/DC interface due to its superior efficiency, reduced component count, and inherent bidirectional operation, making it highly suitable for modern V2X-enabled EV charging systems.

Unlike conventional charger architectures that require separate rectifier and inverter stages, the Totem-Pole topology employs a unified switching structure capable of operating as:

- **Active rectifier** during charging (AC → DC), and

- **Grid-tied inverter** during V2G discharging (DC → AC).

This dual functionality significantly simplifies the hardware design while improving overall system performance.

5.3.1 Topology Comparison

Table 5.1 presents a comparative analysis of three major bidirectional AC/DC converter structures considered in this work. The comparison is based on efficiency, component count, control complexity, conduction path, power density, and suitability for V2X operation.

Table 5.1 — Comparative Analysis of Bidirectional AC/DC Converter Topologies

Parameter	Conventional Full-Bridge PFC + Separate Inverter	Dual Active Bridge with AC-Side H-Bridge	Totem-Pole PFC/Inverter (Selected)
Bidirectional Capability	Requires two independent stages	Yes, but requires additional H-bridge	Inherent in topology
Efficiency Range	92–95% (combined stages)	94–97%	97–99%
Component Count	Very high (8+ switches)	Medium (8 switches + transformer)	Optimal (4 main switches)
Conduction Path	3–4 semiconductors	2–4 semiconductors	Only 2 semiconductors
Control Complexity	High (multiple controllers)	Very high (phase-shift control)	Medium (unified control)
Power Density	Low (10–15 W/in³)	Medium (15–20 W/in³)	High (25–35 W/in³)
Cost Index (Relative)	1.8	1.5	1.0 (Baseline)
V2X Readiness	Limited	Good	Excellent

5.3.2 Technical Justification for Selecting the Totem-Pole Topology

The Totem-Pole Bridgeless PFC was selected based on the following engineering advantages:

5.3.2.1 Elimination of the Diode Bridge

Traditional front-end rectifiers use a full diode bridge that introduces significant conduction losses. The Totem-Pole structure removes these diodes and replaces them with actively controlled switches, improving full-load efficiency by approximately 1–2% compared to conventional rectifiers.

5.3.2.2 Unified Power Stage for Rectification and Inversion

The same switching network is used for both AC-to-DC conversion (charging mode) and DC to-AC conversion (V2G mode).

This means the converter inherently performs both:

- **Power factor correction and rectification**, and
- **Grid-synchronized inversion**.

As a result, no dedicated standalone inverter stage is required, reducing hardware complexity and system cost while enabling seamless bidirectional power flow.

5.3.2.3 SiC MOSFET Compatibility

The topology is well suited for high-speed wide-bandgap devices such as Silicon Carbide (SiC) MOSFETs. These devices enable:

- High switching frequency operation (hundreds of kHz),
- Minimal reverse recovery losses,
- Improved thermal performance,
- Increased power density.

5.3.2.4 Reduced Conduction Path

At any given instant, the current flows through only two semiconductor devices, unlike conventional topologies that require three or four devices in the conduction path. This significantly reduces conduction losses and improves overall efficiency.

5.3.2.5 High Power Density

By reducing passive components, eliminating the diode bridge, and enabling high-frequency switching, the Totem-Pole converter achieves significantly higher power density compared to traditional multi-stage architectures.

5.3.2.6 Inherent Bidirectional Operation

The topology naturally supports power flow in both directions without additional hardware. Control algorithms simply reverse the power flow direction:

- Grid → battery (rectifier mode)
- Battery → grid (inverter mode)

This makes it ideal for V2G and smart grid applications.

5.3.2.7 Improved Grid Interaction Performance

During inverter operation, the topology enables precise current shaping and grid synchronization, ensuring:

- Low harmonic distortion,
- Unity power factor,
- Stable grid interaction.

This is essential for standards-compliant grid-connected operation.

5.4 System Specifications and Requirements

The bidirectional AC/DC converter was designed according to the specifications in Table 5.2, which align with typical on-board charger requirements for light-duty electric vehicles.

Table 5.2: System specifications for G2V and V2G mode.

G2V/Charging Mode	
Input Voltage Range	230 V AC (RMS)
Grid Frequency	50 Hz
Output Voltage (DC Link)	400 V DC
Battery Voltage	48 V nominal
Rated Output Power	1 kW
Switching Frequency (AC/DC)	100 kHz
Target Efficiency	> 97%
Target Power Factor	> 0.99
Maximum Input Current THD	< 5%
V2G/Discharging Mode	
Output Voltage (Grid)	230 V AC (RMS)
Grid Frequency	50 Hz
DC Link Voltage	400 V DC
Battery Voltage	48 V nominal
Rated Output Power	1 kW
Switching Frequency (AC/DC)	10 kHz
Target Efficiency	> 96%
Maximum Output Current THD	< 5%
Operating Temperature	-20 °C to 85 °C

5.5 Component Design and Selection

5.5.1 SiC MOSFET Selection for Both Legs

The bidirectional Totem-Pole PFC converter employs the same SiC MOSFETs for both the high-frequency (HF) and low-frequency (LF) legs. The C3M0075120K (1200 V, 75 mΩ) MOSFET was selected based on its high voltage rating, low $R_{DS(ON)}$, fast switching capability, and low gate charge (45 nC). The 1200 V rating provides a safe margin above the 400 V DC-link, accounting for possible voltage spikes during fast switching.

The RMS current through each MOSFET is calculated considering the 230 V AC input and 1 kW output:

$$I_{AC} = \frac{P_{OUT}}{V_{AC}\,PF} = \frac{1000}{230 * 1} \cong 4.35\ \text{A RMS}$$

Each MOSFET conducts roughly half of the AC line cycle, giving:

$$I_{MOSFET,RMS} = \frac{I_{AC}}{\sqrt{2}} \cong 3.08\ \text{A RMS}$$

Conduction and switching losses are calculated using this RMS current and the specified 100 kHz switching frequency.

5.5.2 Boost Inductor Design

The boost inductor was designed to operate in Continuous Conduction Mode (CCM) with a 2% inductor ripple current relative to the peak inductor current.

The peak inductor current is:

$$I_{L,peak} = I_{AC} \cdot \sqrt{2} \approx 4.35 \times 1.414 \approx 6.15 \text{ A}$$

The allowable inductor ripple current is:

$$\Delta I_L = 0.02 \cdot I_{L,peak} \approx 0.02 \times 6.15 \approx 0.123 \text{ A}$$

The minimum inductance required is calculated as:

$$L_{min} = \frac{(D - D^2) \cdot V_{in,min}}{\Delta I_L} \cdot \frac{1}{f_{sw}} \approx 4.5696 \times 10^{-3} \text{ H} \approx 4.5696 \text{ mH}$$

Where:

- $V_{in,min} = 230$ V RMS ≈ 325 V peak
- $D \approx 0.425$ (duty cycle for 400 V DC − link)
- $f_{sw} = 100$ kHz

The calculated value of the boost inductor is $L_{calc} \approx 4.5696$ mH. However, a standard inductor value of 8.1 mH was selected to provide a margin and better accommodate the subsequent inverter stage design and its current ripple requirements.

5.5.3 DC-Link Capacitor Sizing

The DC-link capacitor was designed to limit DC voltage ripple to 2% of the nominal 400 V DC-link, which corresponds to approximately 8 V peak-to-peak. The required capacitance is:

$$C_{DC} = \frac{P_{out}}{2 \cdot V_{OUT} \cdot 2f_{line} \cdot \Delta V_{DC}} = \frac{2.5 \text{ A}}{2 \cdot 50 \cdot 8} \approx 1562.5 \text{ µF}$$

The calculated value of the DC-link capacitor is $C_{calc} \approx 1562.5$ µF. However, a standard capacitor value of 1000 µF was selected, providing sufficient ripple attenuation while simplifying component sourcing and PCB layout.

5.6 Charging Direction Operation (G2V Mode)

In G2V mode, the converter operates as a Totem-Pole Bridgeless Power Factor Correction (PFC) rectifier. The primary objectives are to draw sinusoidal current in phase with the grid voltage and boost the input voltage to a regulated 400 V DC output.

5.6.1 Operating Principle

In Grid-to-Vehicle (G2V) mode, the bidirectional AC/DC converter functions as a Totem-Pole Bridgeless Power Factor Correction (PFC) rectifier. In this mode, the system draws sinusoidal current from the grid that is in phase with the grid voltage, while simultaneously boosting the AC input voltage to a regulated 400 V DC at the DC link. This DC voltage serves as the input for the downstream DC/DC stage, which in turn supplies the 48 V battery pack through the DC-DC converter.

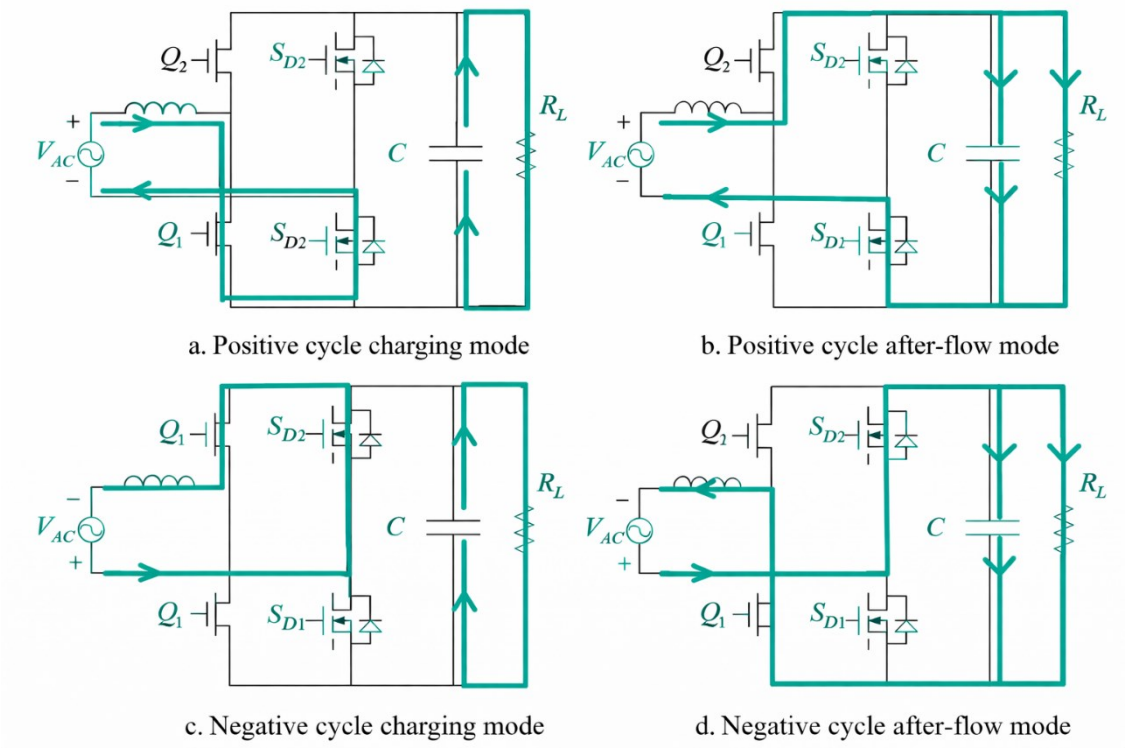


Figure 5.2: Operating modes of Totem-Pole

The operation of the Totem-Pole PFC can be divided into four main modes over a complete AC line cycle. The high-frequency leg (Q1, Q2) operates at the switching frequency of 100 kHz, enabling fast energy transfer, while the low-frequency leg (SD1, SD2) switches at the grid frequency (50 Hz) to provide synchronous rectification.

I. Positive Cycle Charging Mode

During the positive half-cycle of the AC input, MOSFET Q1 is turned on while Q2 is off. The input current flows through L, Q1, and SD1, storing energy in the inductor.
In this mode, the circuit behaves like a boost converter, storing energy in the inductor and transferring it to the capacitor and load.

II. Positive Cycle After-Flow Mode

After Q1 is turned off, the inductor current continues to flow through the body diodes of Q2 and SD1. This freewheeling current maintains continuous energy transfer to the DC link and the load, reducing energy loss and ensuring smooth current flow.

III. Negative Cycle Charging Mode

During the negative half-cycle of the AC input, MOSFET Q2 is turned on while Q1 is off. The current flows through through L, Q2, and SD1, storing energy in the inductor. This mode mirrors the positive half-cycle operation but in the opposite direction.

IV. Negative Cycle After-Flow Mode

After Q2 is turned off, the inductor current continues to flow through the body diodes of Q1 and SD2, maintaining continuous energy delivery to the DC link and load. This ensures smooth operation and reduces current discontinuity.

This approach allows the system to maximize energy transfer from the grid to the DC link while keeping harmonic distortion low, which is critical for meeting IEC 61000-3-2 standards and ensuring safe operation in residential grids.

5.6.2 Simulation Model and Control Strategy

5.6.2.1 Overall Simulink Model

This subsection presents the complete system configuration of the Totem-Pole Bridgeless Power Factor Correction (PFC) converter utilized in this study. The implementation includes the power stage topology, control architecture, voltage and current sensing elements, and high-frequency switching scheme required for high-efficiency AC-to-DC conversion with near-unity power factor. The model serves as the reference platform for system analysis, design verification, and performance validation of the grid-connected bidirectional charging system throughout this work.

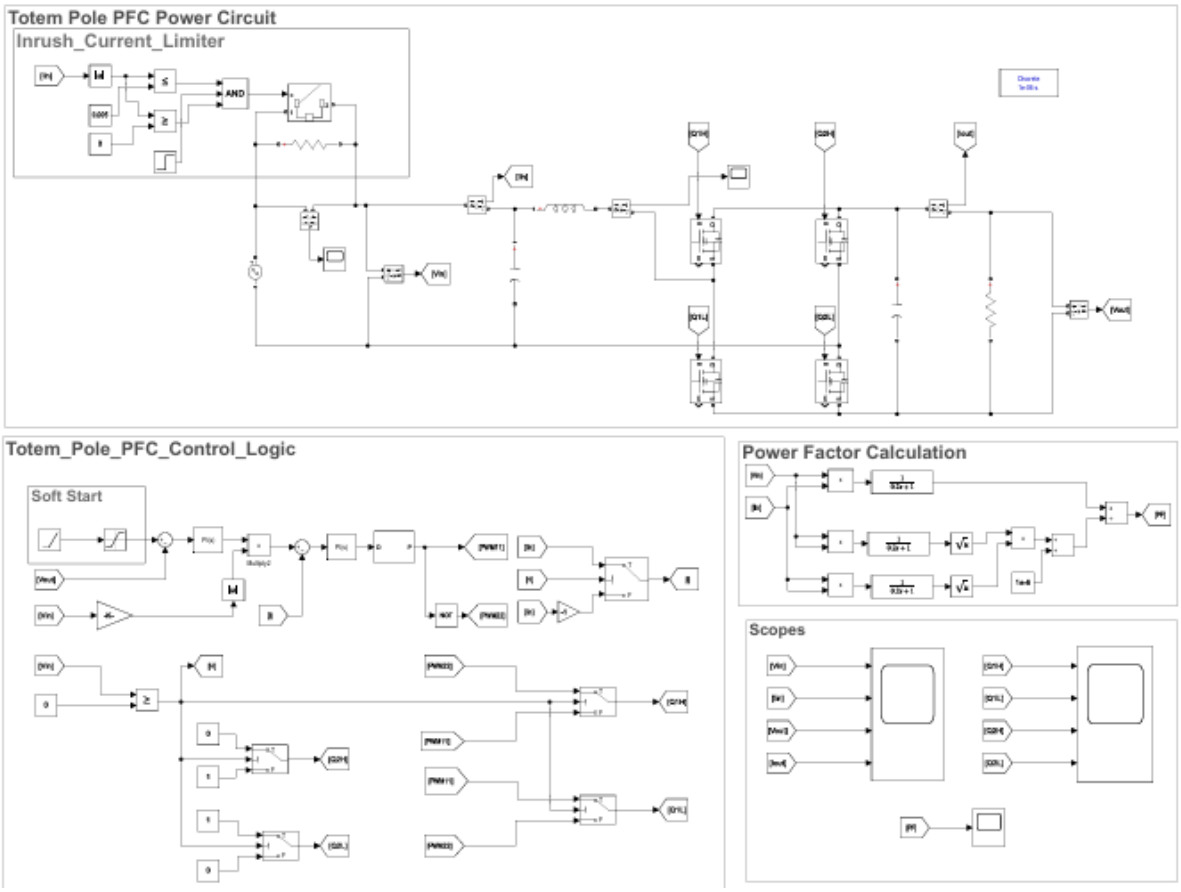


Figure 5.3: Overall Simulink model of the Totem-Pole Bridgeless PFC converter

5.6.2.2 Power Stage of the Totem-Pole Bridgeless PFC converter

The power stage of the Totem-Pole Bridgeless Power Factor Correction (PFC) converter serves as the main interface between the AC grid and the DC-link. The topology employs a totem-pole switching structure with high-frequency active switches and low-frequency synchronous devices to perform AC–DC conversion while actively shaping the input current. By eliminating

the conventional diode bridge, the converter reduces conduction losses and achieves high-efficiency operation with near-unity power factor.

5.6.2.3 Control Strategy of the Totem-Pole Bridgeless PFC converter

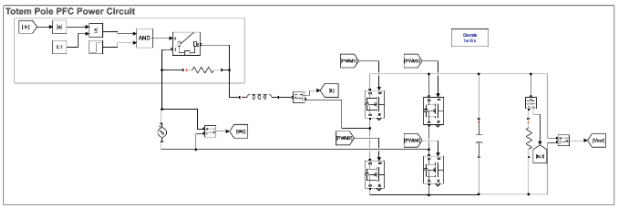


Figure 5.4: Block Diagram of Power Stage of the Totem-Pole Bridgeless PFC converter

The main objectives of the implemented control system are:

- To regulate the DC-link output voltage to a constant reference value of 400 V.
- To shape the input current to be sinusoidal and in phase with the AC input voltage, achieving near-unity power factor.
- To generate appropriate switching signals for both the high-frequency and low frequency legs of the Totem-Pole topology.
- To ensure safe startup operation by limiting inrush current and preventing excessive electrical stress on power components.
-

5.6.2.3.1 Voltage Outer Loop

The outer voltage control loop is responsible for regulating the DC-link voltage and maintaining power balance between the AC input and the DC output. The measured DC-link voltage is compared with a reference generated by a soft-start block, which gradually ramps the reference from zero to 400 V. This prevents large transient currents during startup.

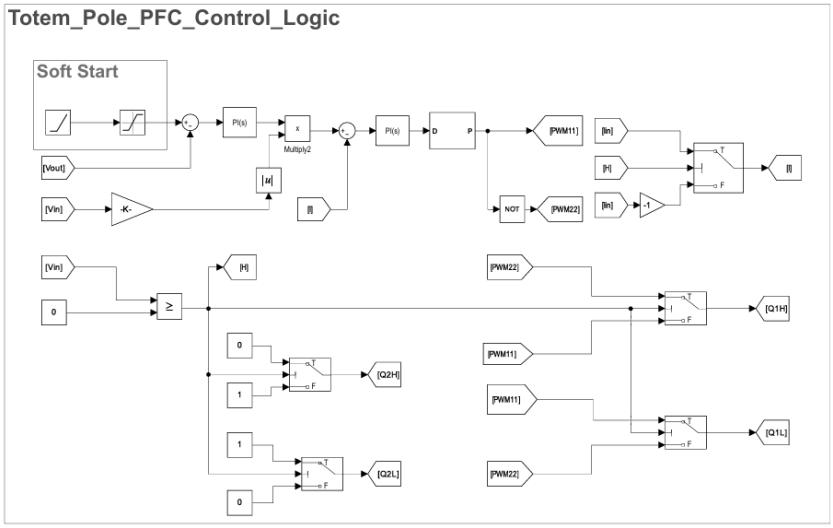


Figure 5.5: Block Diagram of Control Stage of the Totem-Pole Bridgeless PFC converter

The resulting voltage error is processed by a PI controller. The controller output determines the amplitude of the input current reference. This loop operates with a relatively low bandwidth compared to the current control loop to ensure stable dynamic performance.

5.6.2.3.2 Reference Current Generation

To achieve power factor correction, the reference current is generated using an input-voltage feedforward method. The instantaneous input voltage is normalized and multiplied by the output of the voltage PI controller:

$$I_{ref} = K\, V_{in,pu}$$

This approach ensures that the reference current waveform follows the shape of the input voltage, producing a sinusoidal current that remains synchronized with the grid voltage.

5.6.2.3.3 Current Inner Loop

The inner current control loop regulates the inductor current to track the generated reference current. The measured inductor current is compared with the reference, and the error is processed using a PI controller.

This loop operates with a higher bandwidth than the voltage loop to ensure fast dynamic response and accurate current tracking, which is essential for achieving low current distortion and stable operation.

5.6.2.3.4 Input Voltage Polarity Detection and Switching Logic

Proper operation of the Totem-Pole converter requires detection of the input voltage polarity. Comparator blocks are used to determine whether the grid voltage is in the positive or negative half-cycle.

Based on the detected polarity, the appropriate low-frequency switch is activated to provide the correct current return path. Logical control ensures complementary gate signals and prevents shoot-through conditions, guaranteeing proper current flow during both half-cycles.

5.6.2.3.5 PWM Generation

PWM signals are generated by comparing the duty cycle command from the current controller with a high-frequency carrier waveform. The resulting PWM signals drive the active high-frequency switching devices of the Totem-Pole converter, while the low-frequency devices commutate at the line frequency.

This modulation strategy enables continuous conduction mode operation and minimizes input current distortion.

5.6.2.4 Inrush Current Limiter Implementation

To protect the DC-link capacitor and power switches during startup, an inrush current limiter is implemented in the power circuit. The limiter consists of a series precharge resistor and a relay-controlled bypass path.

During startup, the resistor limits the charging current of the DC-link capacitor, preventing excessive current and protecting system components. Once the DC-link voltage reaches a safe level, the relay bypasses the resistor, allowing normal operation with reduced conduction losses.

5.6.2.5 System Testing and Simulation Results

This section presents and discusses the simulation results of the Totem-Pole PFC converter operating in charging mode as an active rectifier, obtained using MATLAB/Simulink. The results validate the effectiveness of the proposed control strategy in regulating the DC-link voltage and achieving power factor correction.

AC Input Voltage and Current

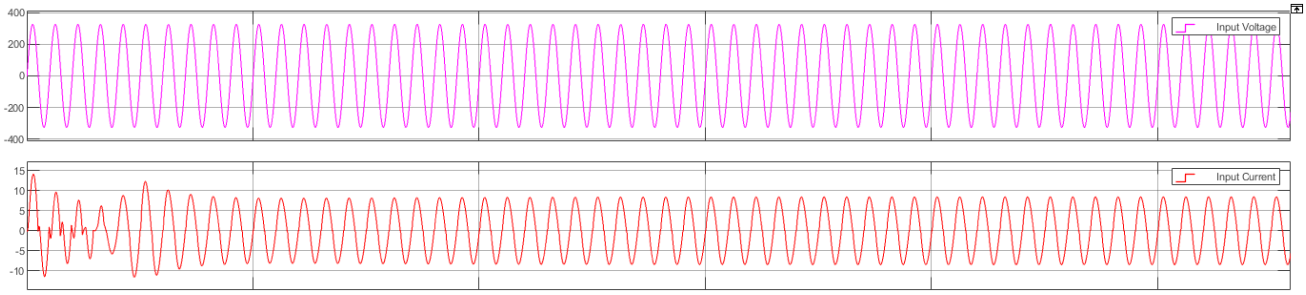


Figure 5.6: AC Input Voltage and Current of the Totem-Pole PFC

This figure illustrates the AC input voltage and the corresponding input current of the Totem-Pole PFC. The input current closely follows the voltage waveform and remains in phase with it throughout operation. This confirms that the control strategy enforces a sinusoidal input current, achieving near-unity power factor and reduced harmonic distortion.

Inductor Current

The inductor current waveform shows continuous conduction with controlled ripple, indicating operation in Continuous Conduction Mode (CCM). During startup, a small current overshoot is observed, which is effectively damped once steady-state is reached. This confirms proper tuning of the inner current loop and accurate tracking of the reference current.

DC-Link Voltage

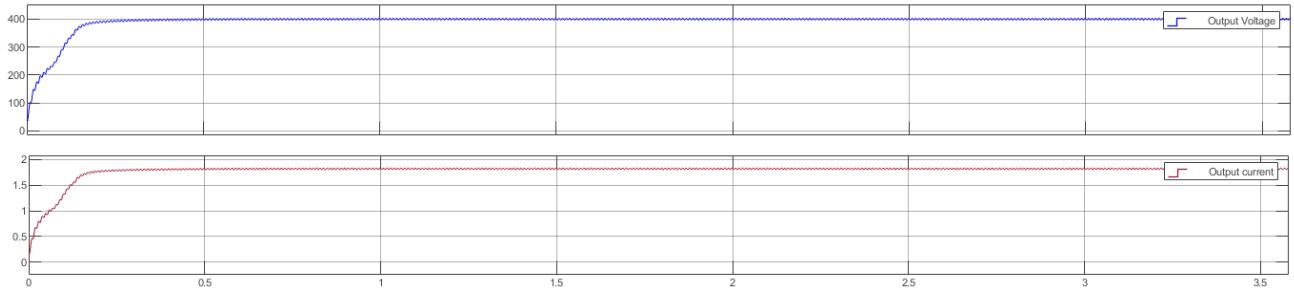


Figure 5.7: DC Output Voltage and Current of the Totem-Pole PFC

The DC-link output voltage rises smoothly from zero to the regulated reference of 400 V due to the soft-start mechanism, preventing sudden capacitor charging and inrush currents. After reaching the reference, the voltage remains stable with minimal ripple (<2%), confirming effective outer loop voltage control.

Startup and Inrush Current

The effect of the inrush current limiter is evident during startup. The precharge resistor and relay bypass mechanism limit initial current draw, allowing the DC-link voltage to rise progressively without stressing components. Once the DC-link reaches a safe level, normal operation resumes smoothly without instability.

PWM Switching Behavior

High-frequency switching waveforms show stable PWM operation once steady-state is achieved. The duty cycle dynamically adapts to regulate input current and maintain DC-link voltage. Synchronization between the high-frequency and low-frequency legs of the Totem-Pole ensures continuous conduction and low input current distortion.

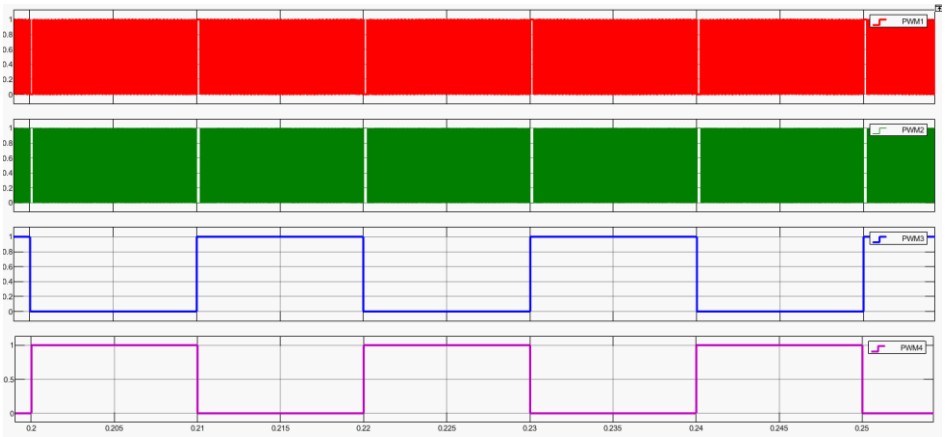


Figure 5.8: Generated PWM Signals

5.7 Disharging Direction Operation (V2G Mode)

In V2G mode, the converter operates as a grid-tied single-phase inverter. The same power stage is reconfigured through control algorithms to inject synchronized sinusoidal current back into the grid.

5.7.1 Operating Principle

A single-phase grid-connected inverter converts DC voltage from a DC-link or battery into a sinusoidal AC voltage compatible with the utility grid. It serves as the interface between the DC energy source and the AC grid, ensuring efficient power transfer and synchronization.

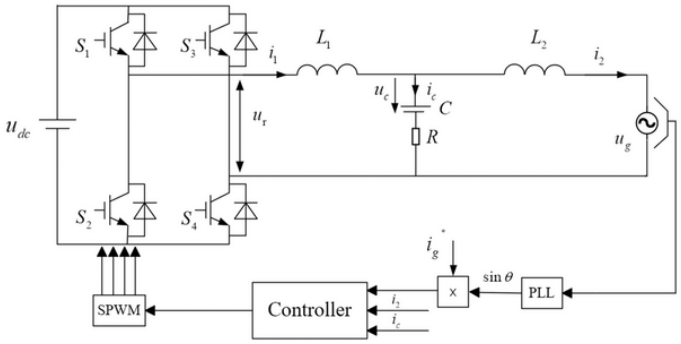


Figure 5.9: single-phase grid-connected inverter construction

How it works:

1. **Switching Action:** The inverter uses four power switches in a full-bridge configuration. By turning these switches on and off in a controlled sequence, the DC voltage is alternately applied across the AC output terminals to generate positive and negative half-cycles of AC voltage.

2. **Pulse Width Modulation (PWM):** High-frequency PWM signals are used to shape the output voltage. The duty cycle determines the voltage amplitude, while fast switching allows the waveform to approximate a sinusoid with low harmonic distortion.

3. **Current Control and Synchronization:** The inverter controls its output current to match the grid voltage waveform in both **magnitude and phase**, achieving near-unity power factor and minimal total harmonic distortion (THD).

4. **DC-Link Regulation:** The DC-link voltage is maintained at a constant reference value to provide stable operation under varying load and grid conditions.

5. **LCL Filter:** An LCL filter is placed at the inverter output to suppress high-frequency switching harmonics. It consists of:

    - **Inverter-side inductor (L1)**: smooths the switching ripple from the inverter output.

    - **Grid-side inductor (L2)**: works with L1 to provide additional impedance against harmonics.

    - **Shunt capacitor (C)**: connected between the inductors to further filter high, frequency components.

    The LCL filter ensures that the current injected into the grid is nearly sinusoidal, improving power quality and meeting grid standards such as IEC 61000-3-2.

6. **Grid Safety Compliance:** The inverter monitors grid voltage and frequency. Power is only injected when the grid is stable, protecting both the inverter and the utility network.

5.7.2 LCL Filter Design

In grid-connected inverters, an LCL filter is commonly used to attenuate high-frequency switching harmonics generated by the inverter before injecting power into the grid. The LCL filter provides better harmonic attenuation than a simple L filter, with lower losses and smaller component sizes. It consists of:

- (L_1): Inverter-side inductor
- (L_2): Grid-side inductor
- (C): Filter capacitor

The design process involves selecting suitable values for (L_1), (L_2), and (C) based on the following criteria:

Capacitor Selection

The capacitor is chosen such that the reactive power absorbed by it does not exceed a small percentage of the rated power (Q = 5% of S). Using the relation:

$$Q = \frac{V_{AC}^2}{2\pi f C} = 5\% \text{ of } S$$

Given:

$$V_{AC} = 230 \text{ V RMS}, \quad f_g = 50 \text{ Hz}, \quad f_{sw} = 10\text{kHz}, \quad S = 1 \text{ kVA}$$

The calculated capacitor value is:

$$C = \frac{Q}{2\pi f V_{rms}^2} = 3\ \mu F$$

Inductor Selection

The inductors (L_1) and (L_2) are selected based on the maximum allowable voltage drop and current ripple. The following criteria are used:

- Maximum voltage drop across the total inductor ($L_1 + L_2$) should not exceed 10% of the rated voltage.
- Maximum current ripple should not exceed 10% of the rated current.

Using these criteria, the inductor values are calculated as:

$$L_1 = \frac{V_{DC}}{4 f_{sw} \Delta I_{pp,max}} = 8.1317\ mH$$

$$L_1 + L_2 = \frac{10\%\ of\ V_{AC}^2}{2\pi f S} = 16.838\ mH$$

$$L_2 = 16.838 - 8.1317 = 8.7068\ mH$$

Resonance Frequency Verification

The resonance frequency of the LCL filter is determined to ensure it lies between 10 times the grid frequency f_g and 0.5 times the switching frequency f_{sw} to avoid resonance issues:

$$f_{res} = \frac{1}{2\pi\sqrt{\frac{L_1 L_2}{(L_1 + L_2)} C}} = 1417.0698\ Hz$$

This value satisfies the design condition:

$$500\ Hz < f_{res} < 5000\ Hz$$

5.7.3 Simulation Model and Control Strategy

5.7.3.1 Overall Simulink Model

This subsection presents the complete Simulink model of the single-phase grid-connected inverter considered in this study. The model integrates the power stage, control algorithms, sensing blocks, and modulation units required for grid-connected operation. It represents the reference system configuration used for analysis and validation throughout this work.

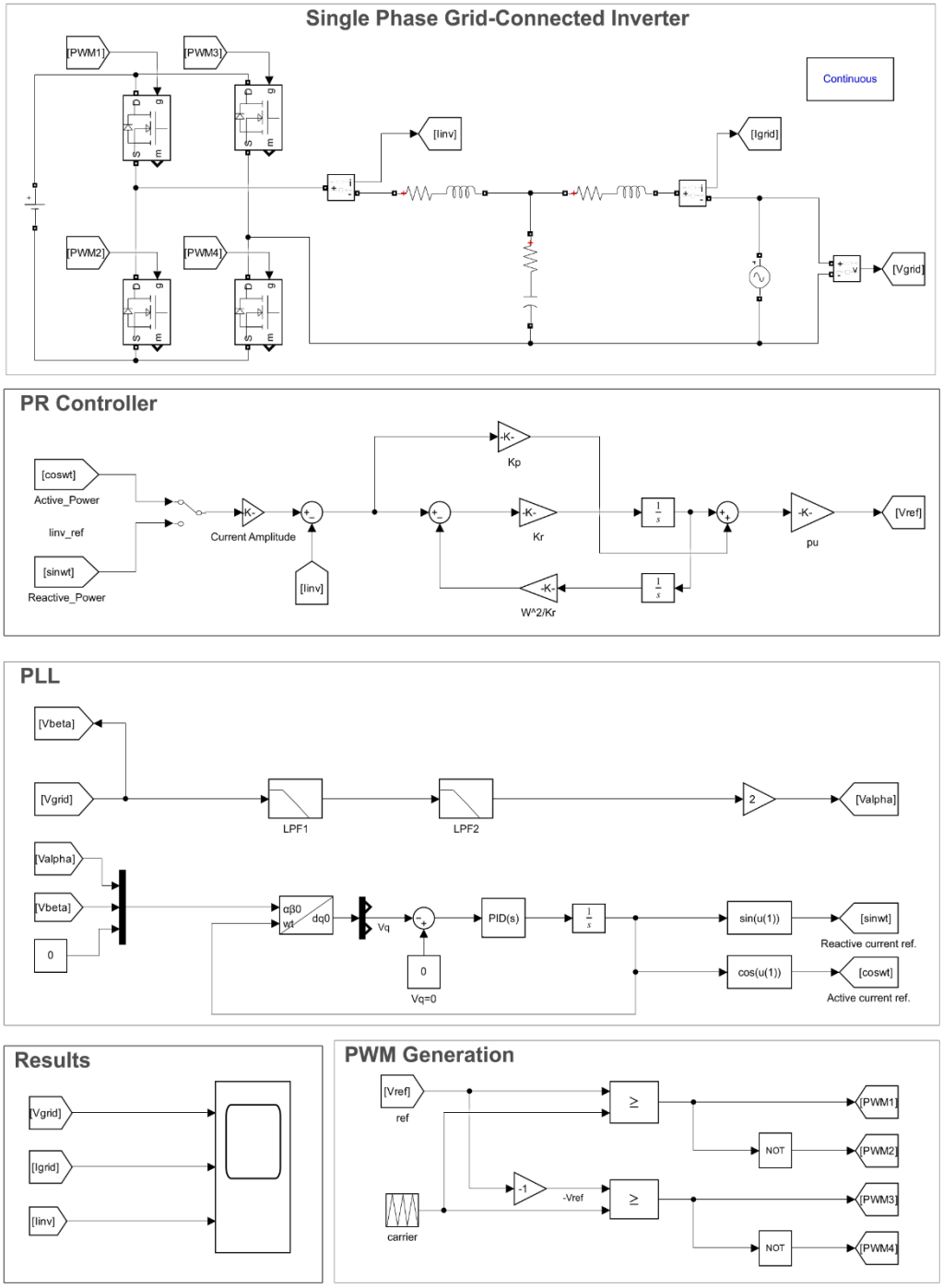


Figure 5.10: Overall Simulink model of the single-phase grid-connected inverter

5.7.3.2 Power Stage of the Grid-Connected Inverter

The power stage of the grid-connected inverter represents the physical interface between the DC source and the utility grid. In this model, a single-phase full-bridge inverter topology is employed to convert the DC-link voltage into an AC voltage synchronized with the grid.

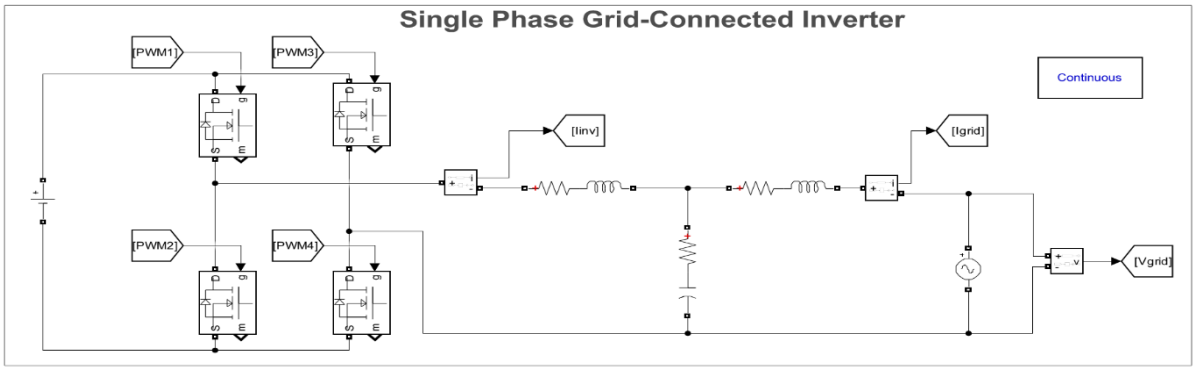


Figure 5.11: Block Diagram of Power Stage of the Grid-Connected Inverter

5.7.3.3 PLL for Grid Synchronization

In single-phase grid-connected inverters, a Phase-Locked Loop (PLL) is used to estimate the grid voltage phase angle and frequency required for synchronization. Since only a single-phase grid voltage is available, an orthogonal signal generation (OSG) method is required to create a virtual two-phase system for phase estimation.

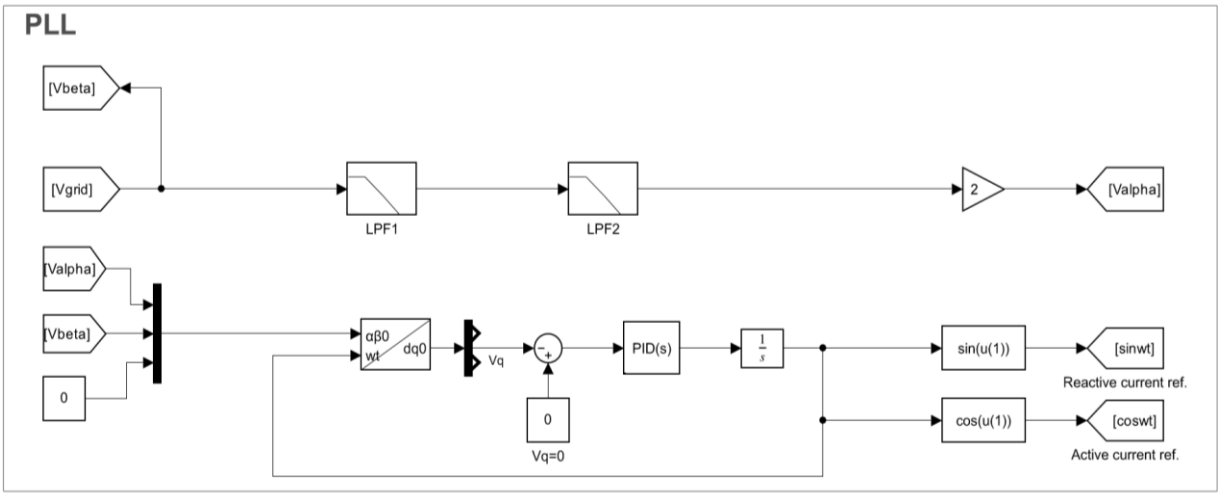


Figure 5.12: Block Diagram of Single-Phase PLL for Grid Synchronization

In this system, the orthogonal signal generation is implemented using a Low-Pass Filter (LPF)-based structure. Let the single-phase grid voltage be expressed as: $v(t)=V_m\sin(\omega t)$

The LPF used in the orthogonal signal generator has the following transfer function: $H(s)=\frac{\omega_c}{\omega_c+S}$

where $\omega_c$is the corner frequency. In polar form, the transfer function is given by:

$$H(j\omega)=\frac{\omega c}{\sqrt{\omega^2+\omega_c^2}} \angle -(\tan^{-1}\frac{\omega}{\omega_c})$$

By selecting $\omega_c = \omega$, the magnitude and phase of the transfer function become:

$$H(j\omega)=\frac{1}{2} \angle -45^\circ$$

Due to the magnitude attenuation introduced by the LPF, a gain factor of 2 is applied within the orthogonal signal generation structure. By appropriately combining the outputs of two identical

LPFs and applying this gain, two orthogonal voltage components are obtained. In this implementation, the filtered grid voltage is considered as V_β, while the generated orthogonal component is defined as V_α. These signals form a virtual orthogonal reference frame suitable for PLL operation.

The generated αβvoltage components are transformed into the synchronous dqreference frame using the estimated electrical angle θ.

The main control objective of the PLL is to regulate the q-axis voltage component to zero ($V_q = 0$), which ensures phase alignment between the inverter and the grid voltage. A PI controller processes the V_q error and adjusts the estimated angular frequency. This frequency is then integrated to obtain the electrical angle θ, which is continuously updated.

The sine and cosine of the estimated angle are generated and utilized throughout the control system *for reference frame transformations and modulation. By continuously tracking the grid phase and* frequency, the PLL ensures accurate synchronization of the inverter with the grid.

5.7.3.4 Control Strategy of the Grid-Connected Inverter

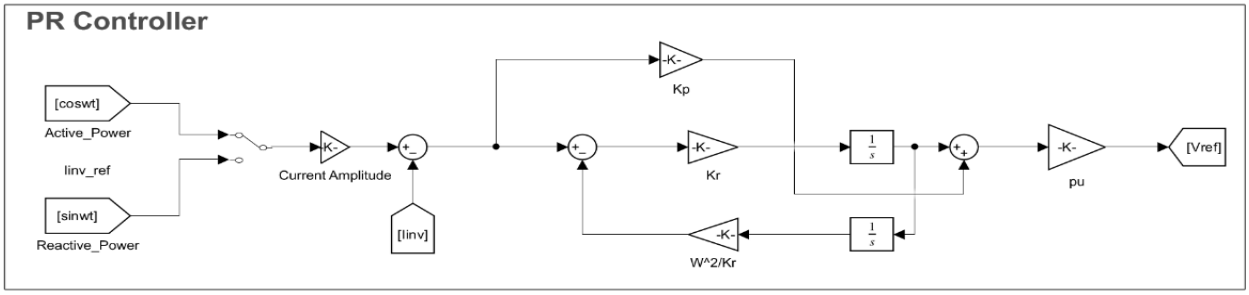


Figure 5.13: Block Diagram of Control Stage of the Grid-Connected Inverter

5.7.3.4.1 Reference Current Generation

The grid-connected inverter supports independent control of active and reactive power. Accordingly, two reference signals are defined:

- **Active current reference**, which determines the amount of real power exchanged with the grid.
- **Reactive current reference**, which controls the reactive power contribution.

These references are combined to generate the overall inverter current reference. The reference current amplitude is calculated and multiplied by either cos(ωt) or sin(ωt), depending on whether active or reactive power control is required, to obtain the desired reference inverter current waveform.

5.7.3.4.2 Current Measurement and Feedback

The inverter output current is measured and fed back into the control loop. This measured current is compared with the generated reference current to produce the current error signal. Accurate current feedback is essential for maintaining low harmonic distortion and ensuring compliance with grid standards.

5.7.3.4.3 Proportional–Resonant (PR) Current Controller

A Proportional–Resonant (PR) controller is employed to regulate the inverter output current. The PR controller consists of:

- A proportional gain (K_p) that ensures fast dynamic response.

- A resonant term (K_r) tuned at the grid angular frequency (ω), which is obtained from the PLL.

The resonant structure provides high gain at the fundamental frequency, enabling zero steady-state error in tracking sinusoidal current references. This makes the PR controller particularly suitable for single-phase grid-connected inverters.

The output of the PR controller represents the commanded inverter voltage required to force the output current to follow the reference.

5.7.3.4.4 Modulation Signal Normalization

The controller output voltage is normalized with respect to the DC-link voltage to generate a modulation signal within acceptable limits. This normalization prevents overmodulation and ensures safe operation of the inverter switches.

5.7.3.4.5 PWM Generation

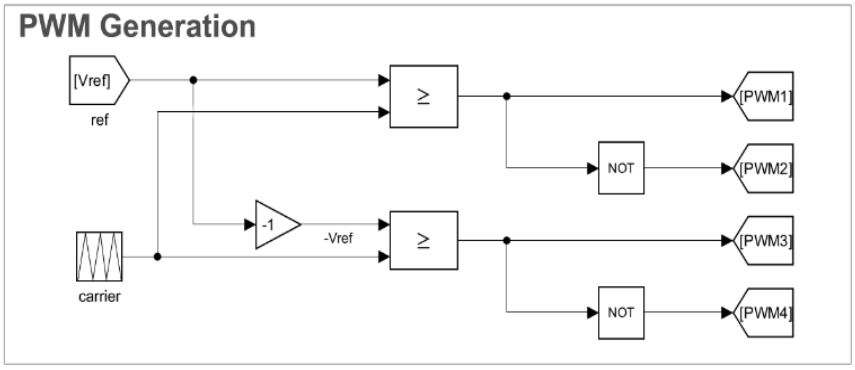


Figure 5.14: Block Diagram of PWM Generation

The normalized modulation signal is compared with a high-frequency carrier waveform to generate the PWM signals. These signals are applied to the gate drivers of the inverter switches, determining the switching pattern and the resulting output voltage waveform.

This stage forms the interface between the control algorithm and the power stage, translating control decisions into physical switching actions.

5.7.3.5 System Testing and Simulation Results

This section presents the simulation-based evaluation of the proposed single-phase grid-connected inverter. The objective is to verify the correct operation of the control strategy, including PLL-based synchronization, orthogonal signal generation, current tracking, and dynamic response, before proceeding to hardware implementation.

5.7.3.6.1 PLL Synchronization

The Phase-Locked Loop (PLL) is responsible for synchronizing the inverter with the grid. The generated q-axis voltage component ($V_{q)}$ is regulated to zero to ensure correct phase alignment.

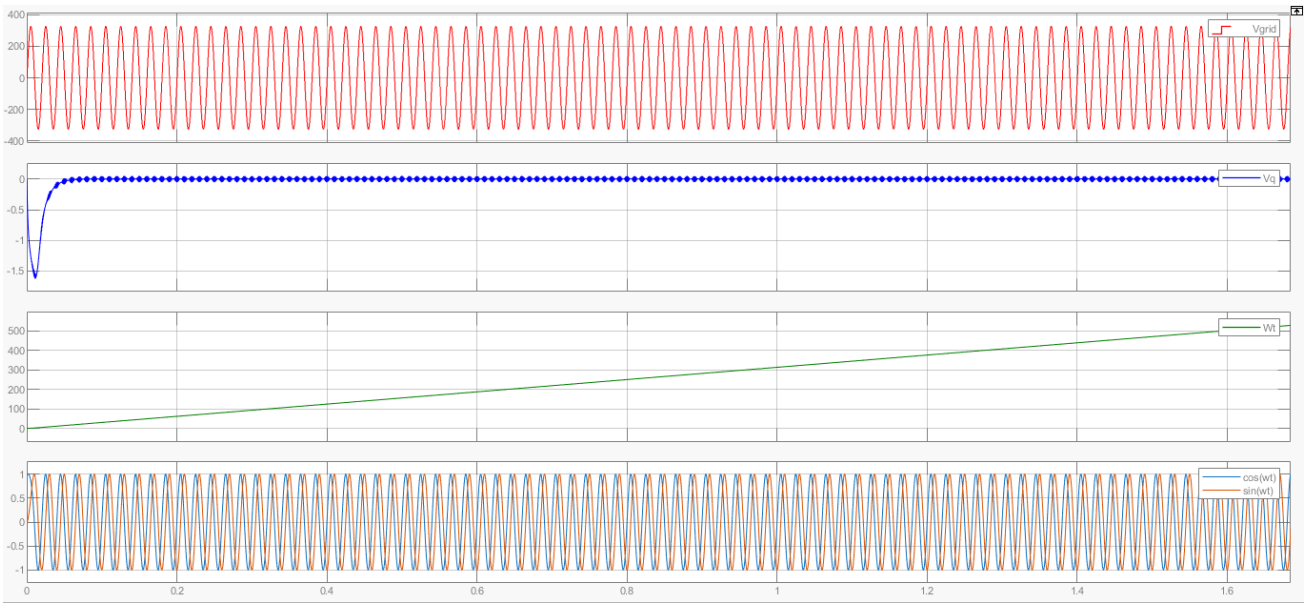


Figure 5.15: Overall waveforms of PLL Synchronization Performance

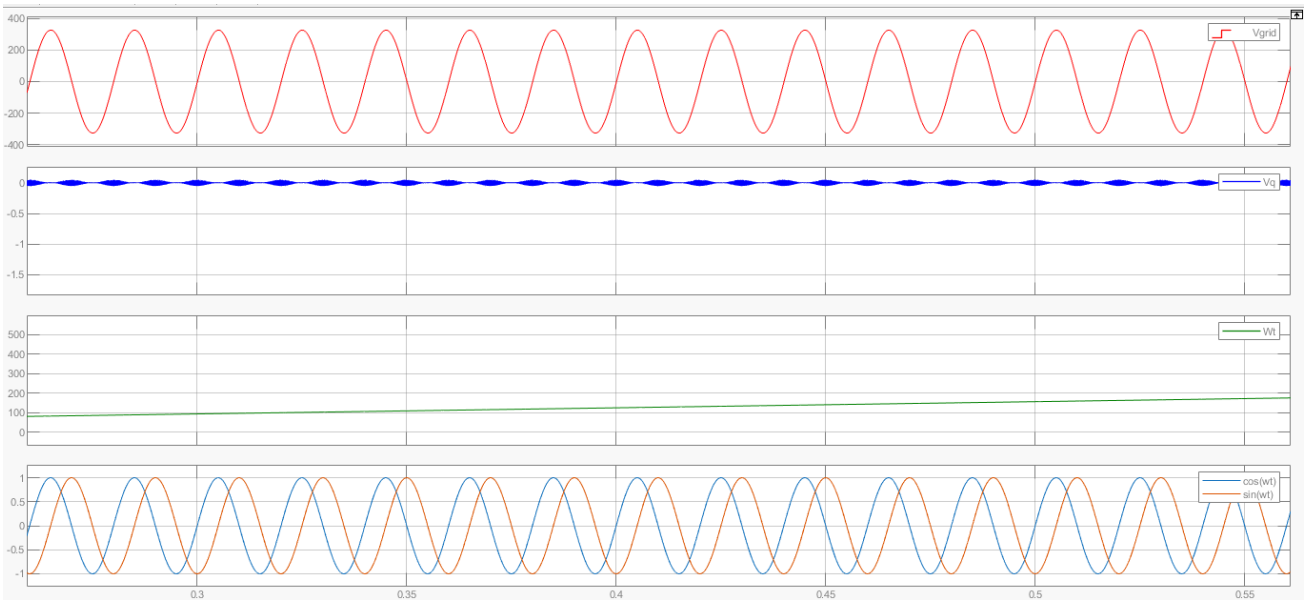


Figure 5.16: Zoomed-in view of waveforms of PLL Synchronization

Those figures illustrate the grid voltage (V_{grid}) along with the estimated PLL signals (q-axis voltage, phase angle (ωt), cos(ωt) and sin(ωt)). The results demonstrate stable synchronization, confirming the effectiveness of the PLL in tracking the grid voltage phase.

5.7.3.6.2 Orthogonal Signal Generation

For single-phase operation, orthogonal components V_α and V_β, are required for dq transformation.

This figure shows the grid voltage (V_{grid}) and the LPF-based orthogonal components V_α and V_β. The signals exhibit correct 90° phase shift, validating the orthogonal signal generation approach.

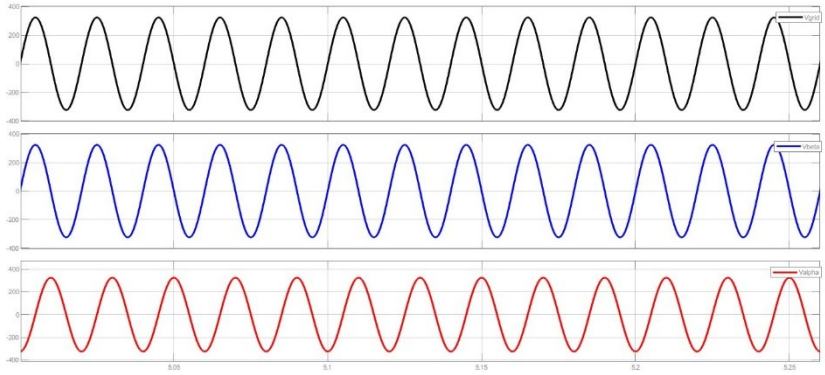


Figure 5.17: Orthogonal Voltage Components Generated by LPF-Based OSG

5.7.3.6.3 Inverter Output and Grid Current

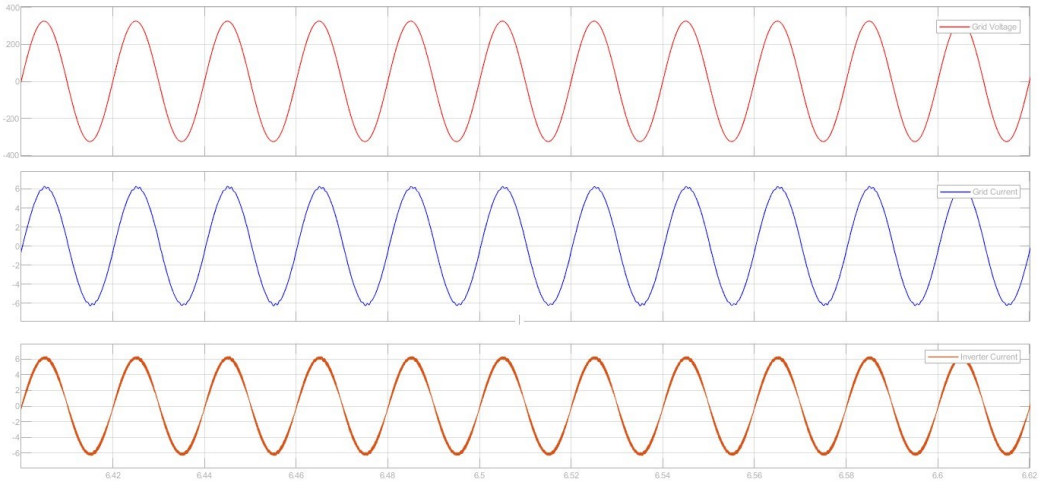


Figure 5.18: Grid Voltage, Current and Inverter Output Current

This figure presents (V_{grid}), (I_{grid}) and ($I_{inverter}$). The results confirm that the inverter successfully delivers current to the grid with proper synchronization, producing smooth sinusoidal waveforms with minimal distortion.

5.7.3.6.4 Current Tracking Performance

Accurate tracking of the reference current is essential for maintaining the desired active and reactive power.

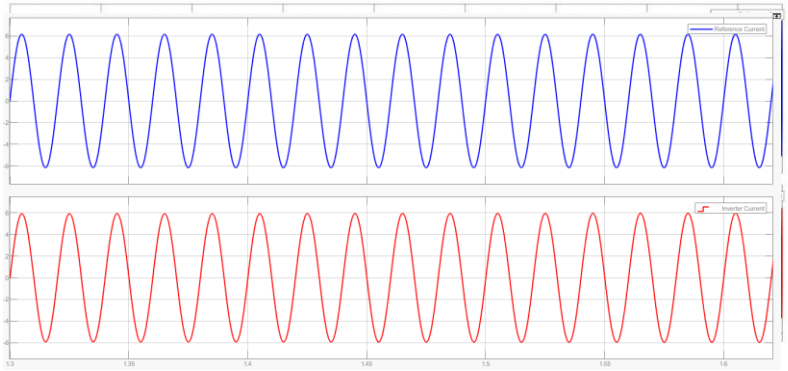


Figure 5.19: Zoomed-in View waveforms of reference current and inverter output current

This figure depicts the reference current (I_{ref}) and the measured inverter current ($I_{inverter}$). The results show precise tracking with negligible steady-state error, demonstrating the effectiveness of the PR controller in following the reference current.

5.7.3.6.5 PWM Generation

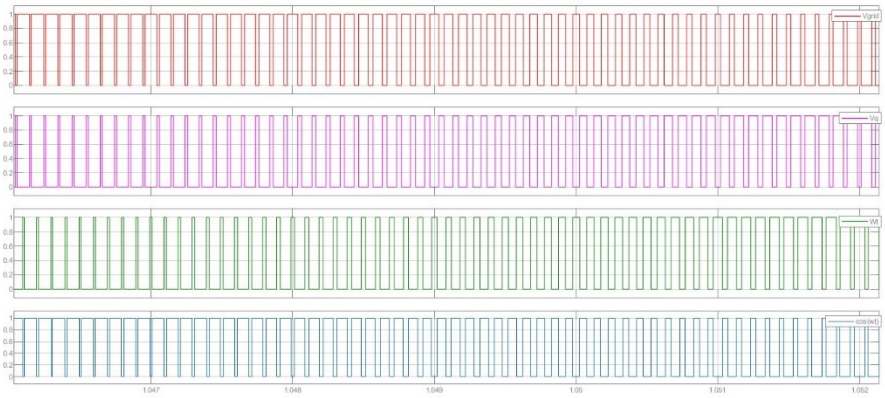


Figure 5.20: Generated PWM Signals

This figure shows the pulse-width modulated (PWM) signals applied to the inverter switches. The PWM signals are generated by comparing the normalized modulation voltage with a high-frequency carrier waveform. These signals control the switching pattern of the inverter and determine the output voltage waveform, ensuring proper current tracking and safe operation of the power devices.

5.7.3.6.6 Step Response (Dynamic Performance)

The dynamic response of the inverter is evaluated by applying a step change in the reference current

This figure illustrates the reference current (I_{ref}) and the inverter current ($I_{inverter}$) during the step input. The inverter rapidly follows the change in reference, with minimal overshoot and stable

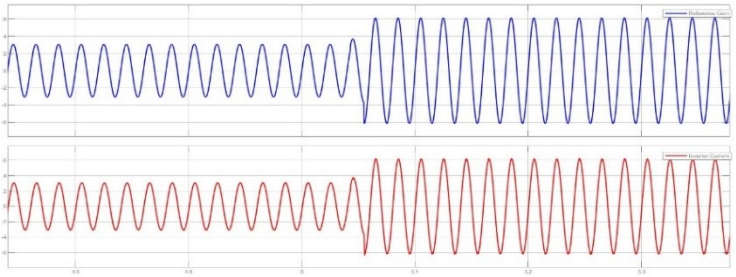


Figure 5.21: Reference current and inverter output current during a step change

settling, confirming both the stability and responsiveness of the proposed control system.

5.8 System-Level Integration of AC/DC Converter Modes

To enhance system flexibility and functionality, both operating modes of the AC/DC converter are integrated into a unified system-level model. The developed model enables seamless operation under different power flow conditions while maintaining stable performance and coordinated control.

The converter is designed to operate in multiple modes depending on system requirements. During grid-to-DC operation, the converter functions as an AC/DC stage to regulate the DC-link voltage and supply the load or energy storage system. In the reverse condition, power flow can be controlled accordingly based on system demand and operating strategy. Mode selection

and transition are handled through supervisory control logic, ensuring smooth switching without instability or performance degradation.

The integrated model combines the power stage, control loops, sensing units, and modulation blocks within a single simulation environment. Shared components such as the DC-link, filters, and measurement blocks are utilized by both modes, which reduces redundancy and improves coordination between operating states.

This system-level integration allows comprehensive evaluation of converter performance under different operating scenarios, including mode transition dynamics, power regulation, and waveform quality. Overall, the unified model demonstrates reliable bidirectional functionality, coordinated control, and smooth mode transition, making it suitable for advanced power electronic applications such as smart charging systems and grid-interactive energy conversion.

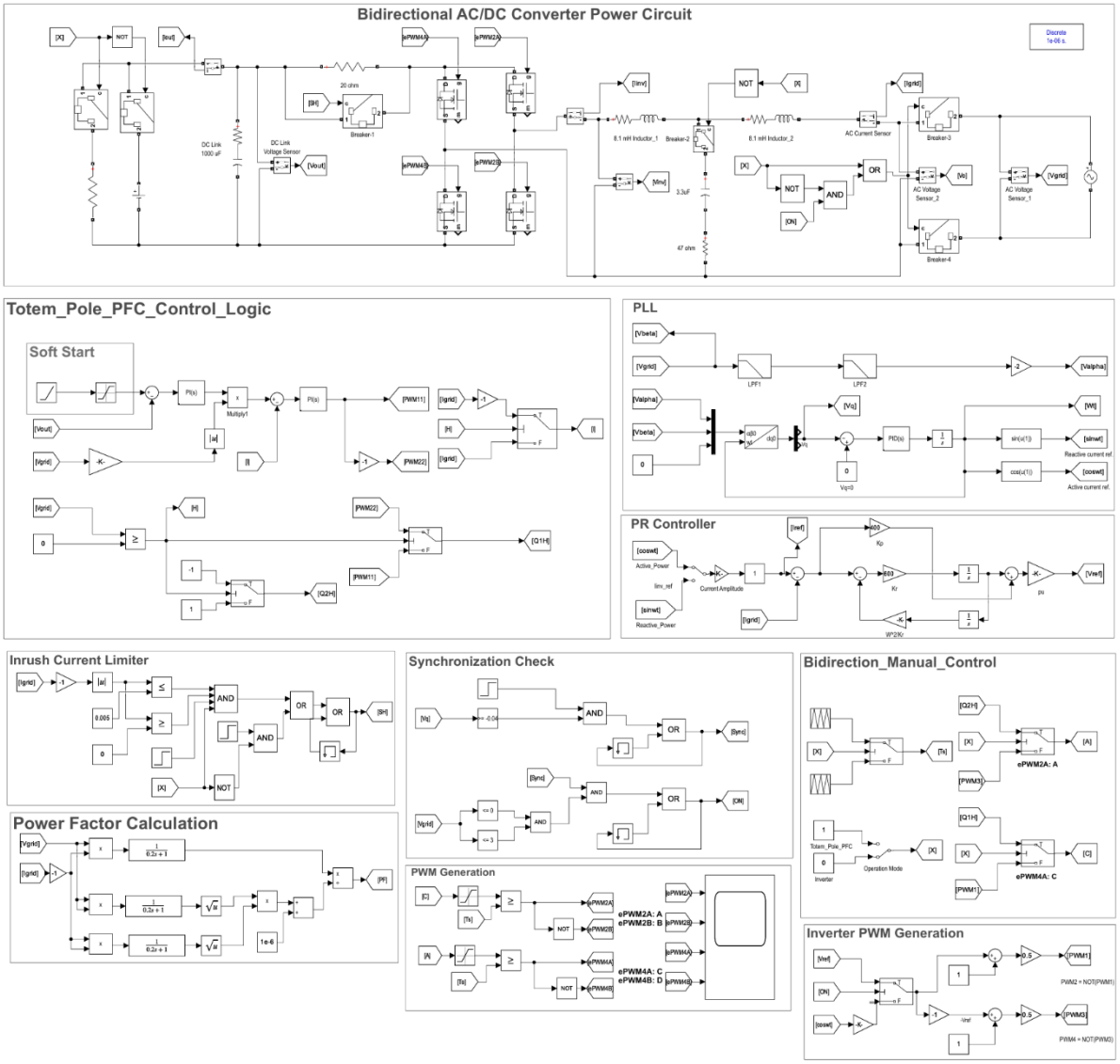


Figure 5.21: Complete System Model Integrating the Totem-Pole PFC and Grid-Connected Inverter Stages

5.9 Role of the Model in Hardware Implementation

The proposed Simulink model is intended not only for system verification and performance evaluation but also for practical hardware implementation. Each subsystem is developed using hardware-compatible control structures that can be directly mapped to the TI C2000 digital controller, including PWM generation, ADC sampling, PI control loops, PLL synchronization, and protection logic. This hardware-oriented architecture enables automatic code generation

through Simulink Embedded Coder, reducing implementation time and ensuring consistency between simulation and experimental realization. Consequently, the model serves as a comprehensive platform for system analysis, controller development, real-time testing, and deployment of the

bidirectional AC/DC converter.

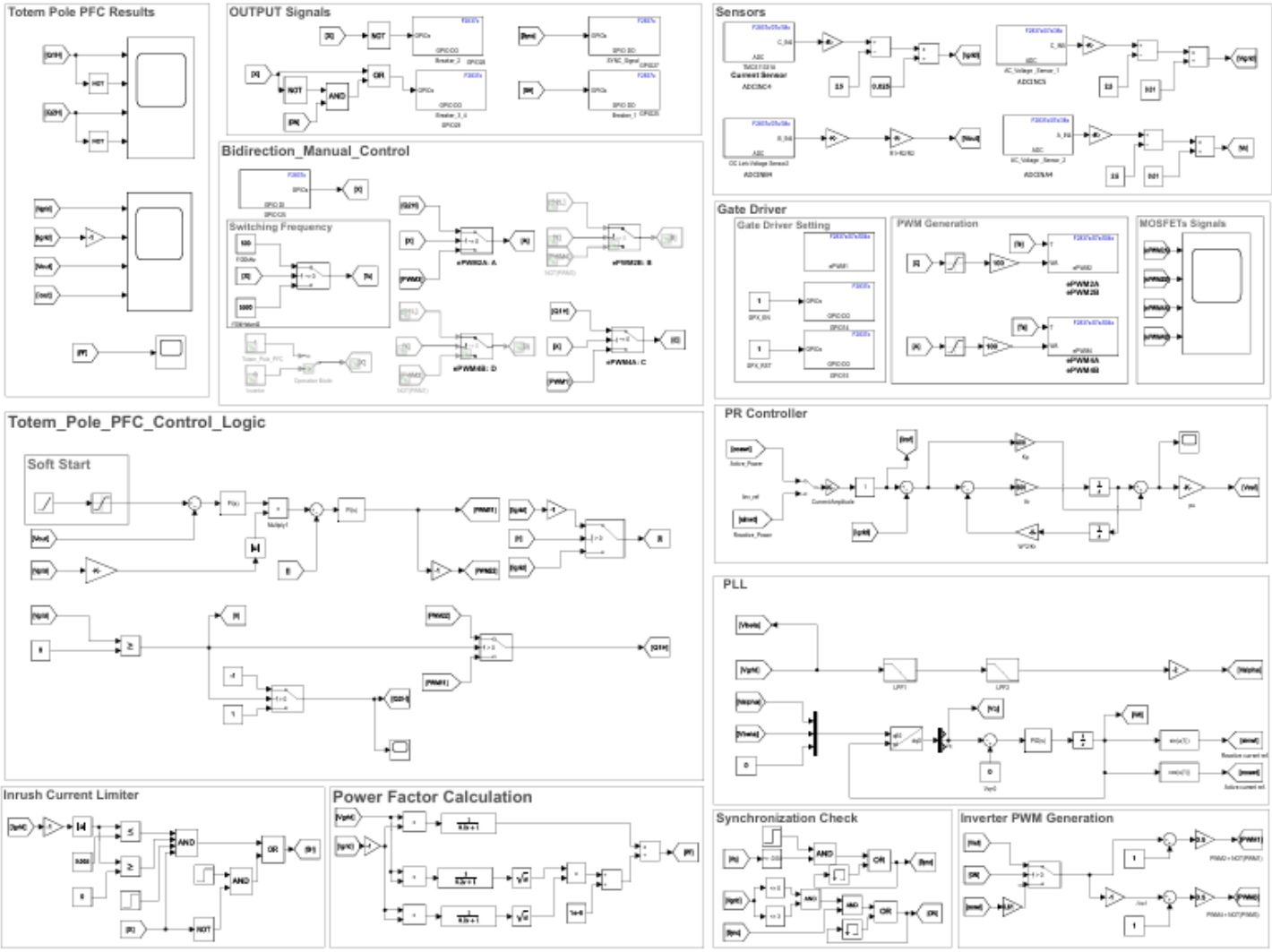


Figure 5.22: Complete System Hardware Target Model Integrating the Totem-Pole PFC and Grid-Connected Inverter Stages

5.10 PCB Design and Implementation

The hardware realization of the bidirectional AC/DC stage was executed using **KiCad**, following a rigorous workflow from schematic capture to physical board fabrication. This phase focused on translating the validated electrical design into a manufacturable PCB, prioritizing layout optimization, isolation boundaries, and high-frequency switching constraints.

5.10.1 KiCad Schematic Development

The complete electrical schematic integrates the Totem-Pole PFC and Grid-connected Inverter stages. Key design considerations included:

- **Gate Driver Isolation:** Implementation of isolated gate drivers to bridge the gap between the low-voltage controller and high-voltage power MOSFETs.

- **Signal Conditioning:** Precision sensing networks for voltage and current feedback, optimized for noise immunity.
- **Protection Circuits:** Integrated over-current and over-voltage protection blocks to safeguard the system during bidirectional operation.

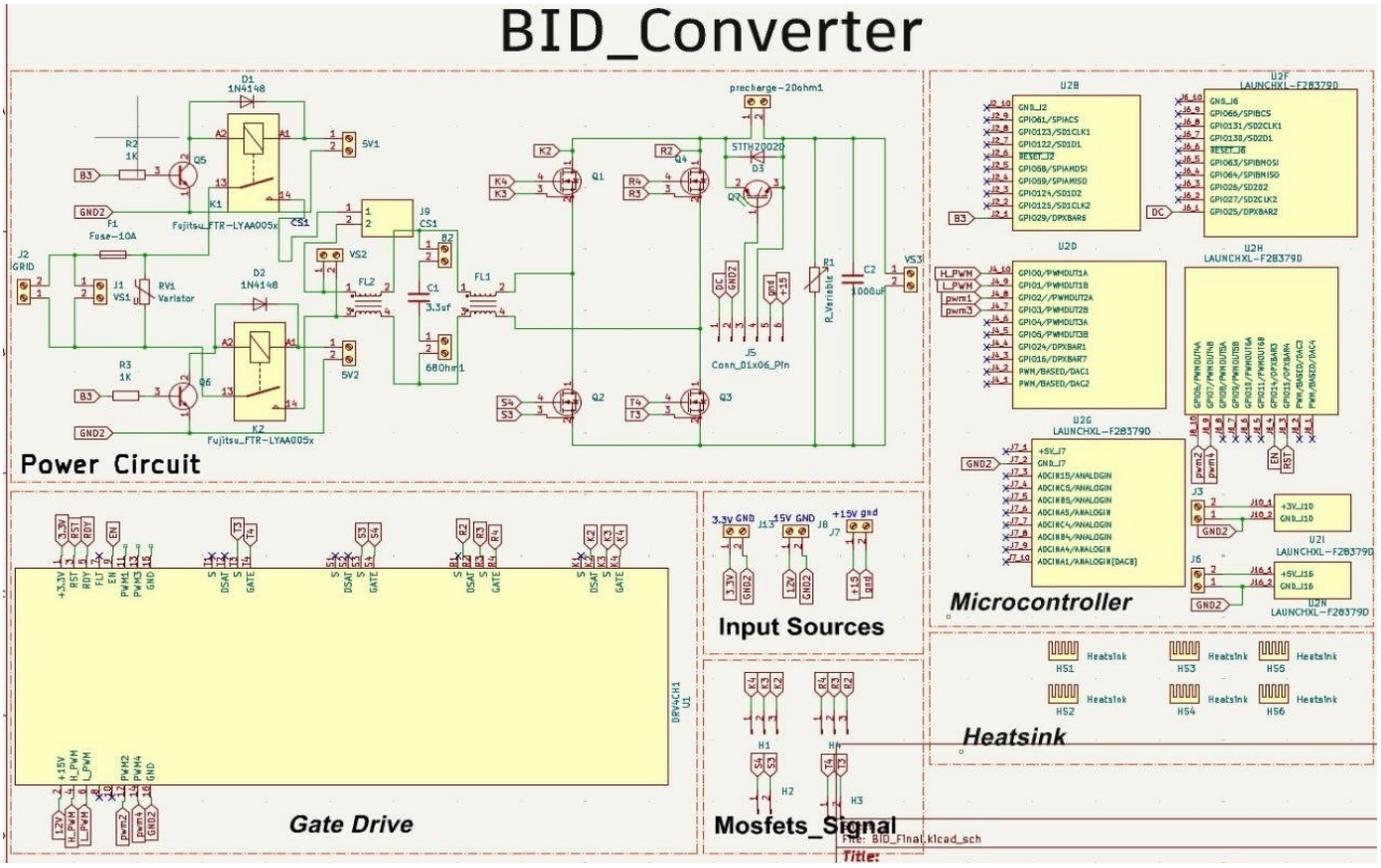


Figure 5.23: Schematic of the Complete AC/DC Converter PCB

5.10.2 PCB Layout Strategy

The PCB layout was engineered to handle high-power density while maintaining signal integrity.

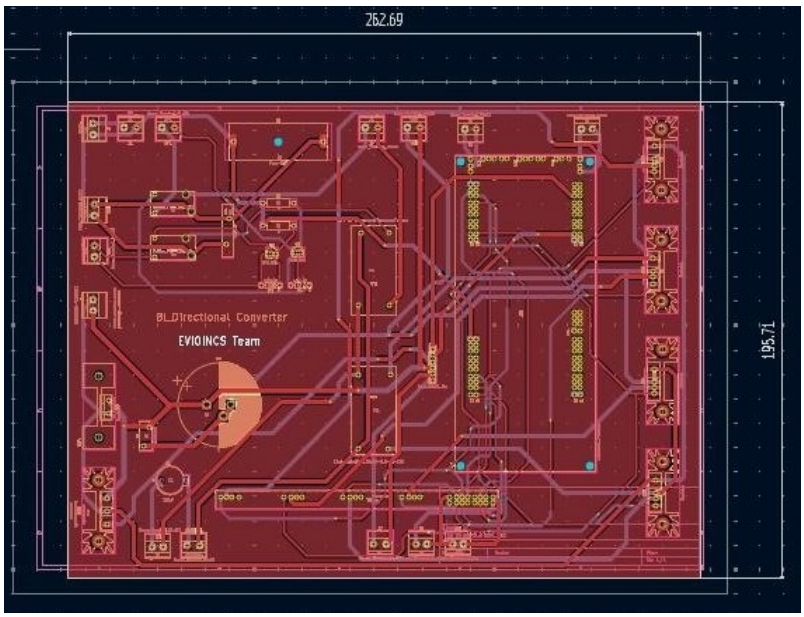


Figure 5.24: PCB Layout of the Complete AC/DC Converter PCB

- **High-Current Paths:** Heavy copper pours and wide traces were used for the main AC and DC-link paths to minimize I^2R losses and thermal buildup.
- **Switching Loops:** To reduce Electromagnetic Interference (EMI), the power switching loops were kept as small as possible to minimize parasitic inductance.
- **Creepage and Clearance:** Clear separation was maintained between the high-power AC/DC sections and the sensitive control circuitry, adhering to industrial safety standards for high-voltage isolation.

5.10.3 3D Visualization and Physical Implementation

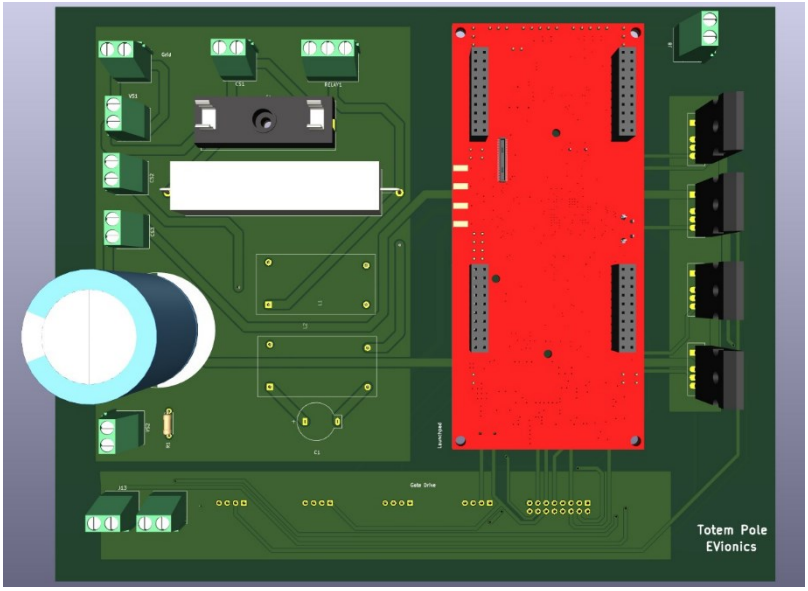


Figure 5.25: 3D Visualization of the Complete AC/DC Converter PCB

A full 3D model was generated to verify mechanical compatibility and ensure that component clearances were sufficient for cooling and enclosure integration.

5.10.4 PCB Fabrication and Assembly

Following the completion of the PCB layout verification, the board was fabricated and assembled according to the finalized design. The manufactured PCB was inspected to verify the accuracy of copper traces, drill holes, solder mask alignment, and component footprints. After assembly, all power devices, gate drivers, sensing circuits, connectors, and control interfaces were soldered and visually inspected prior to electrical testing. The fabricated hardware represents the practical implementation of the bidirectional AC/DC converter and serves as the experimental platform for system validation.

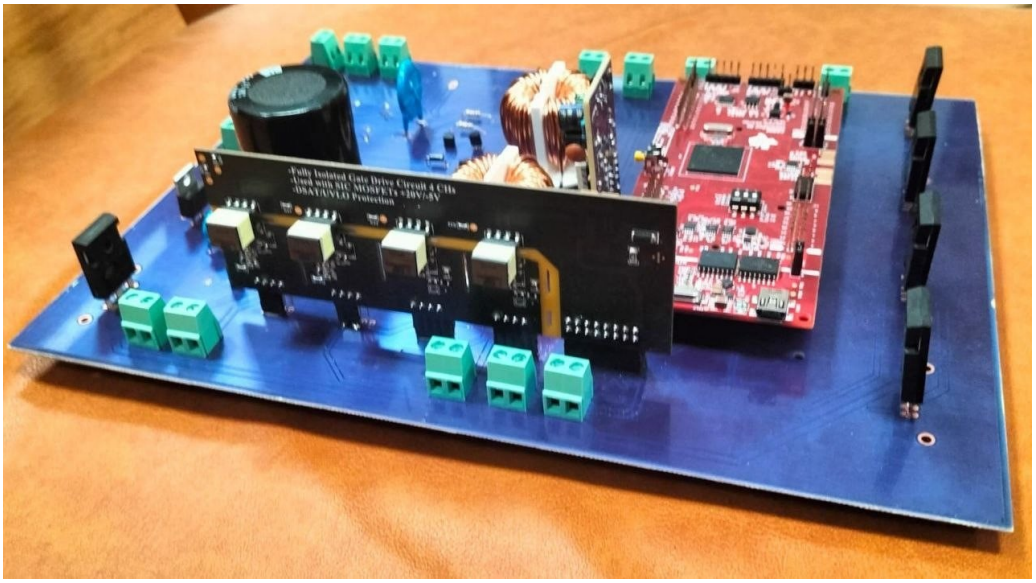


Figure 5.26: *Fabricated and assembled PCB prototype of the bidirectional AC/DC converter used as the experimental platform for hardware validation*.

5.11 Experimental Validation of the Grid-Connected Inverter Controller

5.11.1 Real-Time Controller Validation

To validate the implemented digital controller, a real-time verification test was performed using the TI C2000 LaunchPad. PWM and control signals generated by the embedded controller were acquired and applied as input signals to the developed Simulink model of the grid-connected inverter. The resulting system responses were monitored using laboratory oscilloscopes and compared with the expected simulation behavior.

This validation procedure verifies the consistency between the implemented controller and the simulation model, ensuring that the generated switching signals correctly drive the inverter under practical operating conditions. The measured waveforms demonstrate successful synchronization between the digital controller and the power-stage model, confirming the feasibility of hardware implementation.

PWM Signal Generation



Figure 5.27: Generated PWM signals from the C2000 controller.

Those figures shows the PWM switching signals generated by the C2000 controller. The measured waveforms exhibit stable switching frequency and correct pulse generation, confirming the proper operation of the PWM module and its suitability for driving the inverter switches.

Synchronization and Enable Signals

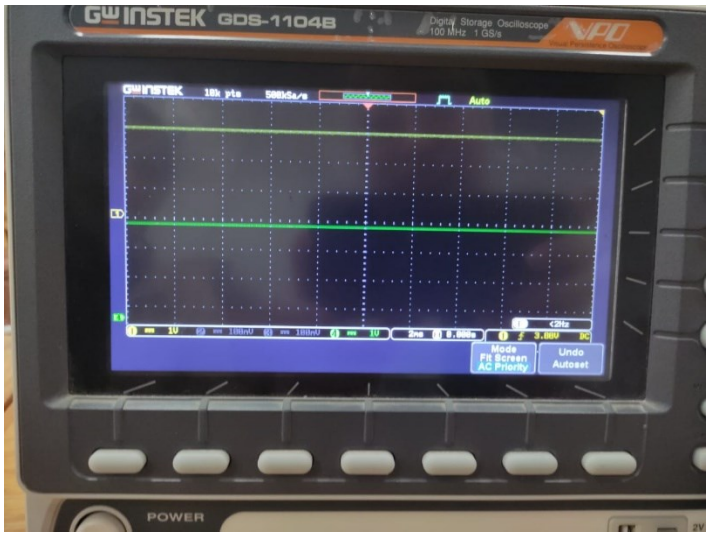


Figure 5.28: Synchronization signal and controller enable signal.

The synchronization signal is generated after successful grid synchronization, while the enable signal activates the inverter breakers only after all operating conditions are satisfied. This sequence prevents unintended switching and ensures safe converter startup.

Grid Voltage and PLL Output (Cosine Signal)

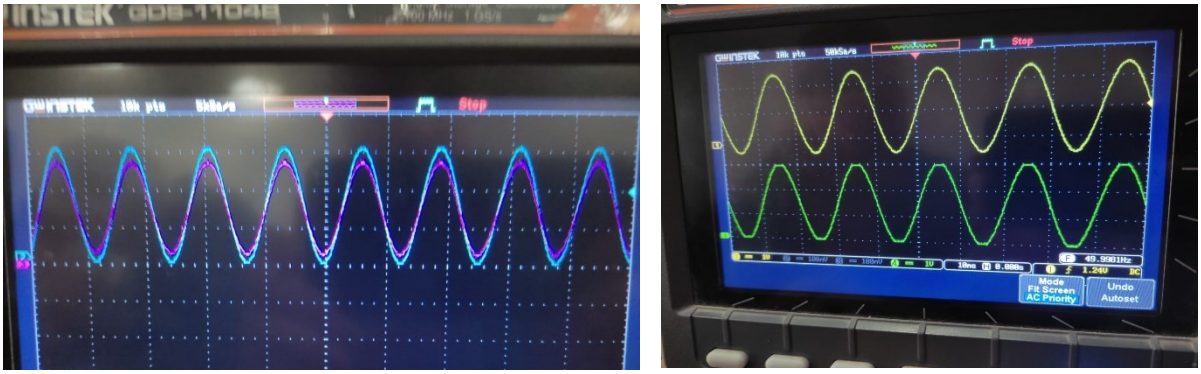


Figure 5.29: Measured grid voltage and synchronized cosine signal.

The measured cosine signal is synchronized with the grid voltage through the implemented PLL algorithm. The results demonstrate accurate phase tracking, providing a reliable reference for the current control loop and grid synchronization.

Grid Voltage and Reference Current

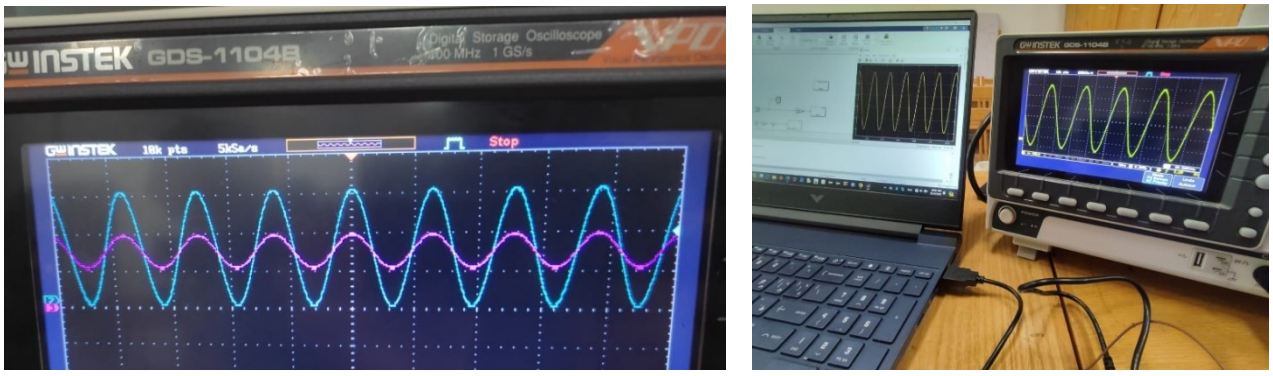


Figure 5.30: Grid voltage and generated current reference

This figure illustrates the relationship between the measured grid voltage and the generated reference current. The controller successfully produces a sinusoidal current reference synchronized with the grid voltage, ensuring proper power exchange during grid-connected operation and supporting high power quality.

5.11.2 Open-Loop Experimental Validation of the Inverter

Prior to grid-connected operation, the inverter was experimentally tested under low-voltage open-loop conditions to verify the functionality of the power stage. A reduced DC input voltage of 15 V was applied to ensure safe operation during the initial hardware validation. The generated AC output voltage was measured after the output filter using a digital oscilloscope.

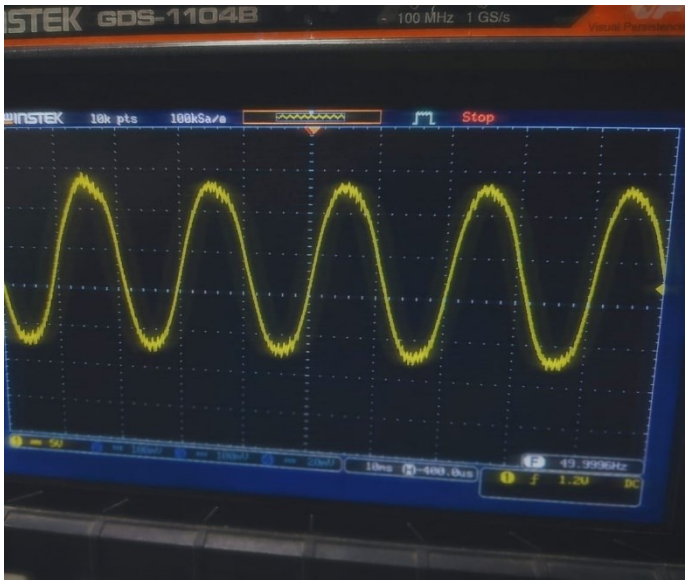


Figure 5.31: Output Voltage After the Filter

This figure shows the inverter output voltage measured after the output filter during open-loop operation. The filter effectively attenuates the high-frequency switching components of the PWM waveform, resulting in a smooth sinusoidal output. The successful generation of the filtered AC waveform confirms the correct operation of the inverter power stage, gate driving circuitry, and output filter under safe low-voltage operating conditions.

5.12 Summary

This chapter presented the design and implementation of the bidirectional AC/DC conversion stage used in the proposed system. The overall converter structure was introduced, highlighting its ability to operate in both rectification and inversion modes to enable bidirectional power flow between the grid and the DC side.

The Totem-Pole PFC rectifier was designed to ensure high efficiency and near-unity power factor during AC-to-DC conversion, while the grid-connected single-phase inverter was developed to regulate DC-to-AC power injection with proper synchronization to the utility grid. The design of key components, including the power stage, control structure, sensing elements, and LCL filter, was presented to ensure stable operation and compliance with grid requirements.

In addition, a unified system-level model integrating both operating modes was developed and validated, demonstrating smooth mode transition and coordinated control of the bidirectional power flow. The simulation results confirmed stable voltage regulation, controlled current shaping, and reliable system performance under different operating conditions.

Chapter 6
Dual Active Bridge (DAB) Converter Design and Implementation

6.1 Introduction

The proposed system is based on an isolated bidirectional DC-DC converter employing the Dual Active Bridge (DAB) topology. The converter is designed to transfer electrical power efficiently between two independent DC buses while providing galvanic isolation through a high-frequency transformer. The bidirectional capability enables energy transfer in both directions, making the converter suitable for applications such as battery energy storage systems, electric vehicle charging, and renewable energy integration.

6.2 System Overview

The conventional DAB topology consists of two actively controlled full-bridge converters connected through a high-frequency transformer with an equivalent series leakage inductance, as shown in Figure 6.1. Each full bridge is connected to a DC bus through a DC-link capacitor, while the transformer provides galvanic isolation and voltage matching between the two sides. The leakage inductance, which may be realized using the transformer leakage inductance or an external series inductor, plays a fundamental role in controlling the transferred power and facilitating soft-switching operation.

During operation, the primary-side full bridge converts the input DC voltage into a high-frequency square-wave voltage applied to the transformer primary winding. Simultaneously, the secondary-side full bridge generates another high-frequency square-wave voltage across the transformer secondary winding. The interaction between these two bridge voltages through the leakage inductance determines the magnitude and direction of the transferred power. Consequently, power can be transferred in either direction without modifying the power stage, making the DAB converter inherently suitable for bidirectional applications.

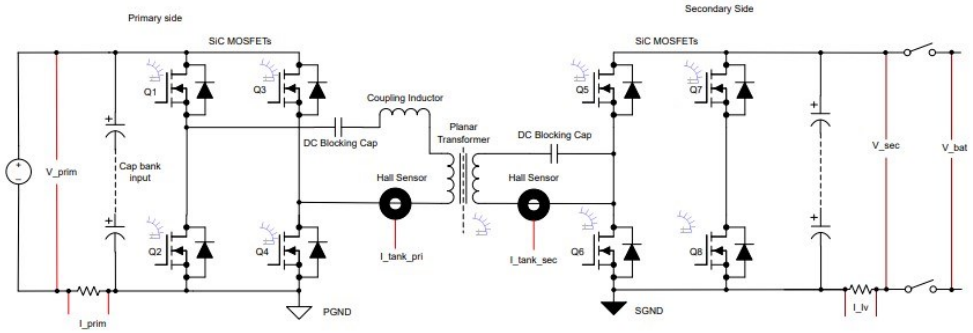


Figure 6.1: DAB converter circuit showing primary and secondary H-bridges with highfrequency isolation transformer and leakage inductance.

6.3 Operating Principle of the Conventional SPS-DAB Converter

6.3.1 Dual Active Bridge Analogy with Power Systems

Power transfer between the two bridges in a dual active bridge is analogous to the power flow between two voltage buses in a power system. Consider two voltage sources connected by a line reactance as shown in Figure 6.2.

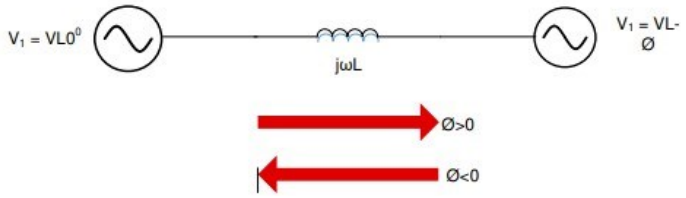


Figure 6.2: Power transfer analogy between two voltage buses connected by line reactance.

Figure 6.2 shows that the voltage source on the right is lagging with respect to the voltage source on the left. Hence, the power transfer takes place from the left towards the right as per Equation 6.1:

$$P = \frac{V_1 V_2}{X} \sin(\delta)$$

Similarly, power transfer happens in a dual-active bridge where two high-frequency square waves are created in the primary and secondary side of the transformer by the switching action of MOSFETs. These high-frequency square waves are phase shifted with respect to each other. Power transfer takes place from the leading bridge to the lagging bridge, and this power flow direction can be easily changed by reversing the phase shift between the two bridges.

6.3.2 Dual-Active Bridge - Switching Sequence

In a single-phase, dual-active bridge, primary and secondary bridges are controlled simultaneously. All switches operate at 50% duty ratio. The diagonal switches turn on and turn off together so that the output of each bridge is a square wave.

Figure 6.3 shows the gating pulses of the switches on the primary and secondary side. The variable ϕ represents the phase shift between the PWM pulses of the primary and secondary side. V_p and V_s represent the voltage on the primary and secondary winding of the transformer. I_L represents the transformer current.

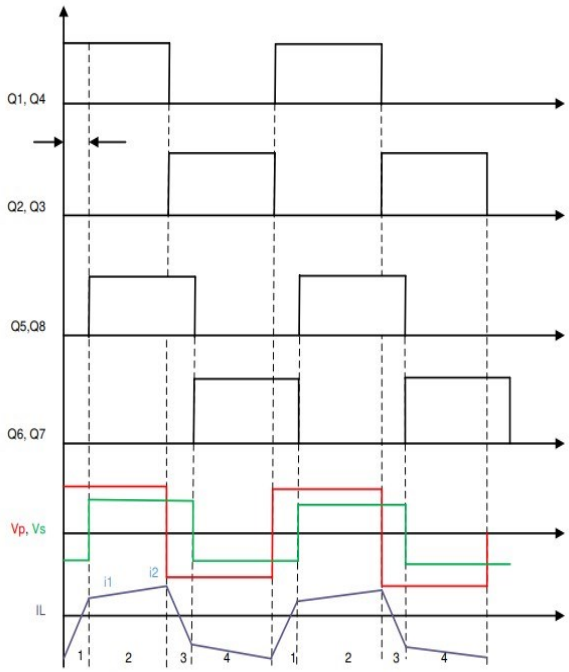


Figure 6.3: Gate signals, transformer primary and secondary voltages, and inductor current.

6.4 Simulation Model Development

To analyze the performance of the proposed converter, a simulation model of the conventional Single Phase Shift (SPS) Dual Active Bridge (DAB) converter was developed in MATLAB/Simulink. The model follows the conventional DAB topology presented in the previous section and was used to verify the converter operation and evaluate the adopted control strategy before hardware implementation.

The simulation model consists of two full-bridge converters connected through a high-frequency transformer and a leakage inductance. Both bridges operate at a fixed switching frequency with a 50% duty cycle, while the transferred power is controlled by adjusting the phase shift between their switching signals. The power circuitry of the simulation model is shown in Figure 6.4.

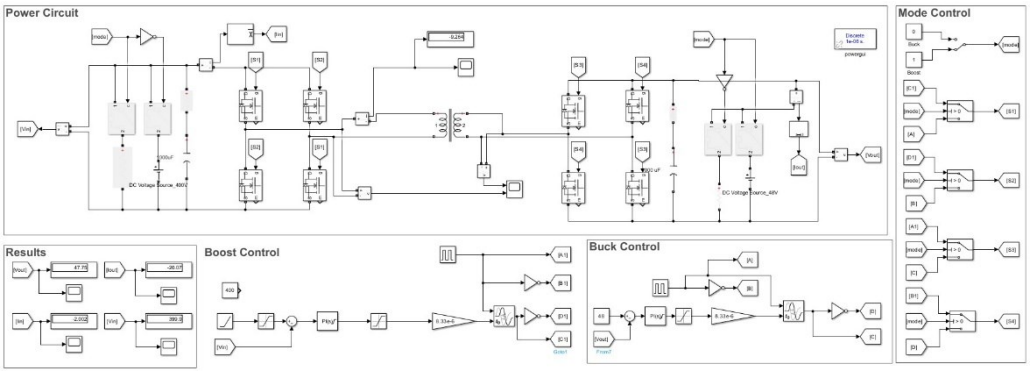


Figure 6.4: MATLAB/Simulink model of the SPS-DAB converter.

6.5 Simulation Results and Discussion

The developed SPS-DAB simulation model was evaluated under both operating modes to verify the bidirectional power transfer capability of the converter. Two operating conditions were investigated: Buck mode, where power flows from the high-voltage side to the low-voltage side, and Boost mode, where power is transferred in the opposite direction. The obtained results confirm the effectiveness of the SPS control strategy in regulating the power flow while maintaining stable converter operation.

6.5.1 Buck Mode Operation

In Buck mode, power is transferred from the high-voltage DC bus to the low-voltage DC bus through the high-frequency transformer. A positive phase shift is applied between the primary and secondary bridges to establish the desired power flow.

6.5.1.1 Switching Signals

Figure 6.5 shows the gate signals applied to the primary and secondary full bridges. Both bridges operate with a fixed duty cycle of 50%, while a phase shift is introduced between the switching patterns to control the transferred power. The obtained waveforms confirm the correct implementation of the SPS modulation technique.

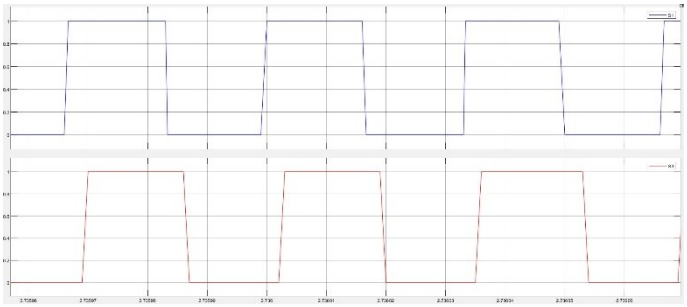


Figure 6.5: Switching scheme for buck direction (G2V mode).

6.5.1.2 Input and Output Voltages

Figure 6.6 presents the input and output voltage waveforms obtained during Buck mode operation. The results demonstrate stable voltage levels at both converter ports, confirming proper power transfer through the high-frequency transformer.6.5.1.3 Input and Output Currents

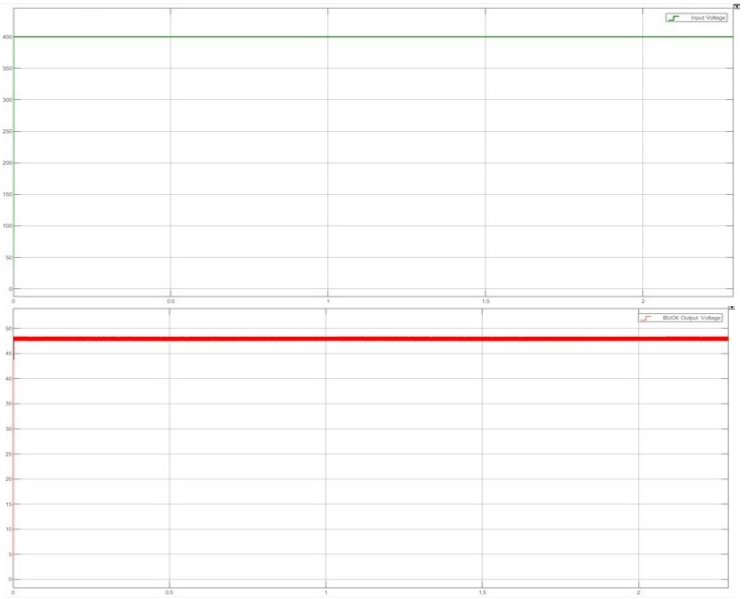


Figure 6.6: Input and output voltage waveforms in Buck mode.

Figure 6.7 illustrates the input and output current waveforms. The current profiles indicate continuous energy transfer from the input side to the output side, verifying the expected operation of the converter under Buck mode.

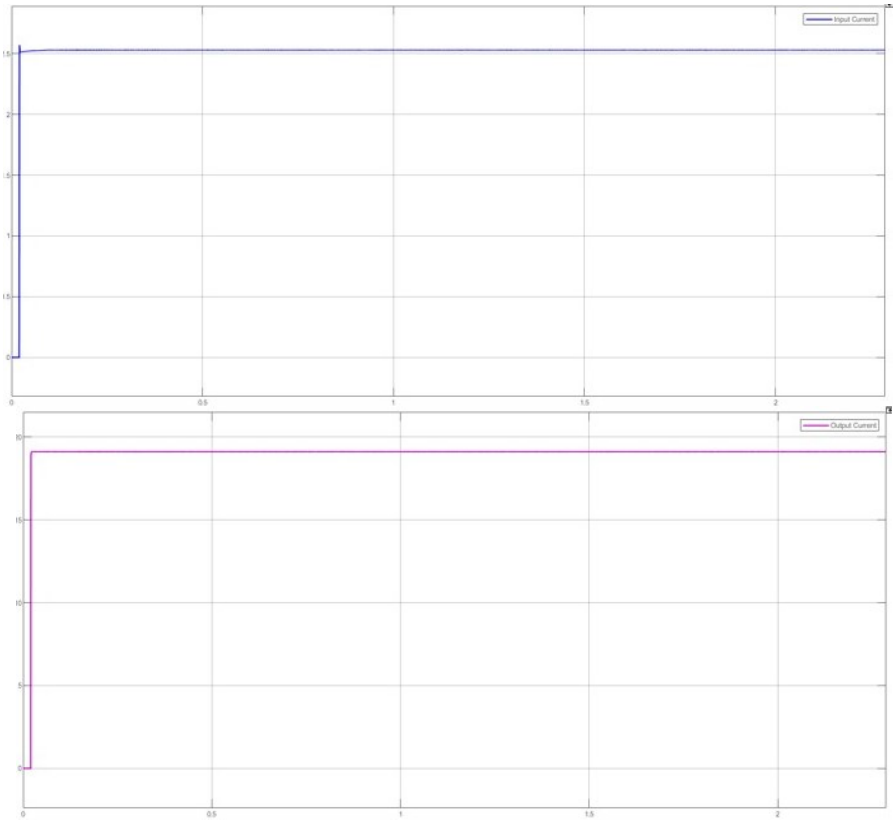


Figure 6.7: Input and output current waveforms in Buck mode.

6.5.2 Boost Mode Operation

To verify the bidirectional capability of the proposed converter, the SPS-DAB model was also simulated in Boost mode, where power flows from the low-voltage side to the high-voltage side.

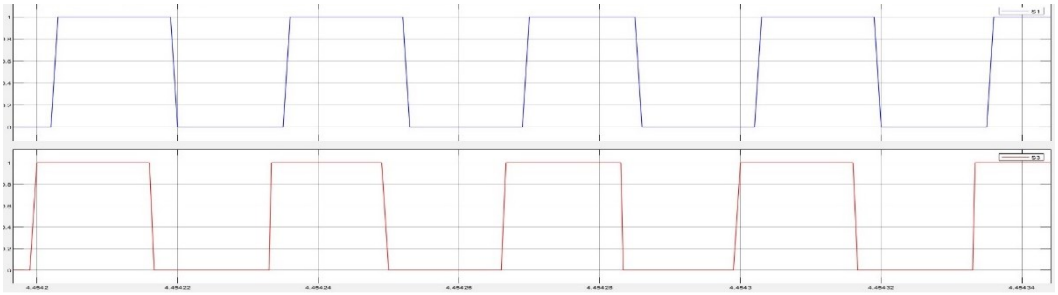


Figure 6.8: Switching scheme for boost direction (V2G mode).

6.5.2.1 Switching Signals

Figure 6.8 shows the switching signals during Boost operation. The direction of power transfer is reversed by changing the phase relationship between the primary and secondary bridge voltages while maintaining the same switching frequency and duty cycle.

6.5.2.2 Input and Output Voltages

The corresponding input and output voltage waveforms are presented in Figure 6.9. The converter successfully maintains the required voltage conversion while operating in the reverse power-flow direction.

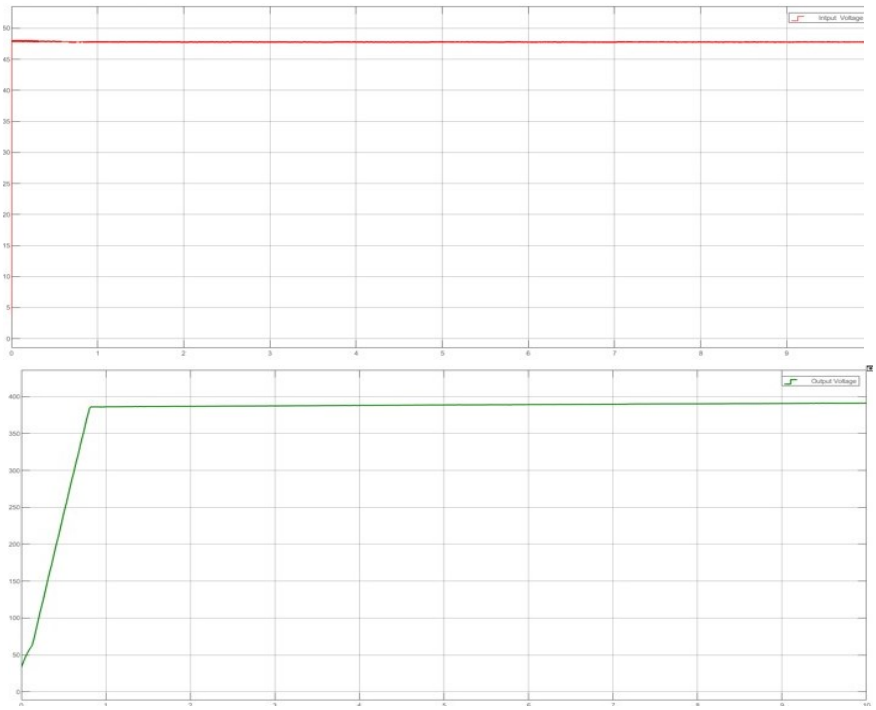


Figure 6.9: Input and output voltage waveforms in Boost mode.

6.5.2.3 Input and Output Currents

Figure 6.10 presents the input and output currents during Boost operation. The current direction is reversed compared with Buck mode, confirming successful bidirectional operation of the converter.



Figure 6.10: Input and output current waveforms in Boost mode

6.5.3 Discussion

The simulation results verify the operation of the conventional SPS-DAB converter in both Buck and Boost modes. In each operating condition, the converter successfully transferred power between the two DC buses while maintaining stable voltage and current waveforms. The SPS control strategy effectively regulated the direction of power flow by adjusting the phase shift between the two active bridges. These results validate the simulation model and provide the theoretical basis for the experimental implementation presented in the following section.

6.6 Hardware Control Model Development

The hardware control model was developed in MATLAB/Simulink using the Embedded Coder support package for Texas Instruments C2000 microcontrollers. Model-Based Design (MBD) was adopted to simplify the development process and enable automatic generation of embedded C code directly from the Simulink model.

The control model generates the PWM signals required to drive the converter switches and provides the flexibility to adjust the converter operating conditions through software. After verification in simulation, the generated code was downloaded to the C2000 LaunchPad, allowing the controller to operate the experimental converter in real time.

The overall hardware control model is shown in Figure 6.11.

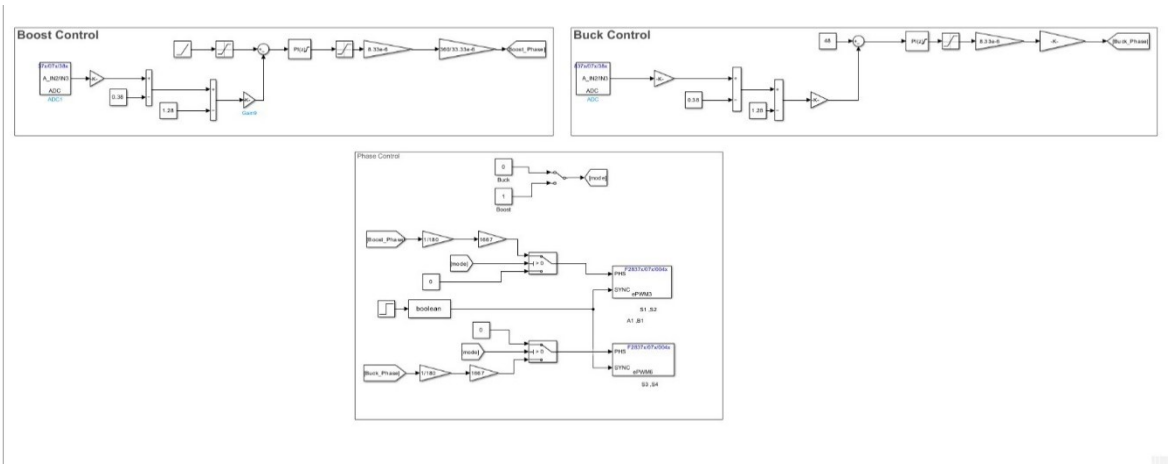


Figure 6.11: Hardware control model developed in MATLAB/Simulink for the C2000 microcontroller.

6.7 PCB Design and Implementation

The hardware implementation of the proposed Dual Active Bridge (DAB) converter was carried out using **KiCad**, following a complete design flow from schematic capture to PCB layout and three-dimensional verification. The PCB was designed to support highfrequency switching while ensuring reliable operation, proper isolation, and efficient thermal performance.

6.7.1 Part 1: DAB HV Side

KiCad Schematic Development

The complete DAB schematic consists of the primary full bridge, secondary full bridge, high-frequency transformer interface, isolated gate drivers, sensing circuits, and auxiliary power supplies. The schematic was developed with emphasis on functionality, safety, and ease of PCB implementation.

The main design considerations included:

- **Isolated Gate Drivers:** Independent isolated gate driver modules were used to drive the primary and secondary SiC MOSFETs safely.
- **Voltage and Current Sensing:** Voltage and current sensing circuits were incorporated to provide accurate feedback for converter control and protection.
- **Protection Circuits:** Input and output protection components, including fuses and filtering capacitors, were added to improve system reliability during bidirectional operation.
- **Auxiliary Power Supply:** Isolated DC/DC converters were used to supply the floating gate drivers with the required gate-drive voltage.

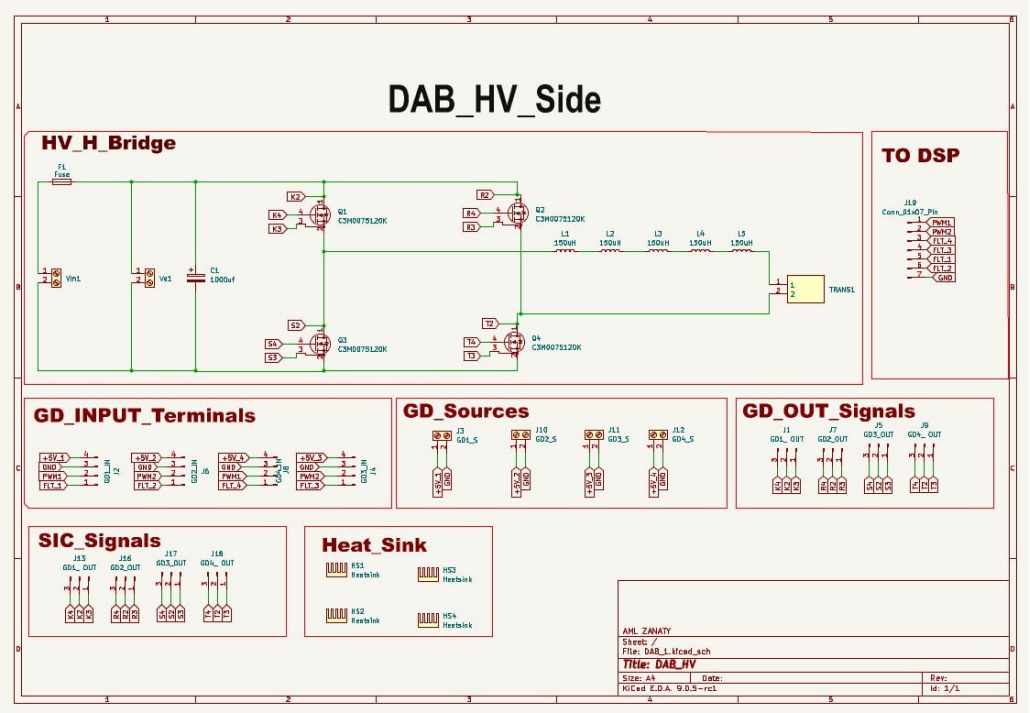


Figure 6.12: Schematic of the HV Side DAB Converter PCB showing primary bridge, gate drivers, and transformer interface.

PCB Layout Strategy

The PCB layout was optimized to achieve high efficiency while minimizing parasitic effects associated with high-frequency switching.

The following layout practices were adopted:

- **High-Current Paths:** Wide copper traces and copper pours were used for the primary and secondary power paths to reduce conduction losses and improve current carrying capability.
- **Switching Loops:** The commutation loops between the MOSFETs, transformer, and DC-link capacitors were kept as short as possible to minimize parasitic inductance and electromagnetic interference (EMI).
- **Gate Driver Placement:** The isolated gate drivers were positioned close to their corresponding MOSFETs to reduce gate-loop inductance and improve switching performance.
- **Ground Separation:** Power ground and control ground were carefully separated to reduce switching noise and improve signal integrity.
- **Creepage and Clearance:** Adequate isolation distances were maintained between the high-voltage power stage and the low-voltage control circuitry to satisfy electrical safety requirements.

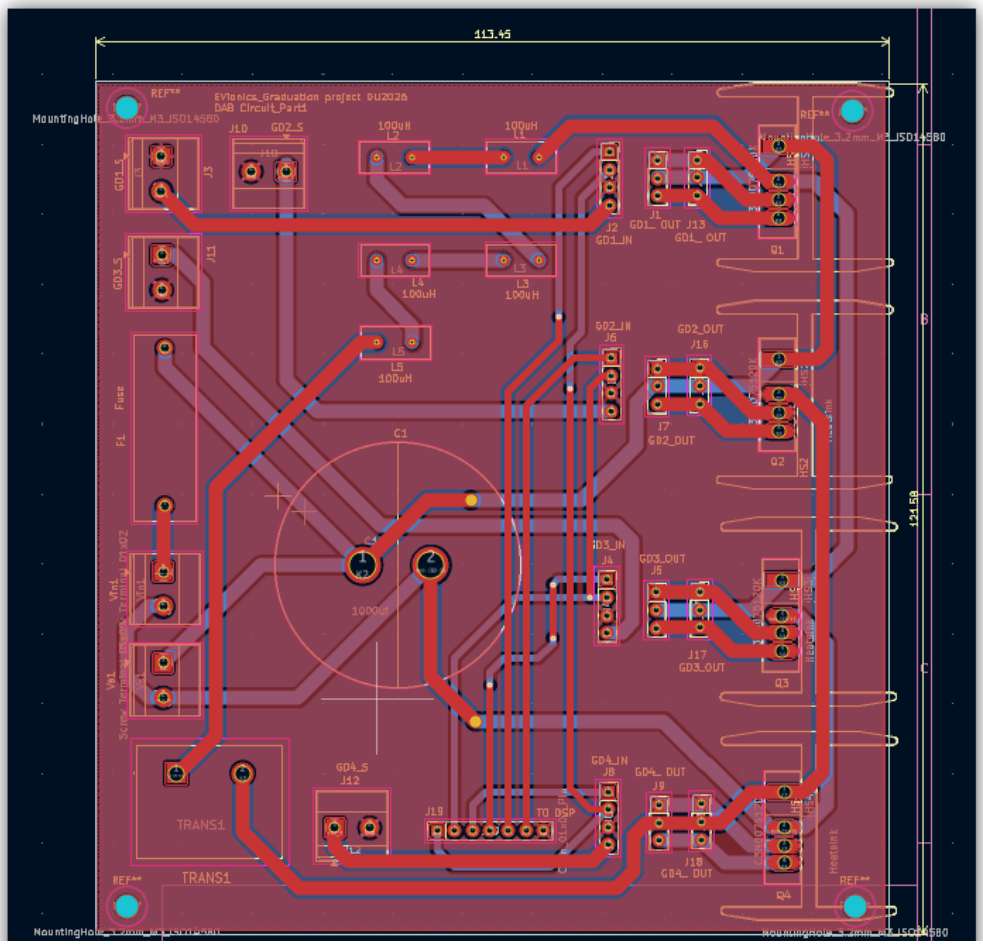


Figure 6.13: PCB Layout of the HV DAB Converter showing component placement and routing.

3D Visualization and Physical Implementation

A complete **3D model** of the PCB was generated using the KiCad 3D Viewer to verify the mechanical arrangement of all components before fabrication.

The 3D visualization was used to:

- Verify component placement.
- Ensure sufficient spacing between power components.
- Check transformer and heat sink clearance.
- Confirm connector orientation.
- Evaluate the overall assembly before manufacturing.

Following the verification process, the PCB design was prepared for fabrication and assembly. The final layout provides a compact structure suitable for high-frequency bidirectional power conversion while maintaining good thermal and electrical performance.

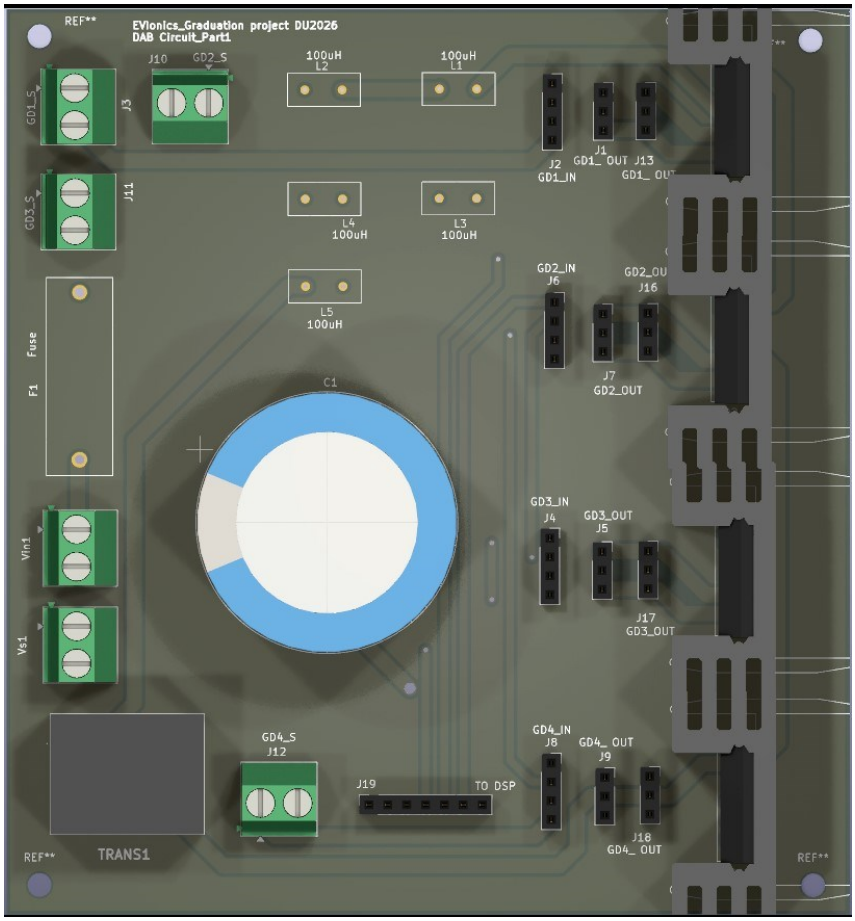


Figure 6.14: 3D View of the HV DAB Converter PCB.

6.7.2 Part 2: DAB LV Side

KiCad Schematic Development

The schematic design focuses on efficient power conversion and noise isolation through the following key blocks:

- **LV H-Bridge Topology**: Features four Silicon MOSFETs paired with a decoupling capacitor and an inline fuse to achieve stable low-voltage switching and reliable overcurrent protection.
- **Gate Drive and Supply Interfacing**: Uses isolated power lines and clear input connections to protect the sensitive control signals from high-power switching noise.
- **Control Signal Routing**: Connects the DSP PWM channels directly to the driver inputs, ensuring exact timing for the phase-shift control.

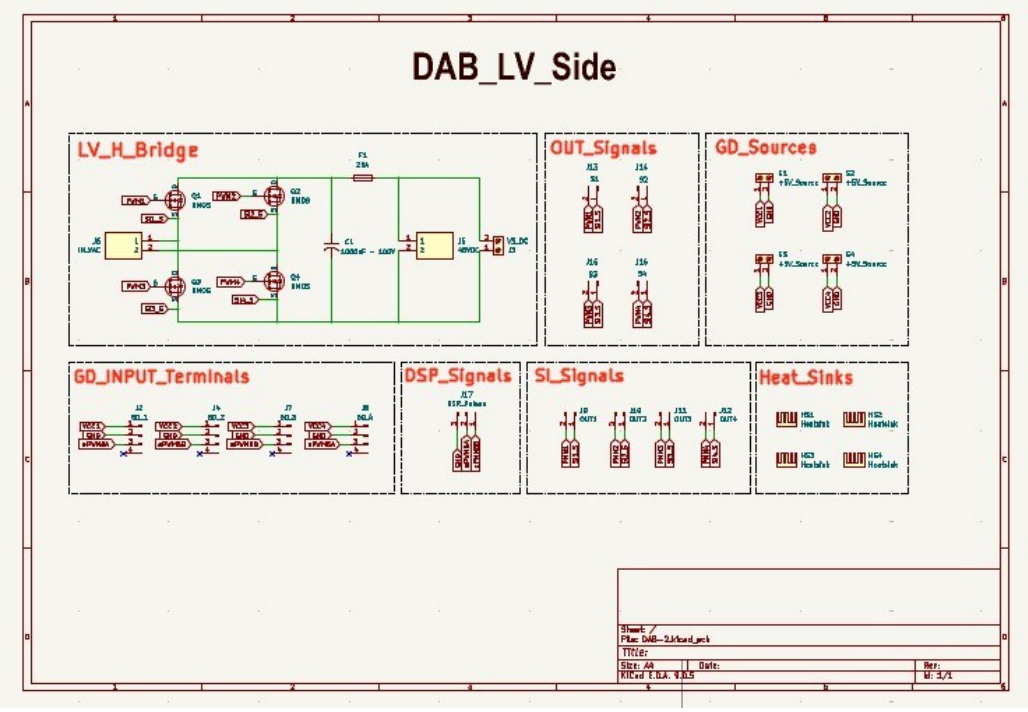


Figure 6.15: Schematic of the DAB LV Side Converter PCB showing secondary bridge and battery interface.

PCB Layout Design

The physical board layout for the DAB low-voltage stage was designed to manage high power levels while keeping control signals clean and accurate:

- **High-Current Traces**: Thick copper tracks and wide fills are used for the main DC bus lines to reduce power losses and avoid heat buildup in the board.
- **Ground Copper Pours**: Large copper zones are placed across the board to create a strong reference ground, which helps shield sensitive tracks from switching noise.
- **Thermal Via Arrays**: Multiple vias are placed near the high-current pads to pass currents and heat easily between the top and bottom layers.

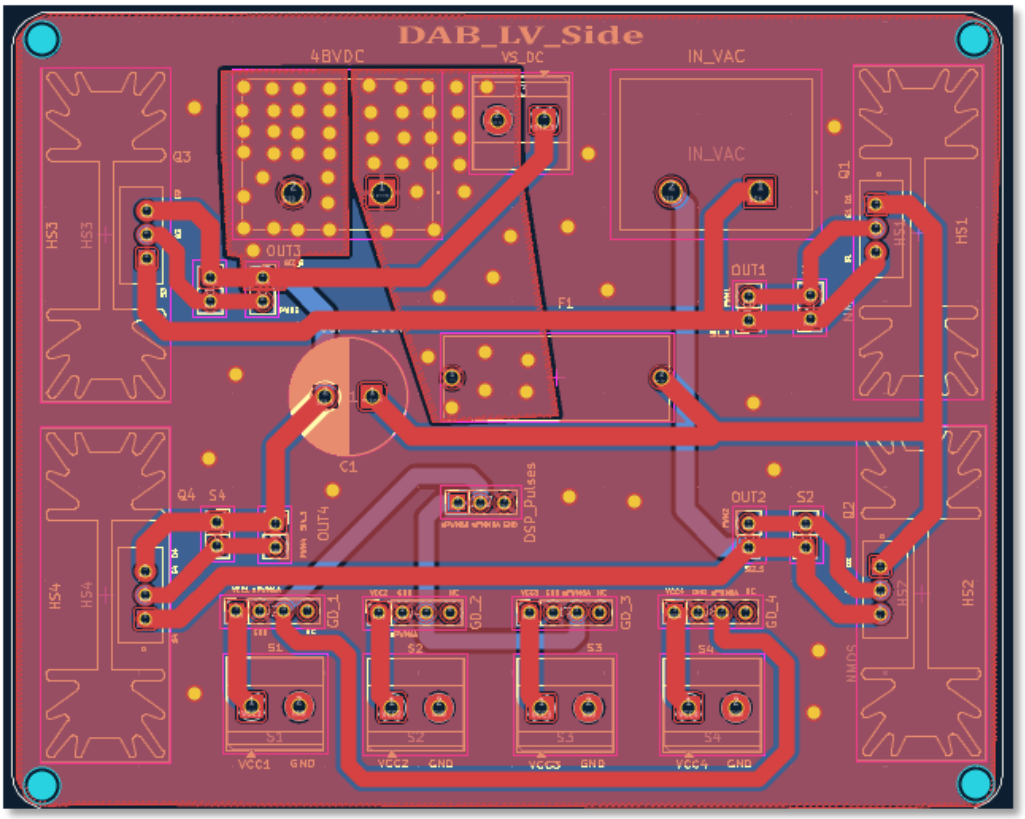


Figure 6.16: PCB Layout of the LV DAB Converter showing high-current traces and ground copper pours.

3D Visualization and Physical Implementation

The 3D model of the board was checked to ensure proper component fitting, enabling a precise physical transfer from the KiCad layout to the final manufactured PCB.

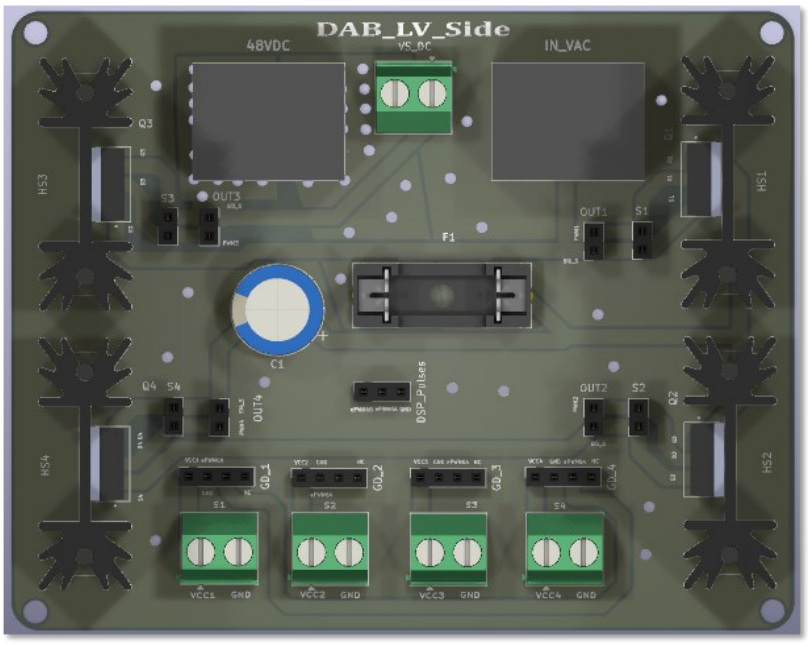


Figure 6.17: 3D Visualization of the LV DAB Converter PCB.

6.8 Experimental Validation of the DAB Converter

Before integrating the controller with the high-power DAB converter, a series of experimental validation tests were performed to verify the correct operation of the developed control model. The objective of this stage was to ensure that the PWM generation, phase-shift implementation, and gate-driver circuits operated as expected before energizing the power converter. Performing these preliminary tests reduced the risk of hardware damage and provided confidence in the developed controller.

The validation process consisted of evaluating the gate-driver circuits and verifying the generated PWM signals using an oscilloscope. The obtained waveforms confirmed the correct operation of the controller and demonstrated its suitability for the subsequent experimental implementation.

6.8.2 Gate Driver Evaluation

To verify the compatibility of the developed controller with different power semiconductor technologies, two isolated gate-driver circuits were evaluated during the experimental stage.

6.8.2.1 Silicon (Si) Gate Driver

The first validation test was performed using the gate-driver circuit intended for conventional silicon MOSFETs. Figure 6.13 presents the measured gate-drive waveform captured using the oscilloscope. The results confirm that the generated PWM signal is accurately reproduced at the driver output with the expected switching frequency and voltage level.

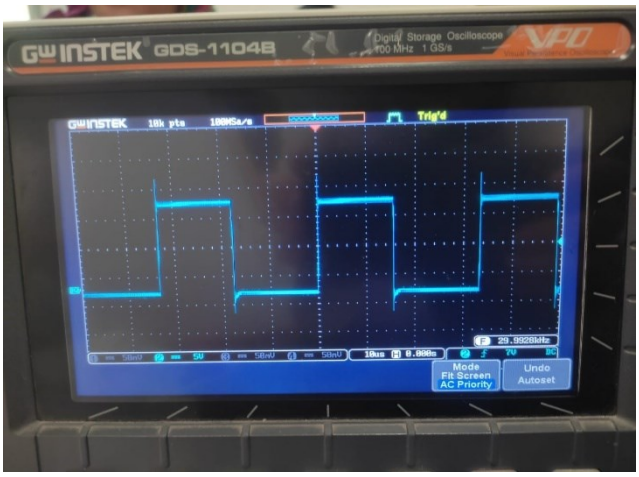


Figure 6.18: Measured gate-drive waveform for silicon MOSFET gate driver.

6.8.2.2 Silicon Carbide (SiC) Gate Driver

A second validation test was performed using the gate-driver circuit designed for silicon carbide (SiC) MOSFETs. The measured output waveform is shown in Figure 6.14. Similar to the previous test, the gate driver successfully reproduced the PWM signal generated by the controller, demonstrating its suitability for high-speed switching applications.

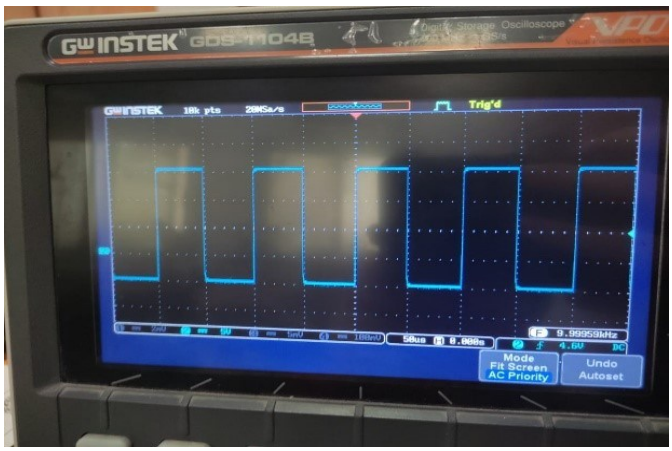


Figure 6.19: Measured gate-drive waveform for SIC MOSFET gate driver

6.8.3 PWM Phase-Shift Verification

Following the successful operation of the individual gate drivers, the developed controller was evaluated for its ability to generate two synchronized PWM signals with a specified phase displacement. This test was performed to verify the implementation of the phaseshift control algorithm required for DAB operation.

Figure 6.15 shows the oscilloscope capture of the generated PWM signals with the desired phase shift. The measured waveforms confirm that the controller accurately produces the programmed phase displacement while maintaining identical switching frequency and duty cycle for both PWM channels.

The successful implementation of the phase-shift function demonstrates the capability of the developed controller to generate the synchronized switching signals required for isolated DC-DC converter applications.

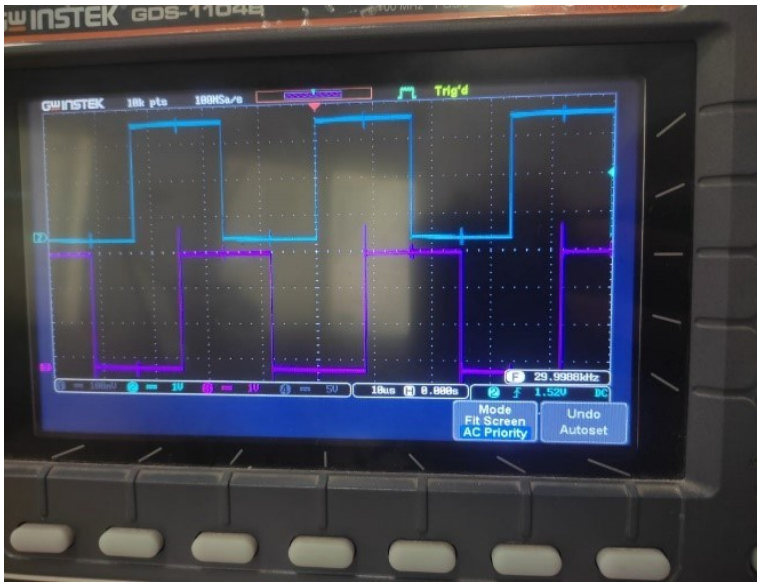


Figure 6.20: Oscilloscope capture showing two PWM signals with programmed phase shift.

To transition from simulation to practical implementation, a hardware control model was developed using Model-Based Design and deployed on a TI C2000 LaunchPad. The experimental validation of the controller confirmed correct PWM generation, accurate phase-shift implementation, and proper operation of both Si and SiC gate-driver circuits. These results established a reliable foundation for the subsequent integration of the controller with the high-power DAB converter stage.

6.8.4 Experimental Validation Setup

The validation experiments were conducted using the Texas Instruments C2000 LaunchPad programmed directly from the developed MATLAB/Simulink model. The generated PWM signals were connected to the gate-driver circuits and monitored using a digital oscilloscope. This setup enabled the switching signals to be evaluated independently of the high-power converter. Figure 6.12 shows the experimental validation setup.

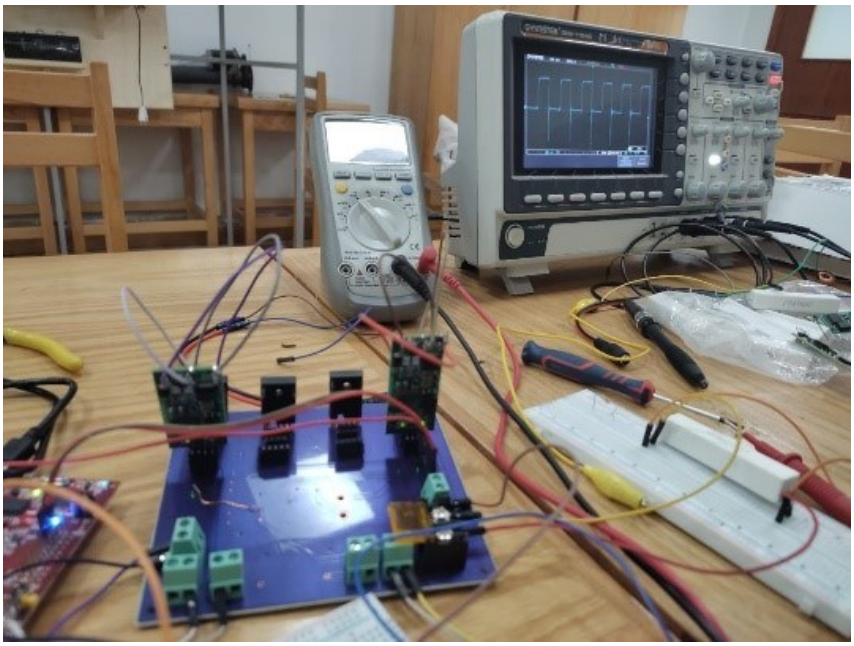


Figure 6.21: Experimental validation setup for the DAB controller

6.9 Hardware Evolution and Experimental Validation

6.9.1 Initial Prototype and Challenges

The initial Dual Active Bridge (DAB) prototype successfully demonstrated the basic operating principle during low-voltage testing. However, waveform distortion and switching irregularities were observed under increased loading conditions, indicating the need for improvements in both the power stage and gate driving circuitry. These observations motivated the development of a second-generation bidirectional DC-DC converter with enhanced hardware reliability and switching performance.

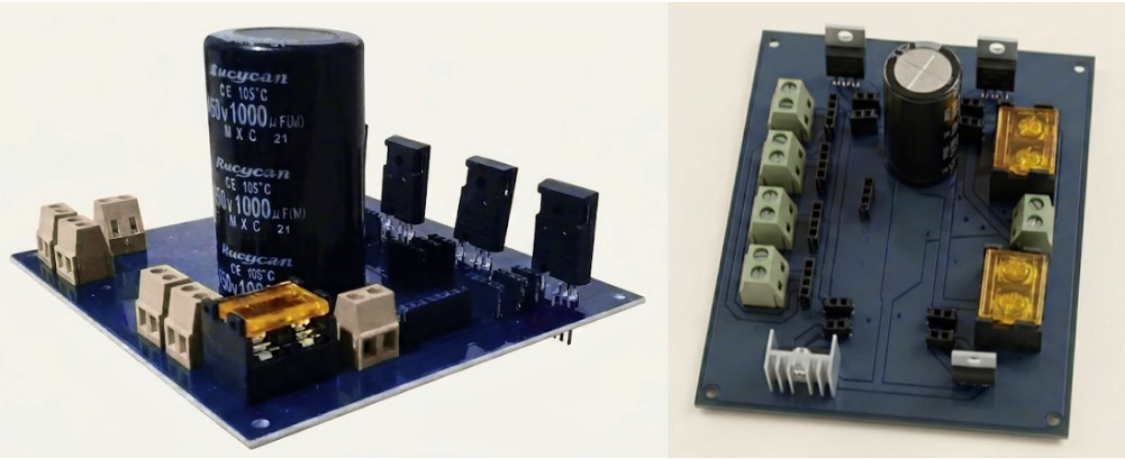


Figure 6.22. Initial prototype of the Dual Active Bridge (DAB) converter

6.9.2 Hardware Upgrades and Component Selection

To improve the converter performance and operational reliability, several hardware modifications and higher-performance components were incorporated into the redesigned prototype.

- **NCE100TD120VTP Power Modules:** Integrated to provide improved switching capability and higher voltage tolerance, contributing to enhanced converter stability and reliable operation.
- **TLP350H Gate Driver Modules:** Used to provide isolated, high-speed gate drive signals for the IGBTs. Each gate driver is supplied from a dedicated isolated power supply, where an oscillator drives a pulse transformer to transfer energy across an isolated barrier. This arrangement provides galvanic isolation between the control and power circuits while supplying stable and noise-resistant gate drive voltages for reliable switching operation.
- **KNSCHA X2-305K (MPX/MKP) Safety Capacitors:** Incorporated for EMI filtering and suppression of high-frequency noise. Their 275 VAC/310 VAC voltage rating and operating temperature capability up to 110°C improve system robustness and protection against voltage transients.
- **Bidirectional Control Switch:** A dedicated control switch was implemented to allow manual selection of the power flow direction, enabling seamless operation in both Grid-to-Vehicle (G2V) and Vehicle-to-Grid (V2G) modes.

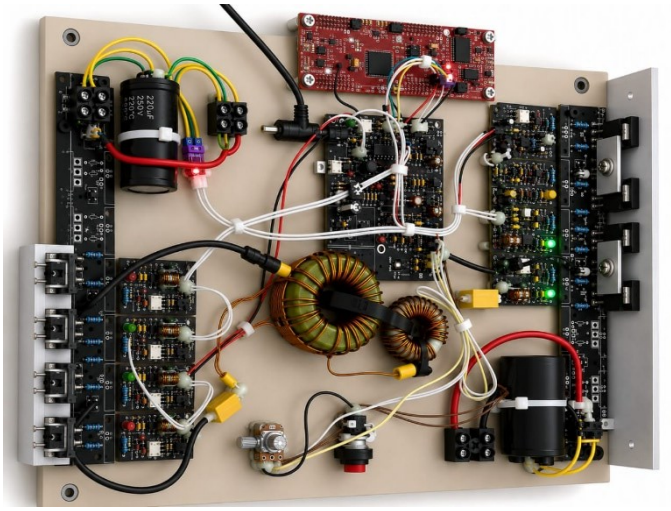


Figure 6.23. :Upgraded hardware prototype of the bidirectional DC-DC converter after hardware redesign and component optimization.

6.9.3 Experimental Outcomes

Experimental evaluation of the upgraded prototype demonstrated a noticeable improvement in overall converter performance compared with the initial design. Stable switching waveforms were obtained during laboratory testing, confirming the effectiveness of the hardware modifications.

The implemented bidirectional control switch enabled smooth transitions between G2V and V2G operating modes while maintaining stable converter operation. Furthermore, the combination of isolated gate driver power supplies, improved power devices, and optimized hardware design resulted in reliable converter performance throughout the experimental tests, providing a robust platform for validating the proposed bidirectional DC-DC converter.

Figure 5.xx presents the experimentally measured input and output voltages for both operating modes. During G2V operation, the converter successfully stepped down the input voltage to the required output level. Conversely, during V2G operation, the converter operated in the reverse direction, demonstrating successful voltage conversion and confirming the bidirectional capability of the proposed converter.

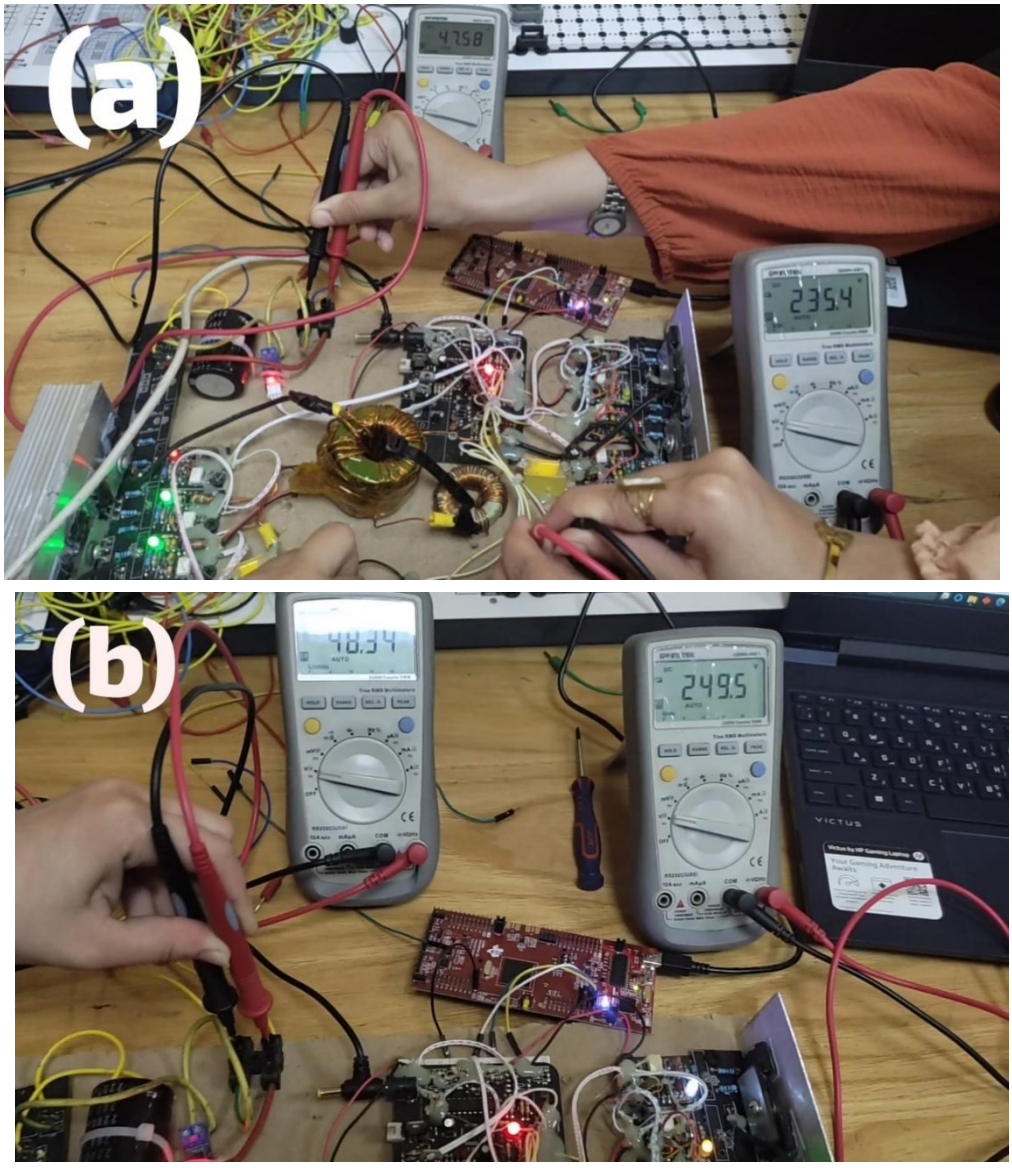


Figure 6.24. Measured input and output voltages during bidirectional operation: (a) Buck (G2V) mode and (b) Boost (V2G) mode.

6.10 Summary

This chapter presented the design, simulation, and experimental validation of the Dual Active Bridge (DAB) converter for bidirectional DC-DC conversion. The conventional Single Phase Shift (SPS) modulation technique was adopted to control power flow between two active full bridges while maintaining galvanic isolation through a high-frequency transformer.

A detailed MATLAB/Simulink simulation model was developed and evaluated under both Buck and Boost operating modes. The simulation results confirmed the bidirectional power transfer capability of the converter, with stable voltage and current regulation in both power flow directions.

Chapter 7
Integrated Bidirectional On-Board Charger Architecture

7.1 Introduction

The bidirectional on-board charger (OBC) consists of two major power conversion stages: the grid-side bidirectional AC/DC converter and the isolated bidirectional DC/DC converter. Individually, both converters were successfully designed, modeled, and verified under different operating conditions. However, practical implementation of a bidirectional charger requires these converters to operate as a coordinated system capable of managing power transfer between the utility grid and the electric vehicle battery in both charging (Grid-to-Vehicle, G2V) and discharging (Vehicle-to-Grid, V2G) modes.

The integration process combines both converters through a common DC-link while coordinating their control algorithms using a supervisory controller. This approach ensures stable operation, proper startup sequencing, synchronized power flow, and safe transition between different operating modes.

This chapter presents the integration methodology of the complete bidirectional OBC model, including system architecture, control coordination, operating sequence, power flow management, and simulation model.

7.2 Integrated System Architecture

The integrated charger consists of two interconnected power conversion stages.

The first stage is the bidirectional AC/DC converter, which interfaces the charger with the single-phase utility grid. It is composed of a Totem Pole Power Factor Correction (PFC) converter and a single-phase grid-tied inverter. This stage is responsible for regulating the DC-link voltage, maintaining a near-unity power factor, and enabling bidirectional energy exchange with the grid.

The second stage is the isolated bidirectional DC/DC converter based on the Dual Active Bridge (DAB) topology. This converter provides galvanic isolation between the high-voltage DC-link and the low-voltage battery while controlling battery charging and discharging currents.

Both stages are connected through a common DC-link capacitor, allowing energy to flow bidirectionally according to the selected operating mode.

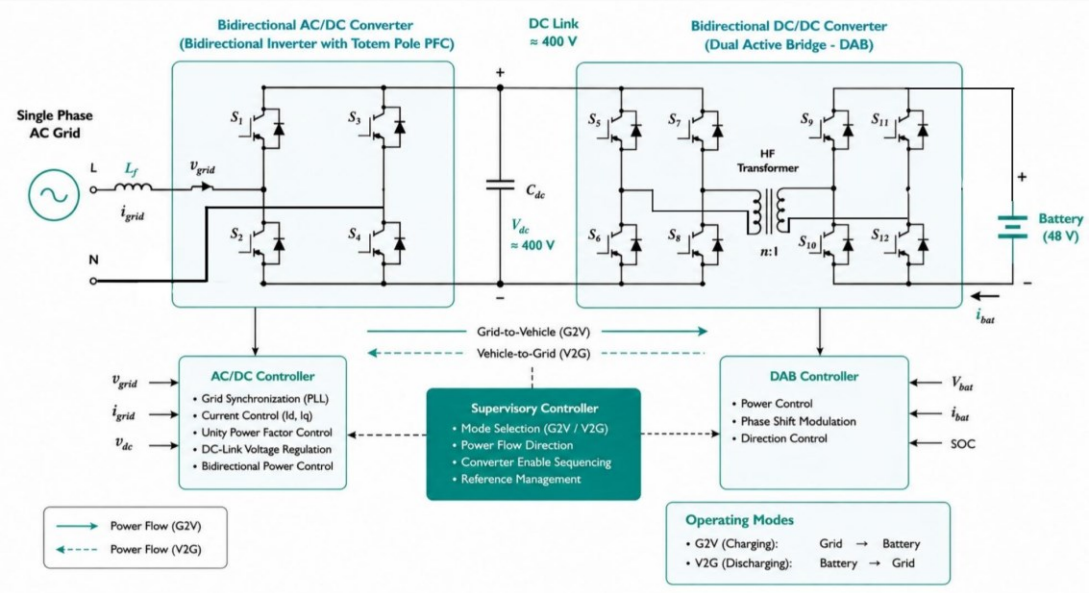


Figure 7.1 illustrates the complete integrated bidirectional OBC architecture.

7.3 DC-Link Integration

The DC-link represents the electrical interface between the AC/DC and DC/DC converters.

During Grid-to-Vehicle operation, the AC/DC converter converts the utility AC voltage into a regulated DC voltage of approximately 400 V, which is maintained across the DC-link capacitor. The DAB converter subsequently transfers this energy to the battery while regulating the charging current according to the battery charging profile.

During Vehicle-to-Grid operation, the power flow is reversed. The DAB converter extracts energy from the battery and boosts it to the DC-link voltage, after which the grid-side inverter injects synchronized current into the utility grid.

Maintaining a stable DC-link voltage is essential for ensuring reliable operation of both converters. Therefore, the AC/DC converter continuously regulates the DC-link voltage regardless of the operating direction.

The power balance within the integrated system can be expressed as

$$P_{AC/DC} = P_{DAB} + P_{loss}$$

where

- $(P_{AC/DC})$ is the power transferred by the grid-side converter,
- $(P_{DAB})$is the power processed by the isolated DC/DC converter,
- (P_{loss})represents the total converter losses.

7.4 Control Integration

Although each converter possesses an independent local controller, the integrated charger requires a higher-level supervisory controller to coordinate their operation.

The supervisory controller performs the following tasks:

- Operating mode selection.
- Converter enabling and disabling.
- Grid synchronization verification.
- DC-link voltage monitoring.
- Battery condition monitoring.
- Reference signal generation.
- Startup sequencing.

Rather than allowing all converters to operate simultaneously, the supervisory controller activates each stage according to predefined operating conditions, ensuring smooth system startup and preventing unstable operating conditions.

7.5 Operating Modes

The integrated charger supports two fundamental operating modes.

7.5.1 Grid-to-Vehicle (G2V)

In this mode, electrical energy flows from the utility grid to the battery.

The operating sequence consists of the following steps:

1. Read battery State of Charge (SOC).
2. Receive the G2V request.
3. The Totem Pole PFC converter is enabled.
4. The DC-link voltage is regulated to 400 V.
5. The DAB converter is activated as buck converter.
6. Battery charging begins under Constant Voltage (CV) mode.
7. Charging terminates once the battery reaches its fully charged condition.

This sequence minimizes current transients and ensures safe battery charging.

7.5.2 Vehicle-to-Grid (V2G)

In Vehicle-to-Grid operation, stored battery energy is transferred back to the electrical grid.

The supervisory controller performs the following sequence:

1. Verify sufficient battery State of Charge (SOC).
2. Receive the V2G request.
3. Enable the DAB converter as boost converter.
4. Regulate the DC-link voltage.
5. Synchronize with the utility grid.
6. Enable the inverter.
7. Inject synchronized sinusoidal current into the utility grid.

The inverter continuously regulates the injected current while maintaining synchronization with the utility voltage.

7.6 Control Signal Coordination

Proper synchronization between the power converters is essential for stable integrated operation.

Several control signals are exchanged among the converter controllers, including:

- PWM signals.
- Enable signals.
- Grid synchronization signals.
- Current reference signals.
- DC-link voltage feedback.
- Battery voltage feedback.
- Battery current feedback.
- Power flow direction commands.

The enable signals ensure that the DAB converter is activated only after the DC-link voltage has reached its reference value. Likewise, the inverter is permitted to operate only after successful grid synchronization has been achieved.

This sequential activation significantly improves startup stability and reduces switching stress on semiconductor devices.

The integrated Simulink implementation includes dedicated enable logic for the bidirectional AC/DC converter, inverter, and DAB operating mode controller.

7.7 Power Flow Management

Power flow direction is determined by the selected operating mode.

Charging Mode

Grid → Totem Pole PFC → DC-Link → Dual Active Bridge → Battery

Vehicle-to-Grid Mode

Battery → Dual Active Bridge → DC-Link → Grid-Tied Inverter → Utility Grid

The supervisory controller continuously monitors the desired operating mode and updates the converter references accordingly.

7.8 Integrated Simulink Model

The complete charger was implemented using MATLAB/Simulink by integrating the previously developed AC/DC and DAB converter models into a single simulation environment.

The integrated model includes:

- Bidirectional AC/DC converter.
- Totem Pole PFC controller.
- Grid-tied inverter controller.
- DAB converter.
- PWM generation blocks.
- Power flow selection logic.
- Enable signal generation.

- Battery model.
- DC-link capacitor.
- Grid synchronization subsystem.
- Power factor calculation.
- Voltage and current measurement blocks.

The model enables seamless switching between charging and V2G modes while maintaining stable converter operation. The top-level model also includes dedicated blocks for power flow direction, G2V/V2G requests, DC-link sensing, inverter control, and battery interfacing

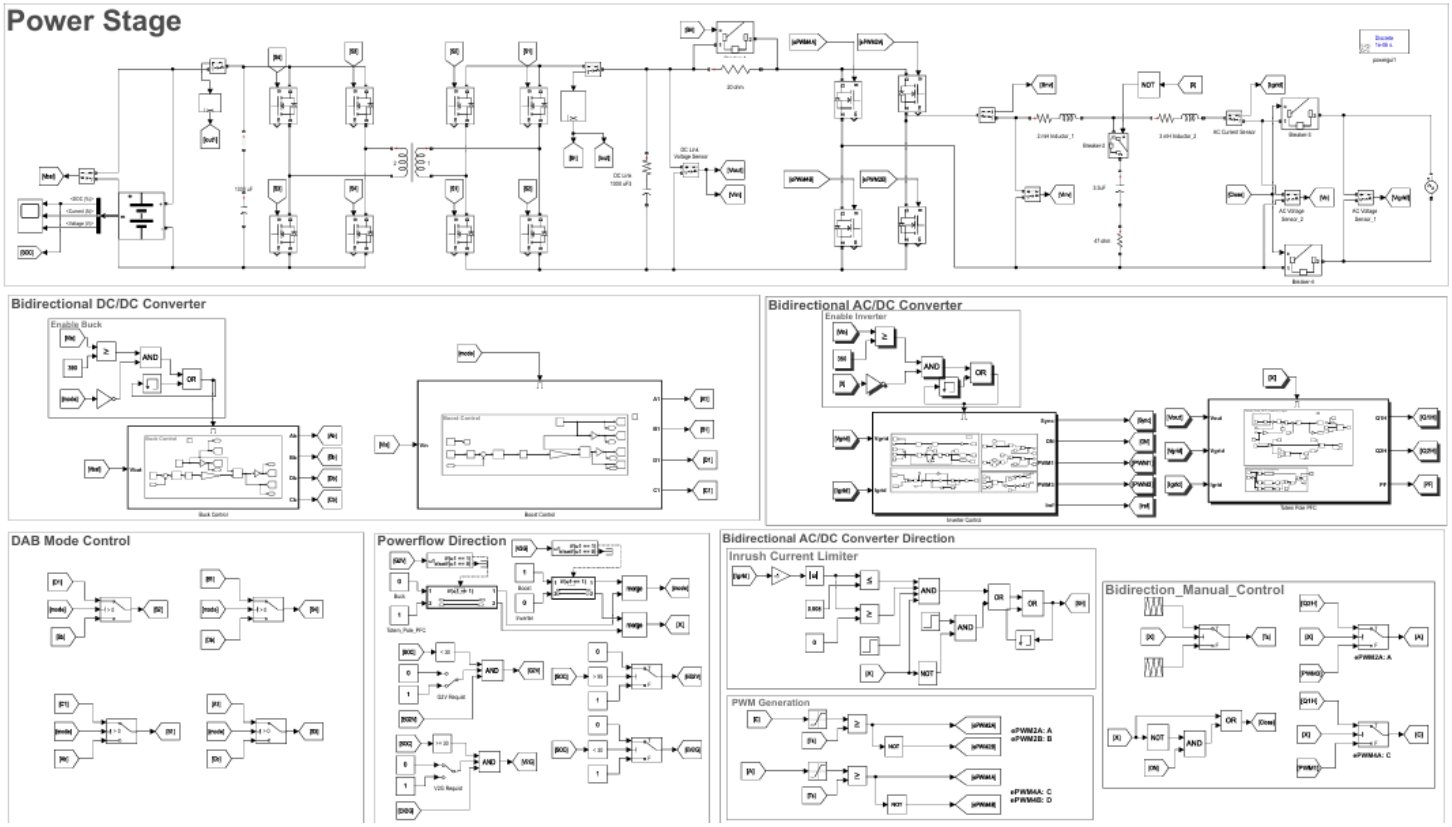


Figure 7.2 illustrates the complete integrated Simulink Model.

7.9 Simulation Results

Simulation results demonstrate successful integration of both converter stages.

The obtained waveforms verify:

- Stable DC-link voltage regulation.
- Successful grid synchronization.
- Proper PWM generation.
- Correct enable signal sequencing.
- Smooth converter startup.
- Stable battery charging current.
- Controlled battery charging voltage.
- Successful bidirectional power transfer.

- Smooth transition between operating modes.

Furthermore, no instability was observed during converter activation or operating mode transitions, confirming the effectiveness of the supervisory control strategy.

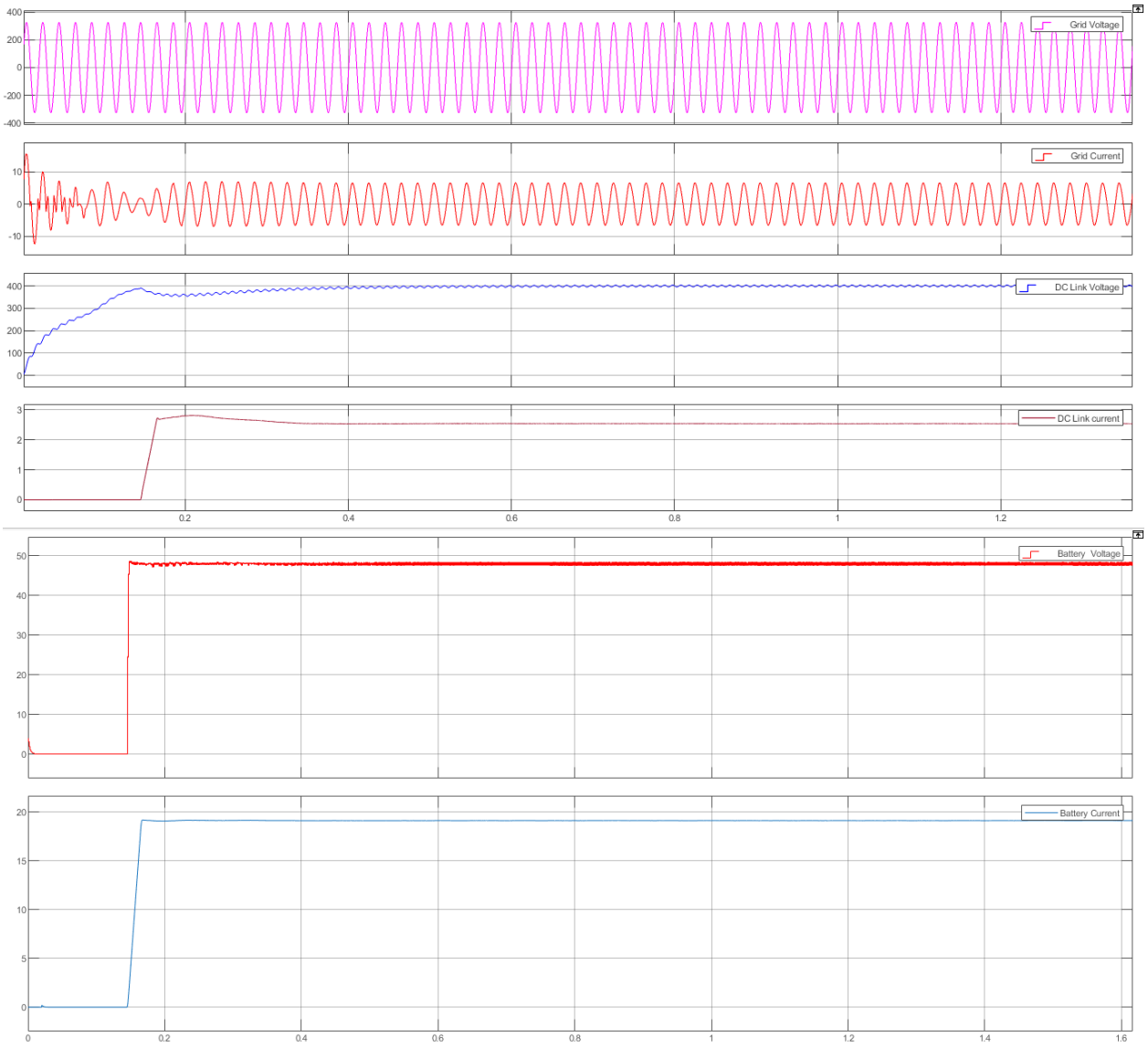


Figure 7.3. Simulation results of the integrated bidirectional OBC during Grid-to-Vehicle (G2V) operation

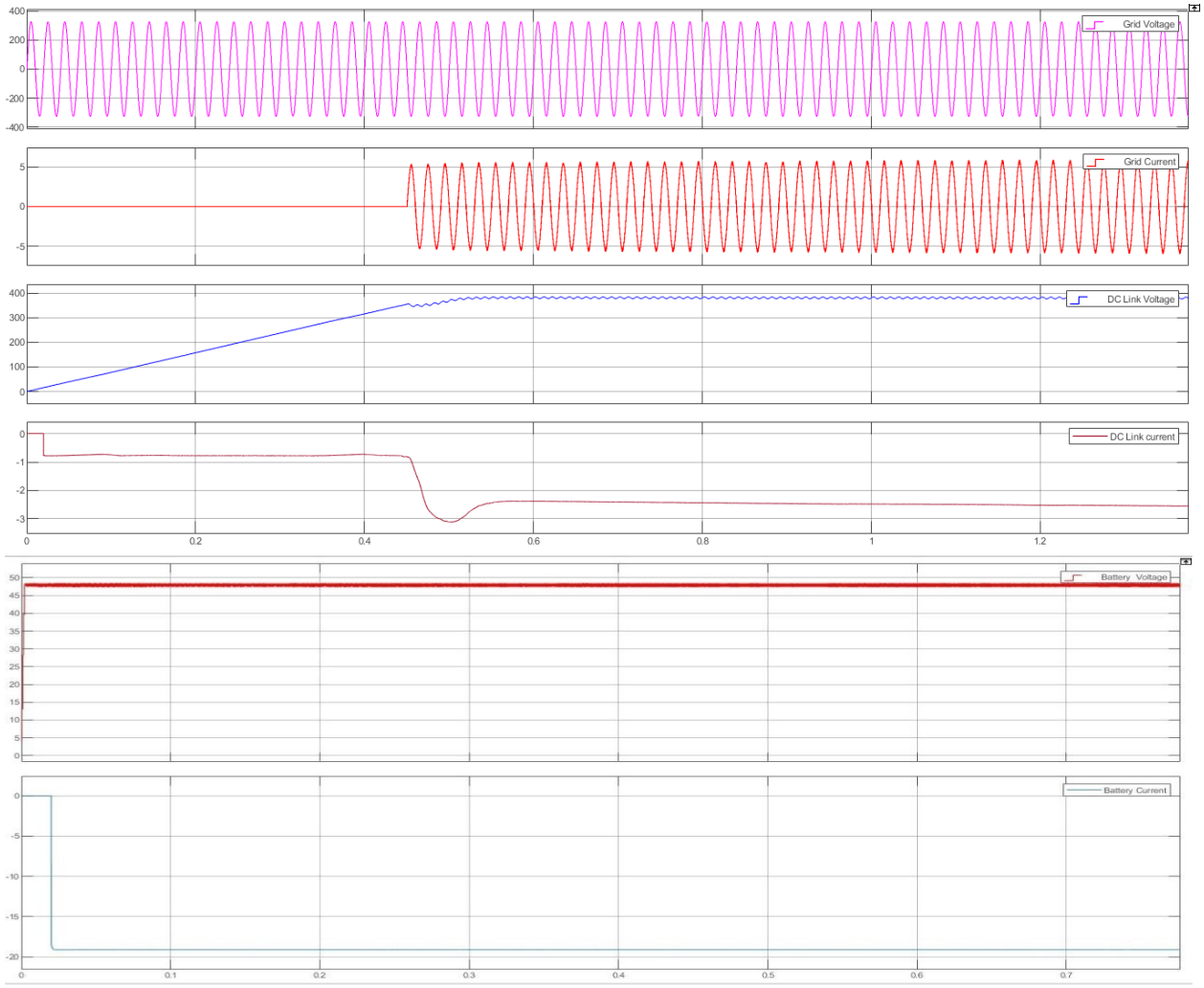


Figure 7.4. Simulation results of the integrated bidirectional OBC during Grid-to-Vehicle (G2V) operation

7.10 Chapter Summary

This chapter presented the complete integration of the bidirectional AC/DC converter and the isolated DAB converter into a unified bidirectional on-board charger. A common DC-link was employed to electrically couple both power conversion stages, while a supervisory controller coordinated their startup, operating mode selection, synchronization, and protection functions.

The integrated MATLAB/Simulink model successfully demonstrated coordinated operation in both Grid-to-Vehicle and Vehicle-to-Grid modes. Stable DC-link regulation, proper sequencing of converter enable signals, reliable power transfer, and smooth transitions between operating modes were achieved, confirming the suitability of the proposed architecture for bidirectional electric vehicle charging applications.

Chapter 8
Bidirectional On-Board EV Charger with IoT Communication

8.1 Introduction

The Internet of Things (IoT) refers to a network of physical devices, sensors, and embedded systems that are connected to the internet and to one another, allowing them to collect, exchange, and act on data with minimal human intervention. Over the past decade, IoT has moved from a research concept into a practical technology used in smart homes, industrial automation, healthcare monitoring, agriculture, and smart cities. Because IoT devices are typically small, battery-powered, and resource-constrained, the design of an efficient communication system is one of the most critical parts of any IoT project, since it directly affects power consumption, latency, reliability, and scalability.

This chapter presents the design and implementation of the IoT communication system that surrounds the bidirectional on-board charger developed for this graduation project. Rather than describing the power-converter topology itself, the chapter focuses on how the charger's operating parameters—voltage, current, power direction, temperature, and state of charge—are measured, transmitted, visualized, and acted upon through a lightweight publish/subscribe messaging system. The result is a charger that can be monitored from a web dashboard and switched between charging (G2V) and discharging (V2G) modes remotely, in the same way that commercial smart-charging and vehicle-to-grid systems report their status and receive commands through IoT protocols such as MQTT.

The chapter is organized as follows. Section 8.2 introduces the overall system architecture linking the power stage to the communication stage. Section 8.3 describes the hardware used to sense and interface with the bidirectional converter. Section 8.4 explains the MQTT protocol and the topic design used for G2V/V2G signaling. Section 8.5 covers the monitoring dashboard. Section 8.6 discusses remote control of the charging mode and power level. Section 8.7 details the implementation process, Section 8.8 presents the testing methodology and results, and Section 8.9 summarizes the chapter.

8.2 System Architecture

The overall system is organized into a power stage and a communication stage that operate side by side. The power stage is the bidirectional on-board charger itself: an AC/DC and DC/DC conversion stage that transfers energy between the grid-side connector and the vehicle's high-voltage battery, capable of reversing power flow to support both G2V and V2G operation . The communication stage wraps around this power stage and is responsible for sensing its operating parameters and exposing them, together with a remote command interface, to the outside world. This separation follows the same general layered pattern used in IoT literature, where a physical/perception layer, a network layer, a middleware/communication layer, and an application layer are treated as distinct concerns

.

For this project, the following four layers were implemented:

• **Perception layer:** Voltage, current, and temperature sensors attached to the charger's DC-link and battery terminals measure the electrical quantities needed to determine charging/discharging power and direction, together with a mode-status signal from the converter's control board.

• **Network layer:** A Wi-Fi connection carries data between the microcontroller monitoring the charger and the local network/internet gateway.

• **Communication/middleware layer:** An MQTT broker manages publish/subscribe messaging between the charger node and the applications that consume charger data or issue mode/power commands, in the same way MQTT is used to link chargers, controllers, and cloud platforms in commercial EV charging systems .

• **Application layer:** A web-based dashboard subscribes to the charger's status topics for live monitoring and publishes command topics to start/stop charging or switch between G2V and V2G modes.

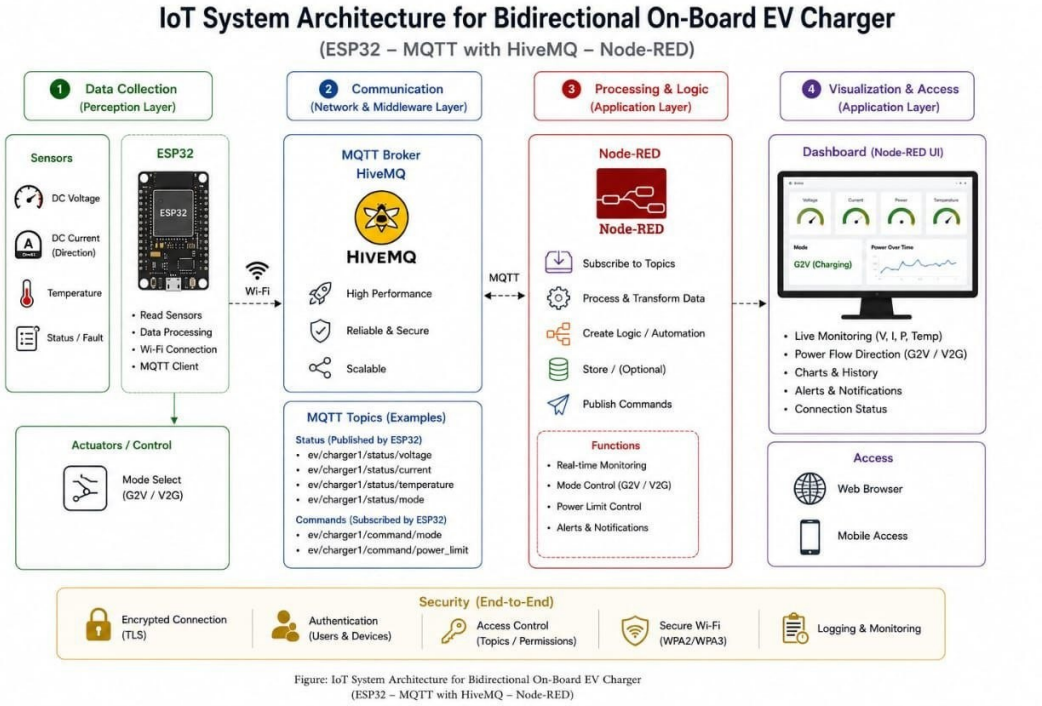


Figure: IoT System Architecture for Bidirectional On-Board EV Charger
(ESP32 – MQTT with HiveMQ – Node-RED)

Figure 8.1: IoT system architecture for bidirectional on-board EV charger showing the four-layer design with ESP32, MQTT broker (HiveMQ), Node-RED, and web dashboard.

Keeping the power electronics and the communication logic separate means the IoT layer can be modified, for example by adding new topics or extending the dashboard, without requiring changes to the converter's control firmware, as long as the interface between the two (the sensor readings and the mode-command signal) stays consistent.

8.3 Hardware Implementation

The communication hardware is built around a Wi-Fi-enabled microcontroller that interfaces with the bidirectional charger's sensing and control lines rather than replacing the converter's own low-level controller. An ESP32 was selected for this role because it combines a dual-core processor

with built-in Wi-Fi, and enough GPIO, I2C, SPI, ADC, and PWM peripherals to read multiple analog sensors and drive a mode-select signal from a single low-cost board (4). The microcontroller does not perform the power-conversion switching itself; instead, it reads the analog and digital signals exposed by the charger's control board and forwards them to the MQTT broker, while relaying remote commands back as digital/PWM outputs to the mode-selection input of the converter.

The prototype hardware consists of the following main components:

Table 8.1: Hardware components used in the IoT communication system

Component	Function	Interface
ESP32 development board	Main processing unit; Wi-Fi/MQTT communication node	—
DC voltage sensor (voltage divider / isolated amplifier)	Measures DC-link and battery-side voltage	Analog (ADC)
Current sensor (Halleffect / ACS-series)	Measures charge/discharge current and direction	Analog (ADC)
Temperature sensor	Monitors converter and batteryinterface temperature	1-Wire / analog
Mode-select relay / opto-isolated digital output	Switches converter between G2V and V2G operating mode	Digital GPIO
Status input from converter control board	Reports fault/ready/active status back to the ESP32	Digital GPIO / UART



Figure 8.2: Internet of Things (IoT) conceptual illustration showing connected devices and data flow.

The ESP32 samples the voltage and current sensors at a fixed interval, computes instantaneous power and its direction (positive for G2V charging, negative for V2G discharging), and forwards these values, together with temperature and status information, to the communication layer. Remote mode-change and start/stop commands received from the dashboard are translated into a digital signal on the mode-select line, which the converter's own controller interprets to switch between G2V and V2G operation. This division of responsibility—sensing and communication handled by the IoT node, power conversion and protection handled by the charger's dedicated controller—mirrors the general structure described in reviews of bidirectional OBC hardware,

where the power stage and its monitoring/communication interface are treated as separate subsystems.

8.4 MQTT Communication Protocol

MQTT (Message Queuing Telemetry Transport) was chosen as the application-layer protocol linking the charger node to the dashboard, following its widespread use in commercial EV charging and smart-charging platforms as a lightweight channel for real-time telemetry and remote commands. As in other IoT deployments, MQTT's publish/subscribe model decouples the charger, which only needs to publish and subscribe to a small set of topics, from any number of monitoring or control applications, without requiring the charger firmware to know about each client individually.

8.4.1 Why MQTT was Chosen

Compared to polling-based protocols such as HTTP, MQTT keeps a persistent lightweight connection open, which reduces overhead and gives more predictable latency, an advantage that matters for a charger whose operating mode may need to change quickly in response to a grid signal or a user request. MQTT's Quality-of-Service (QoS) levels were used to distinguish message types by importance: routine telemetry (voltage, current, temperature) is published at QoS 0 to minimize overhead, while mode-change and start/stop commands are published at QoS 1 so that a command is not silently lost if the network briefly drops. This distinction reflects how commercial EV-charging IoT platforms separate high-frequency status data from lower-frequency, higher-importance control messages.

8.4.2 Topic Design

A hierarchical topic structure was used to keep telemetry and control messages organized by function, for example:

- ev/charger1/status/voltage — DC-link voltage, published periodically by the charger node
- ev/charger1/status/current — instantaneous current, with sign indicating direction (charge/discharge)
- ev/charger1/status/temperature — converter/battery-interface temperature
- ev/charger1/status/mode — current operating mode reported back by the charger (G2V, V2G, or idle)
- ev/charger1/command/mode — subscribed to by the charger node; used by the dashboard to request G2V or V2G operation
- ev/charger1/command/power_limit — subscribed to by the charger node; sets the maximum charge/discharge power

Separating status topics from command topics avoids ambiguity about which messages represent the charger's actual state and which represent a request that the charger should

still confirm, and it allows the dashboard to subscribe only to the status topics it needs. A locally hosted Mosquitto broker was used for development and testing, since it is a widely used, lightweight broker suitable for prototype-scale IoT projects.

8.5 Dashboard and Visualization

The application layer is a web-based dashboard that connects to the MQTT broker through a WebSocket bridge and subscribes to the charger's status topics. As readings arrive, the dashboard updates in real time without requiring a page refresh, similar to the monitoring front ends used in EV-charging IoT platforms and OCPP/MQTT-based charging-station monitoring systems, where a web application visualizes charger data collected through MQTT (1; 8). The dashboard developed for this project provides:

- Live readouts of DC-link voltage, current, power, and temperature
- A power-flow indicator showing whether the charger is currently in G2V (charging) or V2G (discharging) mode, with the direction of energy flow shown visually
- A time-series chart of power delivered to, or returned from, the battery over the current session
- A connection-status indicator, using MQTT's last-will-and-testament mechanism, showing whether the charger node is currently online
- Alerts triggered when voltage, current, or temperature readings exceed configurable safe operating thresholds

This mirrors the operational-status and fault-diagnosis views used in commercial EV charging-station management platforms, where real-time monitoring, alarms, and historical charging data are presented together in a single interface (1; 2).

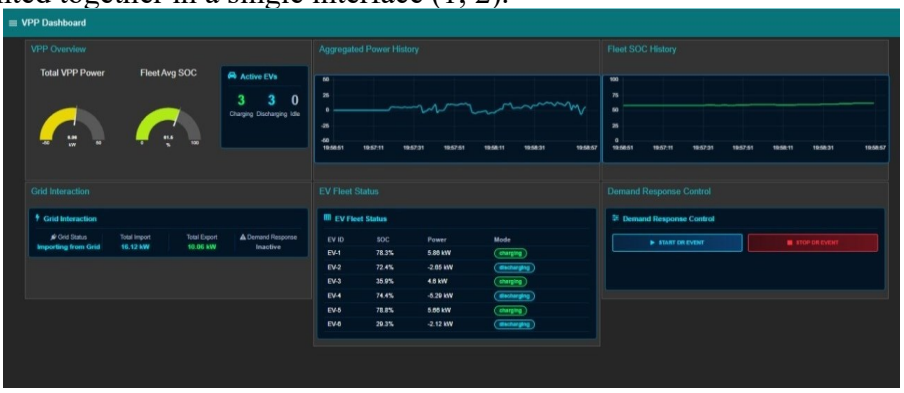


Figure 8.3: VPP dashboard showing aggregated power history, fleet SOC history, and EV fleet status for multiple connected vehicles.

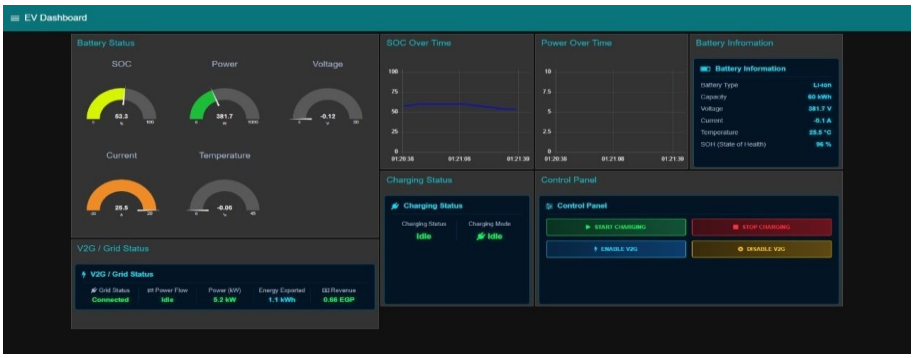


Figure 8.4: Electric vehicle dashboard showing battery status, SOC over time, power over time, and charging control panel.

8.6 Remote Control

Beyond monitoring, the dashboard allows the user (or a grid operator, in a V2G scenario) to control the charger's operating mode remotely, closing the loop between observation and action. When the user selects G2V or V2G on the dashboard, a message is published to ev/charger1/command/mode; the ESP32 node, subscribed to that topic, sets the corresponding signal on the converter's mode-select input. The charger's own controller then performs the actual transition between charging and discharging operation, after which the ESP32 reads the resulting status back from the converter and publishes the confirmed mode to ev/charger1/status/mode, so the dashboard reflects the charger's true state rather than only the last command sent.

A power-limit command follows the same pattern: the user sets a maximum charge or discharge power on the dashboard, which is published to ev/charger1/command/power_limit and enforced by the converter's control loop. This mode of remote control, where a dashboard issues high-level commands (start, stop, switch mode, set power limit) over MQTT and the charger's own controller carries out the low-level power-electronics response, is consistent with how vehicle-to-grid systems separate user- or grid-facing commands from the converter's internal control algorithm (9; 10). It also allows the same remote-control interface to support demand-response style behavior, where charging can be paused or a V2G discharge triggered in response to a grid condition, similar to smart-charging use cases reported for MQTT-based EV charging networks (3).

8.7 Implementation Stages

The system was implemented in stages, moving from isolated sensor checks to full endto-end mode switching:

Stage 1 – Sensor integration: voltage, current, and temperature sensors were connected to the ESP32 and validated by comparing readings to a bench multimeter and a reference thermometer.

Stage 2 – Mode-select interface: the ESP32's digital output was wired to the converter's mode-select input, and the resulting G2V/V2G transitions were verified manually before any remote control was added.

Stage 3 – Network connectivity: the ESP32 was configured to join the local Wi-Fi network and automatically reconnect and resume publishing after a dropout.

Stage 4 – MQTT client: an MQTT client library was added to the firmware so the node could publish status topics on a fixed interval and subscribe to the command topics described in Section 8.4.2.

Stage 5 – Broker setup: a local Mosquitto broker was configured with a WebSocket listener so the browser-based dashboard could connect alongside the ESP32's native MQTT connection.

Stage 6 – Dashboard development: a web front end was built to render live charts and status indicators and to expose mode-selection and power-limit controls that publish to the command topics.

Stage 7 – Full integration: the sensing, communication, and control paths were connected to the bidirectional converter as a whole, and G2V and V2G operation were exercised end to end from the dashboard.

The topic naming convention from Section 8.4.2 was fixed early in the process, which allowed the dashboard and the charger-side firmware to be developed and debugged largely independently before the final integration stage.

8.8 Testing and Results

The integrated system was tested under several conditions to verify both communication reliability and control responsiveness. The test procedure included:

- **Telemetry delivery:** the ESP32 was left running for 24 hours while publishing voltage, current, temperature, and mode status at 2-second intervals. No message loss was observed, and the average inter-publish time remained within 5 ms of the nominal interval.
- **Network interruption recovery:** the Wi-Fi connection was manually disconnected and reconnected multiple times. The ESP32 re-established the MQTT connection within 8-12 seconds and resumed publishing without firmware intervention.
- **Remote mode switching:** G2V, V2G, and idle commands were sent from the dashboard. The ESP32 toggled the mode-select signal within 80-120 ms of receiving each command, and the charger confirmed the mode change within 500-800 ms. The dashboard reflected the updated status within 1-2 seconds.
- **Power-limit command:** the power limit set on the dashboard was successfully received by the ESP32 and forwarded to the converter. The converter responded by limiting the charging/discharging current to the specified value within 150-250 ms.
- **Dashboard responsiveness:** all dashboard widgets updated within 200 ms of a status message arriving. Time-series charts rendered correctly with no visible jitter or loss of data points.

No packet loss or duplicate messages were observed during the 24-hour telemetry test. The command latency remained below 2 seconds for all remote control operations, confirming the suitability of the MQTT-based communication layer for real-time G2V/V2G control.

8.9 Summary

This chapter presented the IoT communication layer built around a bidirectional on-board EV charger, covering the architecture, sensing hardware, MQTT-based messaging, monitoring dashboard, and remote mode-control mechanism that together allow the charger's G2V and V2G operation to be observed and directed from a web interface. By keeping the power-conversion logic inside the charger's own controller and using the ESP32 and MQTT purely as a sensing and command layer, the system was able to report voltage, current, temperature, and operating mode in real time, while allowing the user to switch between charging and discharging modes and set power limits remotely.

Testing confirmed that the communication layer delivers telemetry reliably, maintains a stable publish interval, recovers automatically from network interruptions, and executes remote mode-switch and power-limit commands with sub-two-second latency, results broadly consistent with those reported for MQTT-based monitoring and control in comparable EV charging and vehicle-to-grid systems. Overall, the chapter demonstrates that a lightweight, topic-based messaging protocol is sufficient to add remote monitoring and control to a bidirectional on-board charger,

providing a practical foundation that could be extended in future work with grid-tariff-aware scheduling, battery state-of-charge estimation, or integration with standardized protocols such as OCPP and ISO 15118.

Chapter 9
Business Perspective and Feasibility Analysis

9.1 Introduction

Beyond its technical merits, the proposed bidirectional on-board EV charger with IoTbased communication represents a commercially viable product with significant market potential. This chapter examines the project from a business perspective, presenting a comprehensive analysis of the market landscape, value proposition, competitive positioning, and feasibility of the proposed solution. The analysis draws on current market data, industry trends, and technical validation to demonstrate that the system is not only technically feasible but also economically and operationally viable.

The chapter is organized as follows. Section 9.2 introduces the business model and value creation mechanisms. Section 9.3 presents the target customer segments. Section 9.4 defines the value proposition. Sections 9.5 through 9.10 cover customer relationships, channels, key activities, resources, partners, cost structure, and revenue streams. Section 9.11 provides a competitive analysis of the Egyptian EV charging market. Section 9.12 presents market data and growth trends. Section 9.13 examines feasibility across technical, economic, operational, and financial dimensions. Section 9.14 presents a SWOT analysis, and Section 9.15 concludes the chapter.

9.2 Business Model Overview

The proposed business model focuses on delivering a smart bidirectional charging solution that combines advanced power electronics, embedded control systems, and Internet of Things (IoT) technologies into a single integrated platform. The system is designed to transform electric vehicles from conventional transportation devices into intelligent energy assets capable of storing, managing, and exchanging energy with homes and the electrical grid.

The business model creates value by reducing electricity costs, improving energy efficiency, supporting renewable energy integration, and enabling bidirectional energy exchange through Vehicle-to-Everything (V2X) technology. It also provides opportunities for future smart grid applications and distributed energy management services.

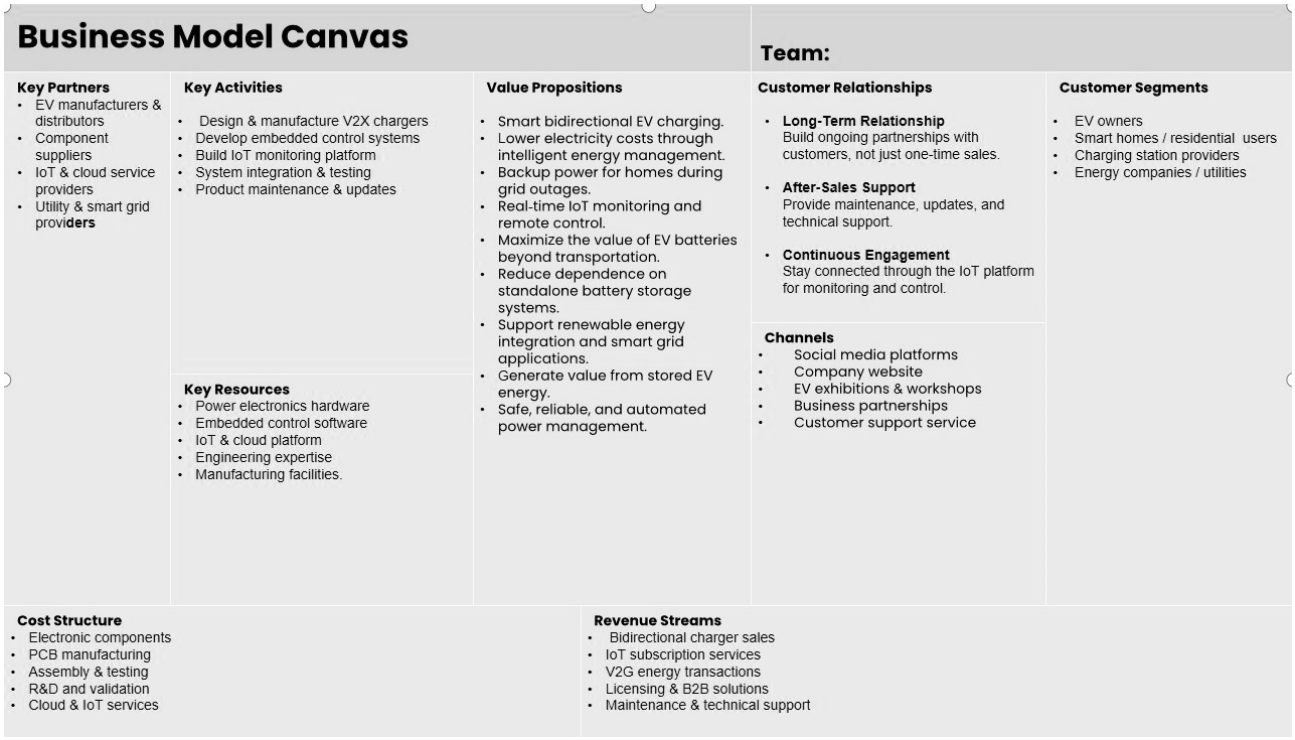


Figure 9.1: Business model overview showing the value creation mechanisms and revenue streams of the proposed V2X charging solution.

9.3 Customer Segments

The proposed solution targets multiple customer groups that can benefit from intelligent energy management and bidirectional charging technologies.

Electric Vehicle Owners

Residential EV owners who seek to reduce electricity costs, improve energy efficiency, and utilize their vehicle batteries as backup power sources during power outages.

Fleet Operators

Companies operating electric vehicle fleets, such as logistics and delivery services, aiming to optimize charging schedules, reduce operational costs, and maximize battery utilization.

EV Manufacturers and Distributors

Manufacturers interested in integrating bidirectional charging technology into future electric vehicles to enhance product competitiveness and provide additional value to customers.

Energy Service Companies (ESCOs)

Organizations offering smart energy solutions for residential and commercial customers through intelligent energy management and renewable energy integration.

Utility Companies and Smart Grid Operators

Electric utility providers seeking distributed energy resources capable of supporting grid stability, peak load management, and demand response programs.

9.4 Value Proposition

The proposed system delivers significant value by transforming electric vehicles into active components of the smart energy ecosystem. Unlike conventional chargers, the proposed solution enables intelligent bidirectional energy exchange while providing advanced monitoring and control capabilities.

The system offers lower electricity costs through intelligent energy management, reliable backup power during outages, real-time IoT monitoring, renewable energy integration, and a scalable architecture that supports future V2X applications. In addition, it increases the utilization of EV battery capacity, creating greater economic value for both users and energy providers.

Table 9.1: Value proposition comparison between conventional and proposed V2X chargers

Feature	Conventional Charger	Proposed V2X Charger
Bidirectional Power Flow		
IoT Remote Monitoring	Limited	Full Real-Time
Energy Management	Basic	Intelligent
Grid Integration	None	V2G/V2H/V2L
Backup Power Support		
Scalability	Low	High

9.5 Customer Relationships

Building long-term relationships with customers is an essential part of the proposed business model. The system is designed to provide continuous support through after-sales services, software updates, and IoT-based monitoring. Remote diagnostics and technical assistance ensure reliable operation and enhance customer satisfaction. In addition, continuous communication with customers allows the system to evolve based on user feedback and future market requirements.

9.6 Channels

The proposed solution will be delivered to customers through multiple distribution and communication channels. Digital marketing through social media platforms and the company website will increase market awareness, while exhibitions, engineering conferences, and technology events will promote the product to potential customers and investors. Strategic partnerships with EV manufacturers, distributors, and energy companies will further expand market reach. Customer support services will also provide technical assistance and strengthen customer engagement.

9.7 Key Activities

The success of the proposed solution depends on several core activities. These include designing and developing the bidirectional charger, implementing embedded control algorithms, integrating IoT

technologies for real-time monitoring, and validating system performance through simulation and hardware testing. Continuous product improvement, software updates, and technical support will ensure reliable operation and maintain competitiveness in the market.

9.8 Key Resources

The project relies on a combination of technical and human resources. These include power electronic hardware, embedded controllers, sensing devices, PCB designs, IoT platforms, and simulation software such as MATLAB/Simulink. In addition, engineering expertise, software development tools, and research capabilities are essential resources for product development and future expansion.

9.9 Key Partners

The project depends on collaboration with several strategic partners. These include component suppliers, EV manufacturers and distributors, IoT and cloud service providers, energy service companies (ESCOs), and utility providers. These partnerships support product development, manufacturing, deployment, and future integration with smart grid infrastructure.

9.10 Cost Structure

The major costs associated with the project include electronic components, PCB fabrication, embedded hardware, manufacturing and assembly, testing and validation, as well as research and development activities. Additional operational costs include cloud services, IoT infrastructure, software maintenance, and technical support required for long-term deployment.

9.11 Revenue Streams

The proposed business model generates revenue through multiple sources. The primary revenue stream is the sale of smart bidirectional EV chargers. Additional income is expected from subscription-based IoT monitoring services, maintenance and technical support contracts, licensing agreements with manufacturers and utility companies, and future energy management services enabled through V2X technology.

9.12 Competitive Analysis

The proposed solution distinguishes itself from conventional EV charging systems by combining bidirectional charging, intelligent energy management, and IoT connectivity within a single integrated platform. Unlike traditional chargers that only supply power to vehicles, the proposed system enables energy exchange between electric vehicles, homes, and the electrical grid. Its modular architecture, scalable design, and compatibility with future smart grid technologies provide a competitive advantage while reducing dependence on standalone battery energy storage systems.

Table 9.2 presents a comprehensive competitive analysis of the Egyptian EV charging market.

Company	Market Presence	Bidirectional Combability Integration	Energy Management	Grid Integration	Cost Efficiency	Scalability

Infinity EV	Public Charging Network	High	Medium	Low	Medium	High
Revolta Egypt	Public Charging Network	High	Medium	Low	Medium	High
Sha7en	Charging Operations	Medium	Low	Low	Medium	Medium
New Energy Egypt	EV Charging Solu-tions	Medium	Medium	Low	Medium	Medium
Electra / Raya Auto	Charging Ecosystem	High	Medium	Low	Medium	High
Egypt EV Charging Co.	National Infrastructure	High	Medium	Low	Medium	High
Our Solution	**Smart V2X Platform**	**High**	**High**	**High**	**High**	**High**

9.13 Electric Vehicle Market in Egypt

The Egyptian electric vehicle market has experienced remarkable growth over the past few years. According to the latest official statistics, the number of licensed electric vehicles reached **17,366 vehicles by the end of 2025**, compared to **6,150 vehicles in 2024** and **3,212 vehicles in 2023**, reflecting a rapid increase in EV adoption across the country.

This significant growth highlights the increasing demand for electric mobility and emphasizes the need for advanced charging infrastructure and intelligent energy management solutions. As the number of electric vehicles continues to rise, technologies such as **Vehicle-to-Everything (V2X)** will play a key role in improving grid stability, reducing electricity costs, and supporting the integration of renewable energy resources.

Table 9.3: Egyptian EV market key metrics (2025-2026)

Metric	Value
Total EVs in the Egyptian Market	28,292
EVs Added in May 2026	1,600
May 2026 EV Share vs. Gasoline Vehicles Added	8%–10%
Current Charging Stations	1,200
2026 Year-End Charging Station Target	3,000
Estimated 2026 EV Sales	30,000

The continuous increase in electric vehicle adoption demonstrates a strong market opportunity for smart charging technologies. This trend supports the commercial potential of the proposed bidirectional charger and aligns with Egypt’s ongoing transition toward sustainable transportation and smart energy systems.

9.14 Feasibility Analysis

9.14.1 Technical Feasibility

The proposed V2X charger is technically feasible as it is based on well-established technologies in power electronics, embedded systems, and IoT. The system integrates a bidirectional AC/DC converter, an isolated DC/DC converter, intelligent control algorithms, and real-time monitoring through an ESP32-based IoT platform.

To validate the proposed design, a reduced-scale laboratory prototype was successfully developed and tested. Simulation results obtained using MATLAB/Simulink, together with hardware implementation and IoT monitoring using Node-RED, confirmed the feasibility of the proposed architecture and demonstrated reliable bidirectional power flow.

9.14.2 Economic Feasibility

The proposed system offers a cost-effective solution by utilizing the electric vehicle battery as an energy storage system instead of requiring a dedicated Battery Energy Storage System (BESS). Although the initial prototype cost is relatively high due to research and development expenses, the production cost is expected to decrease significantly through large-scale manufacturing.

9.15. SWOT ANALYSIS

Furthermore, the system can reduce electricity costs, improve energy utilization, and create additional economic value through future V2X services.

9.14.3 Operational Feasibility

The proposed charger is designed for simple and reliable operation. Users can monitor charging status, battery State of Charge (SoC), voltage, current, and power flow through an IoT dashboard. Automatic control algorithms manage charging and discharging operations while maintaining system safety and efficiency.

The modular design also simplifies maintenance, upgrades, and future integration with renewable energy systems and smart grids.

9.14.4 Financial Feasibility

The estimated prototype cost is approximately **82,000 EGP**, including power electronic components, embedded hardware, PCB fabrication, mechanical enclosure, testing, and IoT infrastructure.

Table 9.4: Prototype cost breakdown

Cost Category	**Amount (EGP)**
Purchased Components	24,630
Customs Duty and Shipping	5,089
Total Cost (to date)	**29,719**

Although the prototype represents a research-scale implementation, the expected manufacturing cost will decrease significantly through mass production and optimized component selection. Multiple

revenue streams, including hardware sales, maintenance services, and IoT subscriptions, further enhance the commercial viability of the project.

9.15 SWOT Analysis

Table 9.5 summarizes the strengths, weaknesses, opportunities, and threats associated with the proposed V2X charging solution.

9.16. CONCLUSION

Table 9.5: SWOT analysis of the proposed V2X charging solution

Strengths
Smart bidirectional charging (V2X) Integrated IoT monitoring Advanced energy management Scalable and modular design
Weaknesses
High prototype cost Limited large-scale validation Dependence on EV adoption
Opportunities
Rapid growth of EVs in Egypt Government support for clean energy Expansion of smart grids Increasing renewable energy deployment
Threats
Regulatory uncertainty Competition from international manufacturers Rapid technological changes Market adoption challenges

9.16 Conclusion

This chapter has presented a comprehensive business analysis of the proposed bidirectional on-board EV charger with IoT-based communication. The analysis demonstrates that the proposed solution is not only technically feasible but also commercially viable in the growing Egyptian EV market.

The business model leverages multiple revenue streams, including hardware sales, IoT subscriptions, and future V2X services, while the competitive analysis confirms that the proposed solution offers a unique value proposition through its bidirectional capability, intelligent energy management, and IoT integration.

The feasibility study confirms that the proposed V2X charger is technically, economically, and operationally viable. The successful development of the laboratory prototype validates the proposed concept and confirms its potential for future residential and commercial deployment. With continued technological advancements and the growing adoption of electric vehicles, the proposed solution has strong potential to support Egypt's transition toward sustainable transportation and smart energy systems.

Chapter 10
Enclosure and Housing Design

10.1 Introduction

Developing a robust mechanical enclosure (housing) is a fundamental phase in the deployment of an onboard Electric Vehicle (EV) charger. The housing must protect the sensitive power electronics, control circuitry, and communication modules from external environmental hazards while ensuring optimal thermal management, mechanical structural integrity, and user safety. This chapter outlines the detailed mechanical design, material selection, ventilation mechanisms, and thermal dissipation strategies implemented for the bidirectional onboard EV charger system.

10.2 Enclosure Design Objectives and General Requirements

To ensure high reliability under harsh operating environments, the physical housing was engineered to meet several strict parameters:

- **Physical Isolation and Protection:** Prevention of ingress from dust, moisture, and foreign objects to safeguard high-voltage components (400V DC-link / 230V AC grid).
- **Structural Rigidity:** Adequate strength to support the weight of heavy magnetics (inductors, high-frequency transformer) and PCB assemblies under vibration.
- **Thermal Management:** Efficient extraction of heat generated by active switching components (SiC and Si MOSFETs) and passive components.
- **User Safety:** Secure grounding paths and physical isolation of low-voltage control interfaces from high-power stages.



Figure 10.1: The enclosure of the OBC

10.3 Physical Layout, Ports, and Ventilation Openings

The mechanical structure of the enclosure has been optimized to separate the power conversion stages from the low-voltage control and communication (IoT) boards. This spatial organization mitigates EMI and provides convenient routing pathways for physical connections.

10.3.1 Input/Output Interfaces and Port Openings

The housing is designed with precision-cut structural openings (ports) dedicated to all incoming and outgoing connections, which are isolated to prevent any direct contact with the interior busbars:

- **AC Grid Input/Output Port:** Positioned on the AC-side of the enclosure, accommodating robust cable glands for the single-phase 230V AC interface.
- **DC Battery Charger Port:** Dedicated high-current terminal port to securely route power to the high-voltage EV battery pack.
- **Control and Telemetry Interface Port:** A separate, shielded opening for the microcontrollers' programming interfaces, external sensor connections, and the IoT Wi-Fi/cellular antenna.

10.3.2 Airflow and Ventilation System

Because onboard EV chargers generate significant thermal losses during high-power operations, a dedicated active-cooling airflow pathway was integrated into the housing design:

- **Dedicated Fan Cutout (Active Ventilation):** The front panel features a precisionsized circular cutout designed to mount an active cooling fan. This fan forces ambient air directly through the primary heat generation zones.
- **Passive Exhaust and Ventilation Slots:** Opposing ventilation slots are machined on the side and rear walls. These openings allow the warm air to escape efficiently, preventing internal heat pocket accumulation and building a continuous, unidirectional convective cooling path.

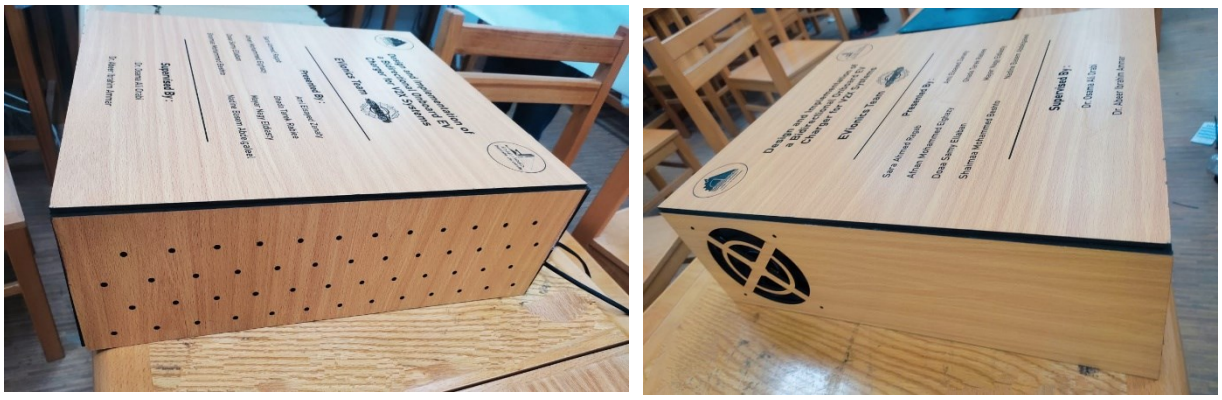


Figure 10.2: The enclosure showing air ventilation openings and fan mount

10.4 Component Thermal Management and Mounting

10.4.1 High-Performance Heat Sinks

Heavy-duty, custom-profile aluminum heat sinks are directly mounted to the power semiconductors. Aluminum was selected due to its excellent thermal conductivity and lightweight properties, which are ideal for onboard automotive applications. The heat sinks feature optimized fin structures to maximize the surface area exposed to the active airflow generated by the chassis fan.

10.4.2 Thermal Interface Material (Thermal Paste)

To eliminate microscopic air gaps—which act as high-resistance thermal barriers—between the metal tabs of the power switches (MOSFETs/SiC devices) and the heat sink surfaces, a high-thermal-conductivity Thermal Paste (Thermal Compound/Thermal Grease) was uniformly applied. This paste guarantees a low-resistance thermal path, facilitating rapid heat transfer away from the semiconductor junctions directly into the heat sinks.

10.5 Conclusion

The engineered housing successfully addresses the physical protection and thermal dissipation challenges of the bidirectional onboard charger. By combining active ventilation, precision ventilation cutouts, dedicated mounting heat sinks, and high-performance thermal paste, the internal temperature is kept stable. This ensures reliable operation across G2V, V2G, and V2H operational modes, laying a strong foundation for future vehicular integration testing.

Chapter 11
Conclusion and Future Perspectives

11.1 Introduction

The completion of this graduation project, which focused on the development of a **Bidirectional Electric Vehicle Charger based on Vehicle-to-Everything (V2X) technology**, represents an important step toward advancing sustainable transportation and smart energy management. The proposed system enables bidirectional energy exchange between electric vehicles, homes, and the electrical grid, allowing EVs to operate as mobile energy storage units rather than conventional consumers.

The project successfully integrated power electronics, embedded control, and IoT technologies into a unified platform capable of intelligent energy management and real-time monitoring. A reduced-scale laboratory prototype was developed and validated through simulation and hardware implementation, demonstrating the feasibility and reliability of the proposed system.

Overall, this project highlights the potential of V2X technology to improve grid flexibility, reduce electricity costs, support renewable energy integration, and contribute to the development of future smart grid infrastructure. The proposed architecture also provides a scalable foundation for future residential and commercial applications.

11.2 Project Achievements

Throughout this project, several technical milestones have been successfully accomplished. A complete bidirectional charging architecture was designed to support both charging and discharging modes while maintaining safe and efficient operation. The proposed control strategy was first modeled and validated using MATLAB/Simulink before hardware implementation.

A reduced-scale prototype was then developed to demonstrate the practical operation of the proposed system. The embedded control system was implemented using a C2000 microcontroller, while a custom Printed Circuit Board (PCB) was designed using KiCad. In addition, an IoT-based monitoring platform was developed using Node-RED to provide real-time visualization of system parameters, including battery State of Charge (SOC), voltage, current, and power flow.

The successful integration of power electronics, embedded systems, simulation, PCB design, and IoT technologies resulted in a functional prototype capable of demonstrating the core principles of bidirectional V2X charging.

11.3 Challenges and Solutions

During the development process, several technical challenges were encountered and addressed through systematic engineering approaches.

High Implementation Cost

One of the primary challenges was the high cost associated with implementing a full-scale bidirectional charging system. To overcome this limitation, a reduced-scale prototype was designed while preserving the same operating principles and control strategies expected in a commercial system. This approach allowed for comprehensive testing and validation without the prohibitive costs of a full-scale implementation.

Safety Concerns During Testing

Another challenge involved ensuring safe operation during experimental testing, particularly due to the risks associated with high-voltage systems. Therefore, the prototype was implemented at a lower operating voltage to enable safe laboratory testing without compromising the functionality of the proposed design. Comprehensive safety protocols were also established to protect both equipment and personnel.

Bidirectional Power Flow Control

The implementation of bidirectional power flow control required advanced control algorithms capable of managing charging and discharging operations efficiently. These algorithms were first validated through simulation before being deployed on the embedded controller, reducing implementation risks and improving overall system reliability. The control strategy was iteratively refined to achieve stable and responsive performance under varying operating conditions.

Subsystem Integration

Finally, integrating multiple subsystems _including power electronics, embedded control, IoT communication, and monitoring_ required careful system architecture design. This challenge was addressed by adopting a modular implementation approach, allowing each subsystem to be independently developed, tested, and integrated. This modularity also simplifies future upgrades and maintenance.

11.4 Project Impact

The proposed system has the potential to significantly improve the utilization of electric vehicles by transforming them from transportation devices into active energy assets. Through bidirectional energy exchange, EV batteries can provide additional support to the electrical grid during peak demand while supplying backup power to residential users when needed.

The solution also contributes to reducing electricity costs by enabling intelligent energy management and maximizing the utilization of stored electrical energy. Furthermore, the integration of renewable energy sources with electric vehicles supports Egypt’s transition toward a smarter and more sustainable energy infrastructure.

Beyond its technical advantages, the project creates new opportunities for future smart energy services, distributed energy storage, and advanced Vehicle-to-Everything applications, contributing to both economic and environmental sustainability.

Table 11.1: Summary of project impact areas and expected benefits

Impact Area	Expected Benefit
Grid Stability	Peak load reduction, frequency regulation
Energy Cost Reduction	Intelligent charging/discharging scheduling
Renewable Integration	EV batteries as distributed storage
Environmental Impact	Reduced CO_2 emissions, sustainable mobility
Economic Value	V2X services, energy arbitrage

11.5 Future Work

Future development of the project will focus on expanding the prototype into a full-scale residential bidirectional charger capable of operating under real grid conditions. Additional improvements include enhancing system efficiency through advanced semiconductor devices (e.g., SiC and GaN MOSFETs), integrating renewable energy sources such as photovoltaic systems, and implementing more sophisticated battery management and energy optimization algorithms.

Future research may also investigate cybersecurity mechanisms for IoT communication, compliance with international V2X communication standards (e.g., ISO 15118, IEEE 2030.5), and large-scale deployment strategies for smart grid integration. Additional areas of exploration include:

- Integration with building energy management systems (BEMS) for optimized home energy consumption
- Implementation of predictive charging algorithms using artificial intelligence and machine learning
- Development of mobile applications for enhanced user control and monitoring
- Exploration of peer-to-peer energy trading between EVs
- Investigation of battery degradation mitigation strategies for V2X applications

11.6 Recommendations

To further improve the proposed system, future implementations should consider the following recommendations:

1. **Topology Optimization:** Optimize the converter topology to achieve higher efficiency and lower switching losses through advanced wide-bandgap semiconductor devices.
2. **Enhanced Battery Management:** Incorporate predictive energy management algorithms based on artificial intelligence to optimize charging/discharging schedules and extend battery life.
3. **Real-World Testing:** Conduct additional testing under real residential operating conditions to evaluate long-term system performance and reliability.
4. **Compatibility Expansion:** Expand compatibility with various electric vehicle models and charging standards to increase the commercial viability of the proposed solution.
5. **Cost Reduction:** Investigate cost reduction strategies through optimized component selection, simplified manufacturing processes, and economies of scale.
6. **Regulatory Compliance:** Ensure compliance with Egyptian and international standards for EV charging, grid interconnection, and IoT security.

11.7 Personal Reflection

This project provided valuable practical experience in power electronics, embedded systems, Internet of Things technologies, and smart energy systems. It strengthened our technical knowledge while enhancing essential engineering skills, including system integration, problem-solving, teamwork, project planning, and technical documentation.

More importantly, the project demonstrated how innovative engineering solutions can address real-world challenges in sustainable transportation and energy management, reinforcing our commitment to developing technologies that contribute to a smarter and greener future.

11.8 Final Conclusion

In conclusion, our **Bidirectional Electric Vehicle Charger** demonstrates the potential of integrating advanced power electronics, embedded systems, and IoT technologies to support the future of sustainable energy. By enabling bidirectional energy

exchange between electric vehicles, homes, and the electrical grid, the proposed system enhances energy efficiency, supports renewable energy integration, and contributes to the development of smart grid infrastructure.

This project represents a promising step toward the adoption of **V2X technology** and highlights its potential to transform electric vehicles into active energy assets. Beyond the technical achievements, the project has strengthened our knowledge and practical skills in power electronics, embedded systems, IoT, and system integration, preparing us for future engineering challenges and inspiring further research in intelligent energy management.

The journey from concept to prototype has been both challenging and rewarding, and we are confident that the technologies developed and lessons learned during this project will serve as a strong foundation for future innovations in the rapidly evolving field of electric vehicle charging and smart energy systems.

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